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Interstellar Precursor Mission to Alpha Centauri

Technical Feasibility Assessment

Minimum-Δv trajectories, solar-electric propulsion closure, vehicle mass budgets, program cost, and alternative targets

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Abstract. This report assesses the feasibility of delivering a vehicle with a payload of at least 1 kg to within 2,600 AU of Alpha Centauri within 500,000 years, launched by the end of 2029 on a rideshare to low Earth orbit (LEO), without gravity assists, for a program cost under $10M. Four conclusions follow, in decreasing order of certainty. Trajectory-class feasibility holds: an explicitly integrated solar-electric trajectory, verified by independent re-integration to 0.29%, reaches the required cruise speed by multi-revolution perihelion pumping without nuclear power. No trajectory found closes the 100 kg concept from a LEO start; this verdict is conditional because the search cannot exclude closure below the 22.26 $\mathrm{km\,s^{-1}}$ heliocentric crossover within the 12-year operational horizon. The reference interval's lower end sits 0.6 $\mathrm{km\,s^{-1}}$ above the crossover, the best trajectory found at the design thrust level (23.98 $\mathrm{km\,s^{-1}}$) sits 1.7 $\mathrm{km\,s^{-1}}$ above it, and the best found at four times the design thrust (22.66 $\mathrm{km\,s^{-1}}$) sits 0.4 $\mathrm{km\,s^{-1}}$ above. The concept meets the specification at 100 kg from a geostationary transfer orbit (GTO) start (a delivery-orbit deviation from the ground rule) or at roughly 150 kg from a LEO start with fragile margins, and the end-2029 launch date is conditional on a propulsion qualification campaign within the demonstrated gridded-ion component class. The program cost exceeds the initial $10M cost target under conventional practice (median near $16M). The gap is programmatic: bus, operations, and margin. It is not physical. An independent implementation recomputed every headline kinematic quantity from first principles; the worst-case discrepancy is below 0.5%.

Contents

Executive Summary 1

Answers to the Mission's Questions 4

1 Introduction 6
- 1.1 Mission Statement . . . 6
- 1.2 Concept Vehicle . . . 7
- 1.3 Scope . . . 7
- 1.4 Assessment Methodology . . . 7
- 1.5 Report Organization . . . 8

2 Intercept Geometry and Arrival-Epoch Optimization 8
- 2.1 Kinematics of the Target . . . 8
- 2.2 The Lead-the-Target Relation . . . 8
- 2.3 Characteristic Epochs . . . 9
- 2.4 Arrival-Epoch Optimization . . . 10
- 2.5 Two Optima, Two Architectures . . . 11
- 2.6 Design Intuitions Examined . . . 12
- 2.7 Miss Tolerance and Aiming Dispersion . . . 14
- 2.8 The 500,000-Year Horizon Is Not Binding . . . 14

3 Departure Energetics from Low Earth Orbit 15
- 3.1 The Impulsive Floor . . . 15
- 3.2 The Low-Thrust Picture . . . 15

4 Solar-Electric Propulsion: the Power Wall and Perihelion Pumping 16
- 4.1 Power Model . . . 16
- 4.2 The Power Wall: Outward-Spiral Saturation . . . 17
- 4.3 Perihelion Pumping . . . 18
- 4.4 The Patience Trade . . . 19
- 4.5 Bounding the Requirement . . . 20
- 4.6 Sensitivity to the Thermal Power Cap . . . 20
- 4.7 Consequences for the Feasibility Verdict . . . 21
- 4.8 Limitations . . . 22

5 Vehicle Mass Closure 22
- 5.1 Mass Model . . . 22
- 5.2 Closure Results . . . 23
- 5.3 Sensitivities . . . 25
- 5.4 Recommendation . . . 26

6 Targeting Confidence Under Execution and Ephemeris Dispersion 26
- 6.1 Method . . . 27
- 6.2 Results . . . 27
- 6.3 Which Error Source Dominates . . . 28
- 6.4 What the Verdict Is Conditioned On . . . 28
- 6.5 A Catalog Systematic, Held Separate . . . 28

7 Program Cost Assessment 30
- 7.1 Cost Model . . . 30
- 7.2 Results: \$16M-Class Medians . . . 30
- 7.3 Comparison with the Mission's Allocation . . . 30
- 7.4 Operations Duration . . . 31
- 7.5 The Path to \$10M . . . 31
- 7.6 The \$50M Tier . . . 32

8 Alternative Targets 32
- 8.1 LSPM J2146+3813: The Energetically Preferred Target . . . 33

8.2 Lambda Serpentis . . . 34
8.3 Alpha-2 Librae: Excluded . . . 34
8.4 Catalog Sweep: No Better Target Found . . . 34

9 Conclusions and Recommendations 35
9.1 Findings . . . 35
9.2 Recommendations . . . 37
9.3 Overall Assessment . . . 38

References 40

A Astrometric Inputs and Intercept Kinematics 41
A.1 Astrometric Inputs and Provenance . . . 41
A.2 Transformation to the Heliocentric Ecliptic Frame . . . 41
A.3 Closed-Form Intercept Kinematics . . . 42
A.4 Consistency with Published Encounter Catalogs . . . 43

B Trajectory Models and Verification 44
B.1 Equations of Motion and Thrust Law . . . 44
B.2 Earth-Escape Spiral . . . 44
B.3 Perihelion-Pumping Control Policy and Optimization . . . 44
B.4 Bounding the Heliocentric Requirement . . . 45
B.5 Independent Verification . . . 47
B.6 A Note on Direct Collocation . . . 47

C Mass-Closure Model 48
C.1 Model Equations . . . 48
C.2 Closure-Boundary Algebra . . . 49
C.3 Closure Grid . . . 50
C.4 Independent Verification . . . 50

D Cost Model Detail 51
D.1 Line-Item Values . . . 51
D.2 Cost-Driver Definitions . . . 51
D.3 Perihelion Thermal-Qualification Increment . . . 52
D.4 Independence Assumption . . . 52
D.5 Scope Exclusions . . . 52

E Computational Campaign and Verification Record 53
E.1 Campaign Structure and Reproducibility Conventions . . . 53
E.2 Perihelion-Pumping Schedule Search . . . 53
E.3 Long-Horizon and Three-Dimensional Trajectory Sweeps . . . 54
E.4 Optimal-Control Solves and the Direct-Collocation Record . . . 55
E.5 Lower-Bound Certificate and Numerical Falsification Harness . . . 56
E.6 Aiming-Dispersion Monte Carlo . . . 57
E.7 Catalog Sweep and Cost Monte Carlo . . . 57
E.8 Arrival-Epoch Re-derivation and Trade Table . . . 58

Nomenclature

a_0	thrust acceleration at 1 AU, full power [$\mathrm{m\,s^{-2}}$]	T	arrival (intercept) epoch from launch [yr]
Δv	velocity increment (propulsive budget) [$\mathrm{km\,s^{-1}}$]	v_e	exhaust velocity, $g_0 I_{sp}$ [$\mathrm{km\,s^{-1}}$]
v_∞	heliocentric hyperbolic excess speed [$\mathrm{km\,s^{-1}}$]	β	aim tilt: departure-asymptote tilt from the ecliptic [deg]
$v_{\infty,E}$	Earth-relative hyperbolic excess speed [$\mathrm{km\,s^{-1}}$]	μ	gravitational parameter [$\mathrm{km^3\,s^{-2}}$]
I_{sp}	specific impulse [s]	P	electrical power at 1 AU [kW]
r	heliocentric distance [AU]	m_0, m_f	initial (wet), final mass [kg]
C_3	characteristic energy, $v_{\infty,E}^2$ [$\mathrm{km^2\,s^{-2}}$]	E	specific orbital energy [$\mathrm{km^2\,s^{-2}}$]

AU = astronomical unit; ly = light-year; LEO/GTO = low Earth / geostationary transfer orbit;
SEP = solar-electric propulsion; PPU = power processing unit;
P10/P50/P90 = cost-distribution percentiles; RV = radial velocity.

Executive Summary

This report assesses the feasibility of a small, low-cost interstellar precursor mission. The vehicle masses approximately $100\,\mathrm{kg}$, uses solar-electric thrusters expelling xenon, reaches low Earth orbit (LEO) as a rideshare payload, and launches before the end of 2029. It must pass within $2600\,\mathrm{AU}$ of the Alpha Centauri AB barycenter within 500,000 years of departure, without gravity assists, for a total program cost below $10M. The verdict is fourfold. The trajectory class is feasible on solar power alone. The $100\,\mathrm{kg}$ concept closes from a transfer orbit; on every trajectory found it fails to close from low Earth orbit. Open-loop targeting succeeds with probability 0.98 under the stated error budget. The program cost exceeds the initial $10M cost target for programmatic reasons; the gap has no physical cause.

The autonomous physics-research platform of Physical Superintelligence (PSI) produced every quantitative result in this assessment end-to-end under staged independent audit. Agentic research systems developed the physics models, the trajectory optimization, and the probabilistic cost model. Independent verification agents replicated the reference analysis, certified a lower bound under adversarial proof audit, carried a fully three-dimensional trajectory optimization to convergence, and ran a seeded, bit-reproducible targeting-dispersion Monte Carlo. They also recomputed the headline numbers from first principles. Section 1 describes the assessment methodology; Appendix E documents the computational campaigns and their acceptance gates. Each result below carries its verification standing in the sentence that states it.

1. The approach geometry and its optimum are settled. Published astrometry for the Alpha Centauri AB barycenter (distance $4.365\,\mathrm{ly}$, space velocity $32.38\,\mathrm{km\,s^{-1}}$) [1] fixes the departure-energy optimum at an arrival epoch of T = 73,012 years after launch; a blind re-derivation that referenced no prior result reproduced the optimum to 1 yr. The required cruise speed (v_∞, the hyperbolic excess speed retained after escaping the Sun) is $23.64\,\mathrm{km\,s^{-1}}$, aimed $2.48°$ below the ecliptic plane. After borrowing Earth's in-plane orbital velocity of $29.78\,\mathrm{km\,s^{-1}}$, the Earth-relative hyperbolic excess is $18.59\,\mathrm{km\,s^{-1}}$, and a single perigee burn from a $400\,\mathrm{km}$ LEO requires an impulsive velocity increment (Δv) of $13.85\,\mathrm{km\,s^{-1}}$. Two optima coexist: minimizing the departure budget selects the 73,012-year arrival, while minimizing the heliocentric cruise speed itself selects an earlier arrival, at 57,854 years and $23.173\,\mathrm{km\,s^{-1}}$. The departure optimum is flat: the departure budget stays within $100\,\mathrm{m/s}$ (under $0.1\,\mathrm{km\,s^{-1}}$) of its floor for arrivals from 66,908 to 80,436 years, and remains under $0.25\,\mathrm{km\,s^{-1}}$ above it out to 85,000 years. Neither mission tolerance binds. The $2600\,\mathrm{AU}$ miss tolerance is worth only about $83\,\mathrm{m/s}$ of cruise speed; at the 500,000-year horizon the impulsive Δv *rises* to $19.93\,\mathrm{km\,s^{-1}}$ because the star is by then receding. The aim-tilt tolerance is loose: $0.41°$, about $168\,\mathrm{m/s}$ of velocity dispersion. Open-loop control meets it without terminal correction. These results agree with the reference analysis of the Fermi Explorer Mission (a 501(c)(3) nonprofit) [14] to within 0.3%; an independent first-principles implementation reproduced them to better than 0.5% in the worst case.

2. The minimum Δv depends on the propulsion regime. A chemical vehicle exploiting the full Oberth benefit at perigee needs the $13.85\,\mathrm{km\,s^{-1}}$ floor. A solar-electric vehicle cannot exploit that benefit and needs 30.5–$31.6\,\mathrm{km\,s^{-1}}$ in total: an assumed $7.6\,\mathrm{km\,s^{-1}}$ Earth-escape spiral (roughly 1.5 years through the radiation belts, with array degradation factored in) plus a heliocentric phase quoted across the 22.9–$24.0\,\mathrm{km\,s^{-1}}$ reference interval. The interval's two ends carry different pedigrees, a verified upper end (independent re-integration agrees to 0.29%) and a best-found lower anchor, established in Section 3. The floor of the argument is a certified lower bound: an adversarially audited proof in the planar model shows that no admissible thrust program (throttleable thrust, *unconstrained* flight time, perihelion no lower than $0.42\,\mathrm{AU}$) departs for less than $16.56\,\mathrm{km\,s^{-1}}$. The bound is a closed-form theorem over that stated class. It does not summarize a search that found no counterexample. That time freedom matters. With unlimited time an explicit trajectory reaches cruise speed for $18.87\,\mathrm{km\,s^{-1}}$ after about 58 years of powered flight; gate-passing schedules span 33–58 yr at 18.9–$20.5\,\mathrm{km\,s^{-1}}$. The best 12-year operational trajectory found requires $23.98\,\mathrm{km\,s^{-1}}$.

3. Solar power suffices for the trajectory class; the mission needs no nuclear source. Because array power falls as the inverse square of solar distance, a conventional *outward* escape spiral saturates: the maximum achievable v_∞ is approximately 0, 3.4, and $17.0\,\mathrm{km\,s^{-1}}$ at thrust accelerations a_0 = 1.5, 5, and $10\times10^{-4}\,\mathrm{m/s^2}$ respectively, and two independent integrators that agree to 2.7% verify the saturation. All three ceilings sit below the $23.38\,\mathrm{km\,s^{-1}}$ cruise floor. Perihelion pumping defeats this wall: retrograde arcs near $1\,\mathrm{AU}$ shed angular momentum and drop perihelion to $0.42\,\mathrm{AU}$; prograde arcs then concentrate near perihelion, where power is one to four times the 1-AU rating. The best-found pumping schedule at $a_0 = 2.5\times10^{-4}\,\mathrm{m/s^2}$ delivers $23.98\,\mathrm{km\,s^{-1}}$ of Δv in 12.0 years of flight time (mostly coasting); independent verification reconstructed the

pumping mechanism from scratch. The result survives halving the assumed thermal power cap at perihelion, at a cost of about $1.9\,\mathrm{km\,s^{-1}}$; the re-optimized schedules under reduced caps need 15.1–19.7 years of powered flight, past the 12-year horizon that anchors mass closure (Section 4.6). For a 2029 launch the vehicle escapes the solar system around 2043; the arrival epoch is unaffected because the optimum is flat, and over 98% of the mission is ballistic coast. Fermi Explorer's reference analysis concluded that pure solar-electric propulsion does not close from LEO *at today's array specific power* (~60 W/kg) and located a pure-solar closing corner above roughly $100\,\mathrm{W/kg}$. The same analysis separately rejected a *single-pass* electric perihelion-burn architecture. Multi-revolution pumping amends both conclusions at once: it answers the single-pass objection, and with that objection answered it closes at today's ~60 W/kg.

4. The 100 kg concept closes from GTO; from LEO it misses on every trajectory found. With the required specific impulse of $2800\,\mathrm{s}$ and present-day component parameters, the minimum closing wet mass from LEO is 113.5–$146\,\mathrm{kg}$ across the 22.9–$24.0\,\mathrm{km\,s^{-1}}$ reference interval: the $100\,\mathrm{kg}$ concept misses by 1.0–2.8% of wet mass for every trajectory found, while $150\,\mathrm{kg}$ closes across the entire interval in the planar accounting (the measured out-of-plane cost overturns this at the conservative end; finding 5). The LEO verdict is conditional: it stands on every trajectory found, and a heliocentric trajectory below the ~$22.3\,\mathrm{km\,s^{-1}}$ crossover inside the 12-year operational horizon would close the $100\,\mathrm{kg}$ vehicle from LEO and overturn it. The margins above that crossover belong to the trajectories found. They do not extend to the trajectory class. The lower end of the reference interval, $22.9\,\mathrm{km\,s^{-1}}$, sits $0.6\,\mathrm{km\,s^{-1}}$ above the crossover; the best trajectory found at the design thrust level, $23.98\,\mathrm{km\,s^{-1}}$, sits $1.7\,\mathrm{km\,s^{-1}}$ above; the best found even at four times the design thrust, $22.66\,\mathrm{km\,s^{-1}}$, remains $0.4\,\mathrm{km\,s^{-1}}$ above. A geostationary transfer orbit (GTO) start via rideshare cuts the departure budget from 7.6 to $4.24\,\mathrm{km\,s^{-1}}$: the $100\,\mathrm{kg}$ concept then closes with margin (+4.1 points at the conservative end; even $80\,\mathrm{kg}$ closes). Under a 30% thrust-acceleration margin the GTO start retains $100\,\mathrm{kg}$ closure at the favorable end of the interval; approximately $110\,\mathrm{kg}$ carries the margin across the whole interval. The same margin drives the LEO closure boundary to 312–$910\,\mathrm{kg}$. The two starts differ in margin under stress: the $100\,\mathrm{kg}$ GTO vehicle retains a +2.3-point margin even under the worst-case out-of-plane increment, whereas the $150\,\mathrm{kg}$ LEO fallback fails at the conservative end of the interval (finding 5). An independent algebraic recomputation verified the closure arithmetic; the LEO closure boundary agrees to 0.2% ($113.3\,\mathrm{kg}$). The recommendation is to procure the GTO drop-off with a 100–$110\,\mathrm{kg}$ vehicle.

5. The out-of-plane cost is measured, and it strengthens the GTO recommendation. The results above are planar; they treat the $2.48°$ aim tilt below the ecliptic as an estimated increment between $22\,\mathrm{m/s}$ and $1.02\,\mathrm{km\,s^{-1}}$. A genuinely three-dimensional optimization measures that increment at $0.58\,\mathrm{km\,s^{-1}}$: the difference between the best-found three-dimensional and planar totals. The value lies inside the estimated band. It is the achieved cost of the best trajectories found; it does not bound the true optimum. The $100\,\mathrm{kg}$ GTO vehicle still closes at both ends of the 22.9–$24.0\,\mathrm{km\,s^{-1}}$ interval (+5.1/+3.1 points; +2.3 even at the $1.02\,\mathrm{km\,s^{-1}}$ worst case); the $150\,\mathrm{kg}$ LEO fallback closes at $22.9\,\mathrm{km\,s^{-1}}$ (+1.0 points) but fails at the conservative $24.0\,\mathrm{km\,s^{-1}}$ end (-0.8 points). The recommendation to procure the GTO drop-off stands with added margin.

6. Open-loop targeting confidence is 0.98 under the stated error budget. A seeded 20,000-trial Monte Carlo propagated execution and ephemeris errors to the encounter and measured the probability of passing within $2600\,\mathrm{AU}$ of the barycenter at 0.982 (95% confidence interval $[0.980, 0.983]$); an independent analytic covariance calculation, sharing no code with the simulation, reproduces the value to 0.011%. Thrust-magnitude knowledge dominates the error budget, ahead of pointing, ephemeris, and departure residuals. The error levels entering the simulation are assumed engineering inputs. Flight-measured performance does not yet exist to replace them. Doubling the assumed thrust-magnitude error alone drops the probability to 0.858; doubling the assumed pointing error drops it to 0.897. Both fall below the 0.9 threshold.

7. The cost gap is programmatic; no physical barrier produces it. A probabilistic parametric model (2026 USD) places the median program cost at $15.7M–$16.6M across the closing architectures, and the probability of meeting the initial $10M target is low under conventional aerospace practice. The structure of the overrun is instructive: launch ($1.0–1.3M) and propulsion ($0.6–1.2M) each come in at roughly one third of their $3M allocations, while the entire gap sits in the bus, operations, and program margin (approximately $12M nominal against $3M allocated). Because the mass-closing trajectory implies about 13 years of powered-flight custody, automation is structural to the cost estimate. It is more than a communications convenience. The path to $10M runs through lean automated operations and compressed integration and test. New physics plays no part in it.

Table 1. Summary of findings and recommendations.

Topic	Finding	Recommendation
Arrival-epoch optimum	T = 73,012 yr (blind re-derivation confirms); v_∞ = $23.64\,\mathrm{km\,s^{-1}}$ aimed 2.48° below the ecliptic; within $0.1\,\mathrm{km\,s^{-1}}$ of the floor over 66.9–80.4 kyr, < $0.25\,\mathrm{km\,s^{-1}}$ out to 85 kyr	Choose the arrival epoch within the window for launch-phasing convenience
Minimum Δv from LEO	$13.85\,\mathrm{km\,s^{-1}}$ impulsive; 30.5–$31.6\,\mathrm{km\,s^{-1}}$ solar-electric ($24.0\,\mathrm{km\,s^{-1}}$ verified constructive bound; $22.9\,\mathrm{km\,s^{-1}}$ best-found single-method anchor; certified lower bound $16.56\,\mathrm{km\,s^{-1}}$, unconstrained flight time)	Size electric propulsion to the top of the reference interval
Solar-electric feasibility	Perihelion pumping delivers the full cruise speed ($23.98\,\mathrm{km\,s^{-1}}$ in 12.0 yr at $a_0 = 2.5 \times 10^{-4}\,\mathrm{m/s^2}$); survives halving the thermal power cap at perihelion for ~$1.9\,\mathrm{km\,s^{-1}}$, with custody extended to 15.1–19.7 yr (Section 4.6)	No nuclear source required; qualify a gridded-ion propulsion string
Vehicle mass closure	$100\,\mathrm{kg}$ misses from LEO for every trajectory found (conditional: ~$22.3\,\mathrm{km\,s^{-1}}$ crossover caveat); closes from GTO, where ~$110\,\mathrm{kg}$ also carries a 30% thrust margin across the interval	Procure a GTO drop-off; fly a 100–110 kg vehicle
Out-of-plane cost	Measured at $0.58\,\mathrm{km\,s^{-1}}$ by three-dimensional optimization, inside the prior 22 m/s–$1.02\,\mathrm{km\,s^{-1}}$ planar estimate; GTO-100 closes at both bracket ends; LEO-150 fails at the conservative end	Keep the GTO drop-off procurement; do not rely on the 150 kg LEO fallback
Targeting dispersion	$P(\mathrm{pass} \le 2600\,\mathrm{AU}) = 0.982$ $[0.980, 0.983]$ under the stated error budget, dominated by thrust-magnitude knowledge; doubling assumed execution errors drops confidence below 0.9	Treat thrust-magnitude knowledge as a driving requirement; revalidate with measured hardware error levels
Program cost	Median $15.7–16.6M; probability of meeting the initial $10M target is low under conventional practice; entire gap in bus and operations	Pursue lean automated operations and compressed integration and test
Alternative target	LSPM J2146+3813: $9.30\,\mathrm{km\,s^{-1}}$ impulsive floor, one-third the cruise speed, same arrival timeline	Prefer this target if the objective is any star's Oort cloud

8. A cheaper star exists. If the objective is a passage through *any* star's Oort-cloud environs, without preference for Alpha Centauri, the M5V red dwarf LSPM J2146+3813 ($22.99\,\mathrm{ly}$) dominates: an impulsive floor of $9.30\,\mathrm{km\,s^{-1}}$ versus $13.85\,\mathrm{km\,s^{-1}}$, a cruise speed of $7.74\,\mathrm{km\,s^{-1}}$ versus $23.64\,\mathrm{km\,s^{-1}}$, and an arrival at 78,017 years, the same timeline. The mission needs no perihelion pumping at all, since the conventional-spiral ceiling exceeds the requirement. Its published future encounter with the Sun, $0.568\,\mathrm{pc}$ at +82.5 kyr, is among the tightest known [6]. A sweep of the same encounter catalog found no higher-confidence target inside the mission horizon (Section 8).

Table 1 condenses the findings and recommendations.

Answers to the Mission's Questions

Each verdict below condenses the finding it draws on; the cited section carries the full analysis.

Requirements Compliance

Table 2. Compliance of the four mission objectives (Table 3).

Objective	Verdict	Basis
Pass within 2600 AU of the Alpha Centauri AB barycenter within 500,000 yr	**Met** (by the recommended architecture)	Arrival at ~73,000 yr; the Fermi Explorer team's planned aim at the ~80,000-year ecliptic-plane crossing is equally acceptable, costing under $0.1\,\mathrm{km\,s^{-1}}$ over the optimum. With all execution and ephemeris error sources acting together under the assumed engineering error budget, the probability of passing inside the sphere open-loop is 0.982 (95% confidence interval $[0.980, 0.983]$); the verdict degrades below 0.9 if either the assumed thrust-magnitude or pointing error doubles (Section 2.7, Section 6).
Payload of at least 1 kg (1U class)	**Met**	Carried explicitly as a fixed line item in every mass budget in this report; the recommended 100 kg architecture closes with the payload aboard at a margin of +3.1 to +5.1 points of wet mass (Section 5).
Launch before the end of 2029	**Schedule-conditional**	Rideshare procurement is routine, but the propulsion qualification campaign (Section 5) and MESSENGER-class thermal qualification, from a mid-2026 start, make an end-2029 launch aggressive. This report performed no full schedule analysis; the verdict is conditional on one.
Program cost below $10M ($3M launch / $3M propulsion / $3M bus and operations)	**Not met** under conventional practice	Median estimate $15.7M–$16.6M. Launch and propulsion under-run their allocations; the entire gap sits in bus, operations, and margin, and it is programmatic (Section 7). No physical constraint sets it. Section 7.5 describes a lean-automation path; it adopts the mission's stated operations posture: the tolerance for mostly-automated operations (a minimal antenna and eventual loss of contact included) is that path's load-bearing assumption (Section 7.4).

Direct Answers

Q1. What is the minimum vehicle Δv from any LEO drop-off, with no gravity assist, arriving within 500,000 years? For impulsive (chemical) propulsion, $13.85\,\mathrm{km\,s^{-1}}$: a single perigee burn from a 400 km orbit that captures the full Oberth benefit (Section 3). For solar-electric propulsion, 30.5–$31.6\,\mathrm{km\,s^{-1}}$: a $7.6\,\mathrm{km\,s^{-1}}$ Earth-escape spiral plus a heliocentric phase whose Δv spans the reference interval 22.9–$24.0\,\mathrm{km\,s^{-1}}$. These are planar figures; the measured out-of-plane increment adds $0.58\,\mathrm{km\,s^{-1}}$ (Section 5). The interval's upper end is a constructive bound: an explicitly integrated, timestep-refined trajectory reaches cruise speed for $23.98\,\mathrm{km\,s^{-1}}$ in 12 years, the best 12-year trajectory found.

The interval's lower end is a single-method optimal-control anchor; its attempted direct-collocation cross-check did not converge, and it marks the best trajectory found. It is not a certified bound. The certified lower bound over the admissible class of throttleable-thrust programs with unconstrained flight time and perihelion at or above 0.42 AU is $16.56\,\mathrm{km\,s^{-1}}$; Section 4.5 carries the full pedigree of both ends. With flight time freed, cheaper admissible trajectories exist, so the 12-year figure is schedule-shaped. It is not the class optimum.

These figures already exercise the offered freedom to choose the drop-off orbit within the low-Earth class: the

injection plane contains the departure asymptote, tilted $2.48°$ below the ecliptic (an assumed launch condition), so no separate plane change is charged, and the altitude lever is weak; it is worth a few tenths of a $\mathrm{km\,s^{-1}}$ across low-Earth altitudes (Section 3). The drop-off choice that matters lies beyond low Earth orbit: a GTO drop-off cuts the departure budget from 7.6 to $4.24\,\mathrm{km\,s^{-1}}$, and this single procurement change closes the 100 kg vehicle (Section 5). No selection among low orbits does.

Q2. What mission profile achieves it? Rideshare to low Earth orbit; an escape spiral of roughly 1.5 years through the radiation belts; then a perihelion-pumped heliocentric phase of about 12 years, most of it coasting. In that phase, retrograde arcs near 1 AU shed angular momentum to drop perihelion to 0.42 AU, and prograde arcs concentrate there, where power peaks. The vehicle departs on an asymptote $2.48°$ below the ecliptic aimed at the 73,000-year intercept and coasts ballistically thereafter; over 98% of the mission is unpowered (Section 4).

Q3. Does electric propulsion plus solar power work, given the inverse-square falloff of sunlight? Yes, for the trajectory class. Outward spirals saturate below the required cruise speed at the design thrust; perihelion pumping does not, because its burns concentrate where power is highest. The mechanism survives halving the assumed thermal power cap at a cost of about $1.9\,\mathrm{km\,s^{-1}}$, and the re-optimized schedules then take 15.1–19.7 years against the 12-year horizon that anchors closure (Section 4). At the vehicle level, closure comes at 100 kg from a GTO start or at approximately 150 kg from a LEO start. The measured $0.58\,\mathrm{km\,s^{-1}}$ out-of-plane increment for acquiring the tilted departure asymptote leaves the first verdict intact at both ends of the reference interval but defeats the second at the interval's conservative end; both closures are conditional on qualifying a propulsion string within the demonstrated gridded-ion component class (Section 5). The solar-electric answer also holds at the targeting level: the open-loop aim delivered at the end of the powered phase passes inside the required sphere with probability 0.982 under the assumed error budget (Section 6).

The Fermi Explorer Team's Five Arrival-Epoch Intuitions

We tested all five against the full optimization (Section 2.6):

1. The budget rises sharply beyond ~80,000 yr: **confirmed** (the impulsive requirement grows from $13.85\,\mathrm{km\,s^{-1}}$ at the optimum to $19.93\,\mathrm{km\,s^{-1}}$ at the 500,000-year horizon, once the star begins to recede);
2. the closest approach (~3 ly) is not the optimum: **confirmed**, with the epoch refined: the approach occurs at 27,955 yr with the star at 3.15 ly, against the anticipated ~20,000 yr at 3 ly, and intercepting there is the costliest option examined;
3. a tangential intercept near 60,000 yr minimizes cruise speed: **confirmed**, now with an exact departure-referred number: the cruise-speed minimum falls at $57{,}854$ yr and $23.173\,\mathrm{km\,s^{-1}}$ (the Sun-only true-intercept identity, with no planetary departure assumed, puts it at $T^* = 58{,}422$ yr, at the star's transverse velocity of $23.38\,\mathrm{km\,s^{-1}}$);
4. the ecliptic-plane crossing minimizes the departure Δv: **very nearly right** (the 79,786-yr crossing costs under $0.1\,\mathrm{km\,s^{-1}}$ more than the optimum; the basin is flat to better than $0.1\,\mathrm{km\,s^{-1}}$ for arrivals between roughly 67,000 and 80,000 yr);
5. the optimum lies near 73,000 yr, aimed a few degrees off-plane: **confirmed almost exactly** (73,012 yr, $2.48°$ below the ecliptic; a blind re-derivation of the arrival-epoch optimization, built without reference to the original solution, recovered the same 73,012-year optimum, with agreement to 1 yr).

The $50M Tier and Better Stars

At $50M. The recommended use of the tier is an escape-injection rideshare: purchasing the departure energy from the launch provider closes the vehicle at approximately 60 kg and removes the belt-transit, array-degradation, and patience risks outright, at a median cost near $15.6M (Section 7.6). A dedicated multi-stage chemical kick stage sized to the $13.85\,\mathrm{km\,s^{-1}}$ impulsive floor buys the same physics less well: it is more expensive and less reliable than purchasing the equivalent injection energy from the launch vehicle. A Jupiter gravity

Table 3. Mission requirements and ground rules.

Requirements	
Payload	At least $1\,\mathrm{kg}$ (1U CubeSat class)
Encounter	Pass within $2600\,\mathrm{AU}$ of the Alpha Centauri AB barycenter within 500,000 years of launch
Launch date	Before the end of 2029
Program cost	Below $10M total: $3M launch, $3M propulsion, $3M bus and mission operations
Ground rules	
Injection	Rideshare drop-off in any low Earth orbit
Trajectory	No gravity assists
Operations	Mostly automated operations acceptable; loss of contact during cruise is tolerable

assist, legitimately back on the table at this tier, offers an assumed 10–$15\,\mathrm{km\,s^{-1}}$ of added cruise speed at scoping fidelity, but it is window-contingent: its value depends on Jupiter phasing against the end-2029 launch date, and it adds transfer years and deep-space operations of exactly the class that dominates program cost (Section 7.6). Radioisotope generators cannot supply the kilowatt-class power the electric propulsion requires, and a fission reactor exceeds even this tier. Among methods beyond those the Fermi Explorer team named, the cheapest sits below the tier entirely: the GTO drop-off of Section 5, which nearly halves the departure budget and closes the $100\,\mathrm{kg}$ vehicle without touching the additional funds. The remaining assist-class option, a solar-Oberth maneuver with a purchased kick stage, demands thermal protection of the Parker Solar Probe class; Section 7.6 assesses it.

Better stars. We adjudicated all three offered candidates. LSPM J2146+3813 is confirmed as the best target on offer: an impulsive floor of $9.30\,\mathrm{km\,s^{-1}}$ at a ~78,000-year arrival, reachable without perihelion pumping (Section 8.1). Lambda Serpentis is a sound solar-type second choice at $11.38\,\mathrm{km\,s^{-1}}$ and a ~152,000-year arrival, and it confirms the anticipated $150{,}000$–$200{,}000$-year window (Section 8.2). Alpha-2 Librae should come off the list: its minimum-cruise-speed intercept lies at 1.03 million years, outside the $500{,}000$-year horizon, and forcing an in-horizon arrival costs $19.74\,\mathrm{km\,s^{-1}}$, above Alpha Centauri's own impulsive floor; the verdict holds across the full $\pm 2\sigma$ range of its poorly determined systemic radial velocity of $-22.0 \pm 5.8\,\mathrm{km\,s^{-1}}$ (Section 8.3). Beyond the offered set, we swept the Bailer-Jones catalog of future stellar encounters [6] for cheaper targets (Section 8.4). None displaces LSPM J2146+3813 as the best high-confidence target: the only entries that formally undercut it are two faint, unnamed Gaia M dwarfs whose catalog radial velocities carry 8–11 $\mathrm{km\,s^{-1}}$ uncertainties, too weak to aim at. Every unflagged catalog candidate inside the 500,000-year horizon is an M dwarf, so among Sun-like stars Lambda Serpentis stands unchallenged.

1. Introduction

1.1 Mission Statement

The Fermi Explorer Mission (a 501(c)(3) nonprofit) seeks the least expensive spacecraft mission that can reach the near environs of another star. The specific formulation under assessment is a vehicle that passes within $2600\,\mathrm{AU}$ of the Alpha Centauri AB barycenter (a distance comparable to the scale of a stellar Oort cloud) within 500,000 years of departure, launched before the end of 2029, on a total program budget below $10M. The Fermi Explorer team published its own reference analysis of the problem [14]. The present report is an independent feasibility assessment by Physical Superintelligence (PSI), built on separately developed models and data. It answers the feasibility question on its own terms and cross-checks the reference analysis. Table 3 states the four mission requirements and the accompanying ground rules.

1.2 Concept Vehicle

The Fermi Explorer concept vehicle has a wet mass of approximately $100\,\mathrm{kg}$ (most of it xenon propellant). It uses solar-electric propulsion to build up the required departure energy and carries the $1\,\mathrm{kg}$ payload. The operational posture is deliberately expendable: the mission succeeds once tracking establishes the intercept trajectory. Continued contact through the multi-millennial cruise is optional. The central question of this report is whether this concept closes in trajectory physics, in vehicle mass, and in cost.

1.3 Scope

The assessment covers the interstellar approach geometry and the choice of arrival epoch; departure budgets from low Earth orbit in the impulsive (chemical) and low-thrust (solar-electric) regimes; the feasibility of solar-electric escape given the inverse-square decline of array power with solar distance; vehicle mass closure at the level of component parameters (thruster specific impulse and efficiency, array specific power, tankage, and power processing); a probabilistic program cost estimate against the $10M target and its internal allocations; and a survey of alternative stellar targets evaluated by identical methods. It does not cover detailed spacecraft bus design, launch-vehicle selection, or subsystem-level thermal and communications design. Where such a discipline constrains feasibility (for example, the thermal qualification implied by repeated close perihelion passes), the report identifies the constraint and carries its cost or mass consequence. The report attempts no design.

1.4 Assessment Methodology

Four legs carry the analysis; none shares models or code with Fermi Explorer. The first is independent physics models: we compute the kinematics of the encounter from published astrometry of the Alpha Centauri system [1], cross-check them against independent orbit and parallax determinations [2, 3], and require closed-form and numerical treatments to agree. The second is trajectory simulation and optimization: power-limited thrust models propagate the low-thrust escape trajectories, and the key heliocentric velocity budget is quoted as the 22.9–$24.0\ \mathrm{km\,s^{-1}}$ reference interval. The interval's two ends carry different pedigrees: a verified upper end (independent re-integration agrees to 0.29%) and a best-found lower anchor (Section 3). The interval carries the quantitative conclusions; no single point solution does. The third is a probabilistic cost model: ten parametric line items with triangular uncertainty distributions, sampled by Monte Carlo, so cost conclusions take the form of percentile ranges.

The fourth leg is independent verification. PSI's staged, artifact-bound verification framework indexes the manuscript's claims and audits them in independent passes for mathematical soundness, physical support, literature positioning, and proof integrity; each pass writes machine-validated artifacts. Under that framework an independent verification pass re-derived every headline quantity in this report from the cited astrometry and the stated models. The pass had no access to the originating code. The worst kinematic discrepancy is below 0.5%; two independent integrators confirmed the low-thrust saturation result, agreeing to 2.7%; a separate implementation, given no access to prior results, re-derived the arrival-epoch optimum blind, with agreement to 1 yr (Section 2.4); and an algebraic check confirmed the mass-closure arithmetic to 0.2% (Section C.4). One standard technique failed: direct collocation does not converge in the low-thrust, multi-revolution regime. The failure is recorded (Appendix B) and no conclusion depends on it.

PSI's autonomous physics-research platform executed all four legs end-to-end. Autonomous research agents carried the trajectory-optimization, mass-closure, cost, and dispersion campaigns from model construction to converged results, and separate verification agents re-derived every headline number from the indexed claims. The verification agents did not read the originating code. Claims advanced at theorem grade, principally the certified lower bound on the heliocentric velocity budget, passed an adversarial proof audit in which a dedicated agent attempted to defeat each step of the argument before the proof was accepted. Every Monte Carlo result in this report is seeded and bit-reproducible; Appendix E documents the seed discipline and acceptance gates. The platform's role is confined to the campaign this report documents; no conclusion rests on unaudited machine output.

Three findings carry the most weight; each keeps its full pedigree at its primary site. The heliocentric velocity budget is anchored from below by a certified lower bound: no admissible thrust program in the assessed class (planar model) can depart the solar system for less than $16.56\,\mathrm{km\,s^{-1}}$ without a time-of-flight restriction

(theorem and adversarial audit in Appendix B). The out-of-plane increment of the tilted departure asymptote is measured at $578\,\mathrm{m\,s^{-1}}$, computed as the difference between the best trajectories found with and without the out-of-plane requirement (Section 5). The probability of meeting the $2600\,\mathrm{AU}$ encounter criterion is measured by a seeded 20,000-sample Monte Carlo dispersion analysis under assumed engineering error levels, marked at their point of use (Section 6). The verification pass also reproduces the geometric optimum of the Fermi Explorer reference analysis to 0.3% (Section 2.4 gives the per-quantity comparison); the two analyses share a common baseline before their conclusions diverge.

1.5 Report Organization

The report proceeds from kinematics to program cost. It first establishes the approach geometry and the arrival-epoch optimum, then develops the departure budgets from low Earth orbit in each propulsion regime. The solar-electric feasibility question follows, including the power-limited saturation of conventional escape spirals and the perihelion pumping schedule that overcomes it. Vehicle mass closure and its sensitivities come next, then an execution-dispersion analysis that propagates departure, pointing, thrust-knowledge, and ephemeris uncertainties through the multi-millennial cruise to the probability of meeting the encounter criterion, and then the probabilistic cost assessment against the $10M target. The report then evaluates alternative stellar targets under identical methods, compares the present results with the Fermi Explorer reference analysis, and closes with conclusions and recommendations. Appendices document the independent verification of each headline result.

2. Intercept Geometry and Arrival-Epoch Optimization

Three kinematic quantities govern the mission: where the target will be, when it is cheapest to meet it, and how tightly the departure must be aimed. All results here follow exactly from the target's measured space motion. An independent verification pass recomputed every headline quantity to better than 0.5% (Section 1), and the departure optimum agrees with the Fermi Explorer reference analysis [14] to 0.3% or better (Section 2.4 gives the per-quantity comparison).

2.1 Kinematics of the Target

The mission aims at the barycenter of the Alpha Centauri AB binary, and the target's measured space motion fixes the whole intercept problem: distance $d = 4.365\,\mathrm{ly}$ and heliocentric space velocity $32.38\ \mathrm{km\,s^{-1}}$, resolved into a radial component $v_r = -22.40\ \mathrm{km\,s^{-1}}$ (approaching the Sun) and a transverse component $v_t = 23.38\ \mathrm{km\,s^{-1}}$. These values are assumed inputs, adopted from the orbit-corrected barycentric astrometry of Kervella et al. [1]. Over the 10^4–10^5-year horizons of interest we model the barycenter in uniform straight-line motion. This idealization costs almost nothing: a certified closed-form bound, valid over the linear-motion class, limits the differential Galactic tidal perturbation between the Sun and the target over the 73,000-year flight to about $1.2\,\mathrm{AU}$ in position and under $0.2\ \mathrm{m\,s^{-1}}$ in velocity (negligible against the $2600\,\mathrm{AU}$ miss tolerance), and published orbit-integrated encounter solutions validate the linear model externally (Section A.4). Appendix A details the astrometric inputs, the coordinate transformation, and the closed-form results quoted below.

Figure 1 shows the resulting approach geometry. Today the barycenter lies roughly 3 ly below the ecliptic plane and closes on the Sun; in the tens of millennia ahead it makes its closest natural approach, crosses the plane, and then recedes. Four epochs on this path organize the analysis; Table 4 collects them together with the 500,000-year requirement horizon.

2.2 The Lead-the-Target Relation

Once beyond the Sun's gravitational influence, the vehicle cruises in a nearly straight line at its cruise speed v_∞ (its hyperbolic excess speed). Let $\mathrm{r}_0 = d\,\hat{\mathrm{n}}$ be the target's present position ($\hat{\mathrm{n}}$ the unit line-of-sight vector) and $\mathrm{v}_\star$ its space velocity. Meeting the star at time T after departure requires the cruise velocity

$$\mathrm{V}(T) = \frac{\mathrm{r}_0}{T} + \mathrm{v}_\star, \tag{1}$$

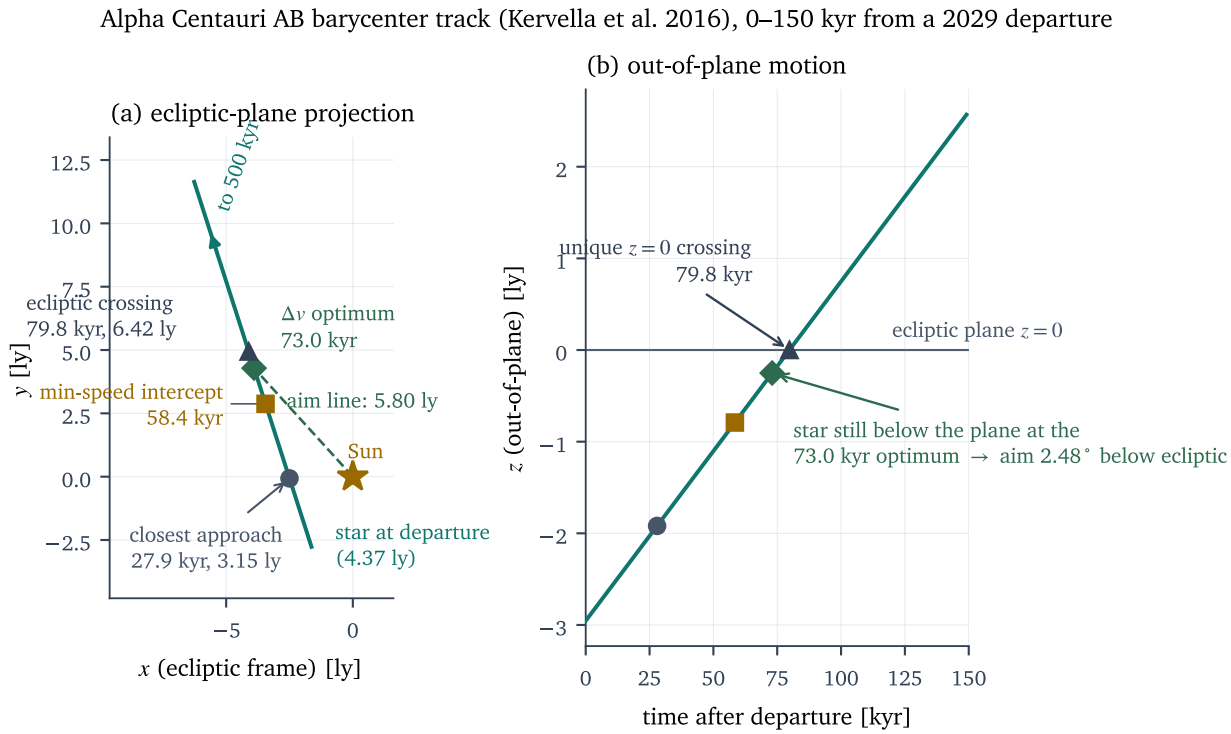


Figure 1. Approach geometry of the Alpha Centauri AB barycenter over the next 150,000 years, from the astrometry of [1]. (a) Projection of the star's path onto the ecliptic plane, with the Sun, the aim line to the 73.0 kyr intercept point, and the four characteristic epochs of Table 4 marked. (b) Out-of-plane coordinate z versus time: the star currently sits about 3 ly below the ecliptic and crosses the plane 79,786 years from launch. This geometry (a closing, below-plane target that crosses the ecliptic and then recedes) is why patience pays until roughly 80,000 years and no further.

i.e. the vehicle must fly toward where the star will be at time T. The star's present position is the wrong aim point. The required velocity is the sum of a *closing* term r_0/T, which shrinks as the mission accepts a longer flight time, and the star's own velocity, which the vehicle must match in full no matter how patient the mission is. Decomposing $\mathrm{v}_\star$ into its radial and transverse parts gives the scalar form

$$v_\infty^2(T) \;=\; \left(\frac{d}{T} + v_r\right)^2 + v_t^2. \tag{2}$$

Because the star approaches the Sun ($v_r < 0$), the radial term vanishes identically at $T^* = d/|v_r| = 58{,}422$ years: at that arrival epoch the star's own approach closes the line-of-sight distance for free, and the required cruise speed reaches its absolute floor, $v_\infty = v_t = 23.38\ \mathrm{km\,s^{-1}}$, exactly the star's transverse velocity. The identity is exact under linear motion for a *true* intercept (Appendix A), and the floor it sets is the single most important number in the study: *no trajectory, however patient, can truly intercept Alpha Centauri at a cruise speed below the star's transverse velocity.* The mission's $2600\ \mathrm{AU}$ miss tolerance lowers the effective floor to $23.17\ \mathrm{km\,s^{-1}}$ at an earlier re-optimized epoch (Section 2.5), and the headline optimum of Section 2.4 already includes that tolerance. For earlier arrivals the closing term dominates and the requirement rises steeply; for later arrivals the star recedes and the requirement rises again, gently at first.

2.3 Characteristic Epochs

The first four epochs of Table 4 are distinct because they minimize different quantities. The closest *natural* approach (27,955 years, 3.152 ly) minimizes the distance flown and is by far the costliest intercept examined: the vehicle must close 4.365 ly in under 28,000 years, driving the departure budget to more than twice the

Table 4. Characteristic intercept epochs for the Alpha Centauri AB barycenter, from launch. The impulsive Δv is a single perigee burn from a 400 km low Earth orbit with full Oberth benefit (Section 2.4). The minimum-cruise-speed row is the exact tangential intercept; with the 2600 AU miss tolerance the minimum re-optimizes to 57,854 years at 23.17 $\mathrm{km\,s^{-1}}$ (Section 2.5).

Event	Arrival T (yr)	Cruise v_∞ ($\mathrm{km\,s^{-1}}$)	Comment
Closest natural approach	27,955	n/a	star at 3.152 ly; costliest intercept examined
Minimum cruise speed	58,422	23.38	$v_\infty = v_t$ exactly (tangential intercept)
Departure-Δv optimum	73,012	23.64	aim 2.48° below ecliptic; $\Delta v = 13.85$ $\mathrm{km\,s^{-1}}$
Ecliptic-plane crossing	79,786	24.14	star at 6.425 ly; within 0.1 $\mathrm{km\,s^{-1}}$ of optimum
Requirement horizon	500,000	30.61	star receding; $\Delta v = 19.93$ $\mathrm{km\,s^{-1}}$ (not binding)

optimum (Figure 2). The tangential intercept (58,422 years) minimizes the cruise speed. The departure-Δv optimum (73,012 years) minimizes the quantity that matters, the departure budget for leaving Earth, and falls between the tangential intercept and the ecliptic-plane crossing (79,786 years) for the reason developed next.

2.4 Arrival-Epoch Optimization

The departure budget depends on more than the cruise speed. A vehicle departing the Earth–Moon system borrows Earth's orbital velocity of 29.78 $\mathrm{km\,s^{-1}}$, but the loan extends only to the component of the departure asymptote that lies *in* the ecliptic plane: we choose the departure epoch so that the in-plane projection of the asymptote aligns with Earth's motion, and the vehicle buys any out-of-plane component with propellant. Because the intercept point sits below the ecliptic for arrivals earlier than the plane crossing, the optimizer faces a trade (Figure 3): wait until the star crosses the plane (no tilt, but a more distant, faster-receding target) or depart slightly out of plane earlier (a small tilt penalty, but a cheaper intercept).

We therefore minimized the impulsive departure Δv from a 400 km low Earth orbit jointly over the arrival epoch T and the aim direction, using patched-conic departure kinematics (Appendix A gives the closed-form chain: cruise $v_\infty \rightarrow$ Earth-relative hyperbolic excess $v_{\infty,E} \rightarrow$ perigee-burn Δv, with reference speeds $v_{\mathrm{circ}} = 7.67$ $\mathrm{km\,s^{-1}}$ and $v_{\mathrm{esc}} = 10.85$ $\mathrm{km\,s^{-1}}$ at 400 km, and solar escape speed 42.12 $\mathrm{km\,s^{-1}}$ at 1 AU). The result:

- optimum arrival $T = 73{,}012$ years after launch;
- required cruise speed $v_\infty = 23.64$ $\mathrm{km\,s^{-1}}$, only 0.26 $\mathrm{km\,s^{-1}}$ above the absolute floor;
- departure asymptote tilted $\beta = 2.48°$ *below* the ecliptic;
- Earth-relative hyperbolic excess $v_{\infty,E} = 18.59$ $\mathrm{km\,s^{-1}}$;
- impulsive departure $\Delta v = 13.85$ $\mathrm{km\,s^{-1}}$ from a single perigee burn.

The same epoch and tilt minimize all three departure metrics examined (impulsive Δv, Earth-relative hyperbolic excess, and a low-thrust departure proxy), so the optimum is not an artifact of the impulsive assumption. A blind re-derivation then hardened its provenance: a separate implementation, given no access to any prior epoch value, scanned arrival epochs from 50,000 to 100,000 years at one-year resolution, refined the minimum locally, and converged to the same optimum of 73,012 years (73,012.17 after refinement) at $\Delta v = 13.85$ $\mathrm{km\,s^{-1}}$, agreement to 1 yr. Independently of both, Fermi Explorer's reference analysis [14] (computed with entirely separate code) places its optimum at 13.88 $\mathrm{km\,s^{-1}}$ at 72,800 years with a 2.4° tilt. The departure Δv, the arrival epoch, and the Earth-relative hyperbolic excess agree with ours to 0.3% or better; the optimum aim tilt agrees to 0.09° absolute. The tilt is ill-conditioned along the flat valley of Figure 2, where a small absolute spread is expected.

The flatness of the optimum has a practical consequence: the impulsive floor of Figure 2 stays within 100 $\mathrm{m\,s^{-1}}$ of the optimum from 66,908 to 80,436 years, and within 0.25 $\mathrm{km\,s^{-1}}$ out to 85,000 years. Arrival-epoch selection inside this window costs almost nothing and can be ceded entirely to launch-window phasing and operational convenience. In particular, the multi-year low-thrust escape examined in later sections delays solar-system departure to roughly 2043 for a 2029 launch. That delay is negligible against a 73,000-year cruise and a flat optimum.

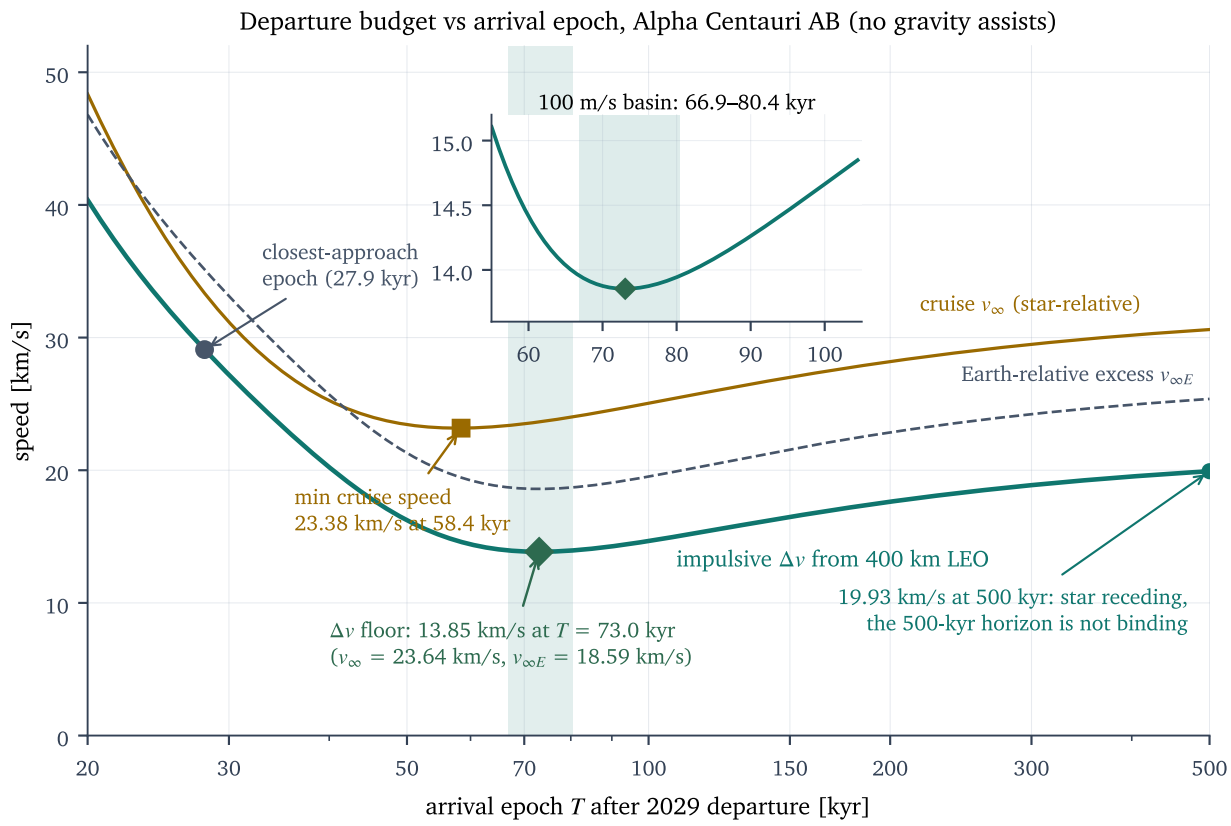


Figure 2. Departure budget versus arrival epoch. Impulsive Δv from a 400 km low Earth orbit and required cruise v_∞ as functions of arrival epoch on a logarithmic axis from 20 to 500 kyr, with the optimum, minimum-cruise-speed, and closest-approach epochs and the 500 kyr endpoint marked. The inset expands 55–105 kyr: the impulsive floor stays within $100\ \mathrm{m\,s^{-1}}$ of the optimum from 66,908 to 80,436 years, and within $0.25\ \mathrm{km\,s^{-1}}$ out to 85,000 years, so the arrival-epoch choice across this basin is free for other considerations. Past ~80 kyr the star recedes and the requirement rises monotonically, reaching $\Delta v = 19.93\ \mathrm{km\,s^{-1}}$ at the 500 kyr horizon.

2.5 Two Optima, Two Architectures

The blind re-derivation also mapped the arrival-epoch trade across the full 55,000–85,000-year window (Figure 4), and the map exposes a structural fact: the problem has *two* optima, and which one governs depends on what the architecture is charged for. A vehicle that pays for the Earth-relative departure (as the present concept does) borrows Earth's orbital velocity in plane and collects the Oberth benefit of a deep-in-the-well burn, and both effects favor an asymptote lying nearly in the ecliptic; its joint optimum therefore accepts a higher cruise speed ($23.64\ \mathrm{km\,s^{-1}}$) in exchange for a nearly flat aim ($2.48°$) and falls at 73,012 years. An architecture charged only for the heliocentric cruise speed (one departing from a platform already free of Earth, or pricing its propulsion after escape) faces neither the in-plane borrowing nor the Oberth geometry, and its optimum moves to 57,854 years, where the required cruise speed reaches its minimum of $23.173\ \mathrm{km\,s^{-1}}$. That epoch is the practical form of the exact tangential identity of Section 2.2: a true intercept is cheapest at $T^* = 58{,}422$ years, where v_∞ equals the transverse velocity of $23.38\ \mathrm{km\,s^{-1}}$, and the $2600\ \mathrm{AU}$ miss tolerance re-optimizes that epoch earlier. Table 5 gives a curated subset of the underlying 31-row scan; the full table, including the best pumping schedule found at each arrival epoch, appears in Appendix E.

The early-arrival branch is therefore tempting: between roughly 58,000 and 65,000 years the required cruise speed sits nearly on its floor, and the best pumping schedules found there (23.74 and $23.49\ \mathrm{km\,s^{-1}}$ of heliocentric Δv at 58,000 and 65,000 years) cost no more than the $23.98\ \mathrm{km\,s^{-1}}$ found at the departure-Δv optimum. Two

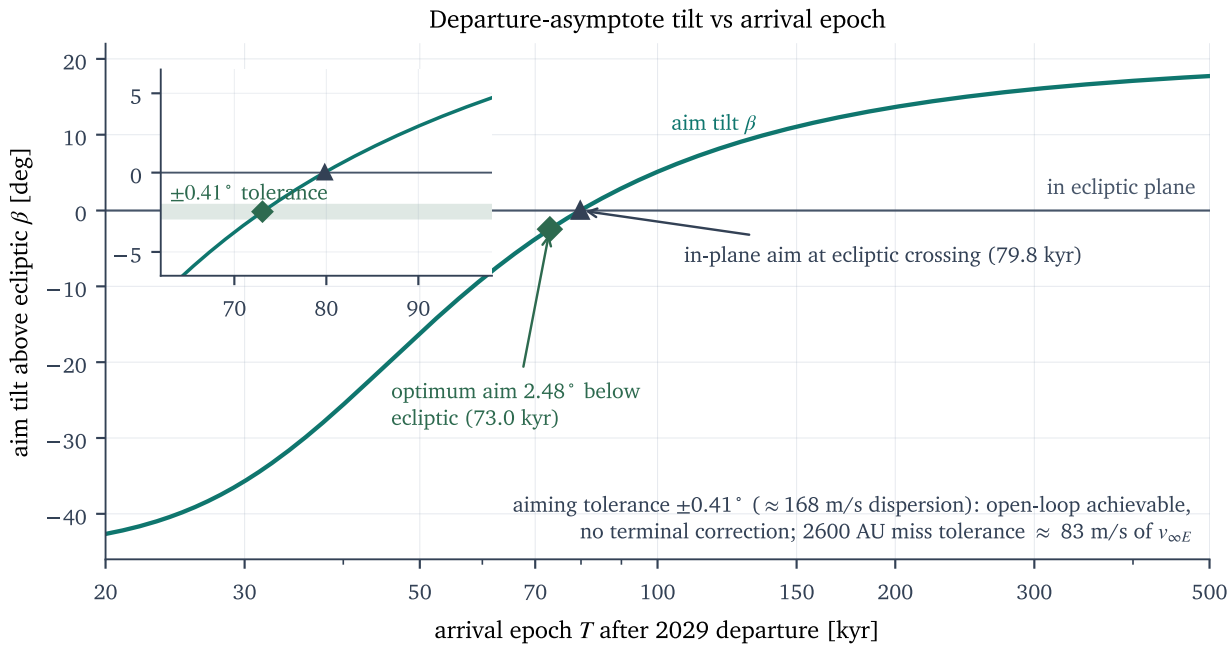


Figure 3. Departure aim tilt β versus arrival epoch; the inset shows the $\pm 0.41°$ open-loop aiming tolerance band around the optimum. The vehicle borrows Earth's orbital velocity of $29.78\ \mathrm{km\,s^{-1}}$ in plane only, so the tilt directly prices the out-of-plane component of the departure asymptote. The tilt passes through zero at the ecliptic-plane crossing (79,786 yr); the joint optimum accepts a $2.48°$ tilt to intercept the star several thousand years earlier, when it is closer and slower to chase.

caveats keep this report's design point at 73,012 years. First, the pumping-schedule figures are best-found upper bounds. They are not certified optima, and they scatter by roughly $\pm 0.2\ \mathrm{km\,s^{-1}}$ between search seeds, a scatter comparable to the 0.2–$0.5\ \mathrm{km\,s^{-1}}$ row-to-row differences under comparison, so the comparison does not resolve the ordering. Second, and more important, those schedules are planar: they exclude the cost of acquiring the out-of-plane tilt. We measured that cost only at the design point's $2.48°$, where it is $0.58\ \mathrm{km\,s^{-1}}$; it grows steeply with angle, and the required tilt steepens to about $-10°$ at the cruise-speed minimum (Figure 4b). Whether the cruise-speed saving survives the tilt bill is an open question; the early-arrival branch is promising, unsettled, and left as future work.

2.6 Design Intuitions Examined

The Fermi Explorer team advanced several geometric intuitions about where the optimum should lie. We tested each against the full optimization, and the scorecard endorses the team's physical reasoning.

1. *The departure budget rises sharply for arrivals much beyond ~80,000 years.* **Confirmed.** The required cruise speed grows from $24.14\ \mathrm{km\,s^{-1}}$ at the plane crossing to $30.61\ \mathrm{km\,s^{-1}}$ at 500,000 years, and the impulsive departure Δv from 13.9 to $19.93\ \mathrm{km\,s^{-1}}$: patience stops paying once the star begins to recede.
2. *The star's closest natural approach (~3 ly) is not the Δv optimum.* **Confirmed**, with the epoch refined: the closest natural approach occurs 27,955 years from launch at 3.152 ly, and intercepting there is the costliest intercept examined; the short flight time more than cancels the shorter distance.
3. *A tangential intercept near 60,000 years minimizes the cruise speed.* **Confirmed exactly:** $T^* = 58{,}422$ years, where the required v_∞ equals the star's transverse velocity of $23.38\ \mathrm{km\,s^{-1}}$ by the identity of Section 2.2.
4. *The ecliptic-plane crossing minimizes the departure Δv.* **Very nearly right:** arriving at the crossing epoch of 79,786 years costs less than $0.1\ \mathrm{km\,s^{-1}}$ more than the true optimum. The intuition correctly identifies

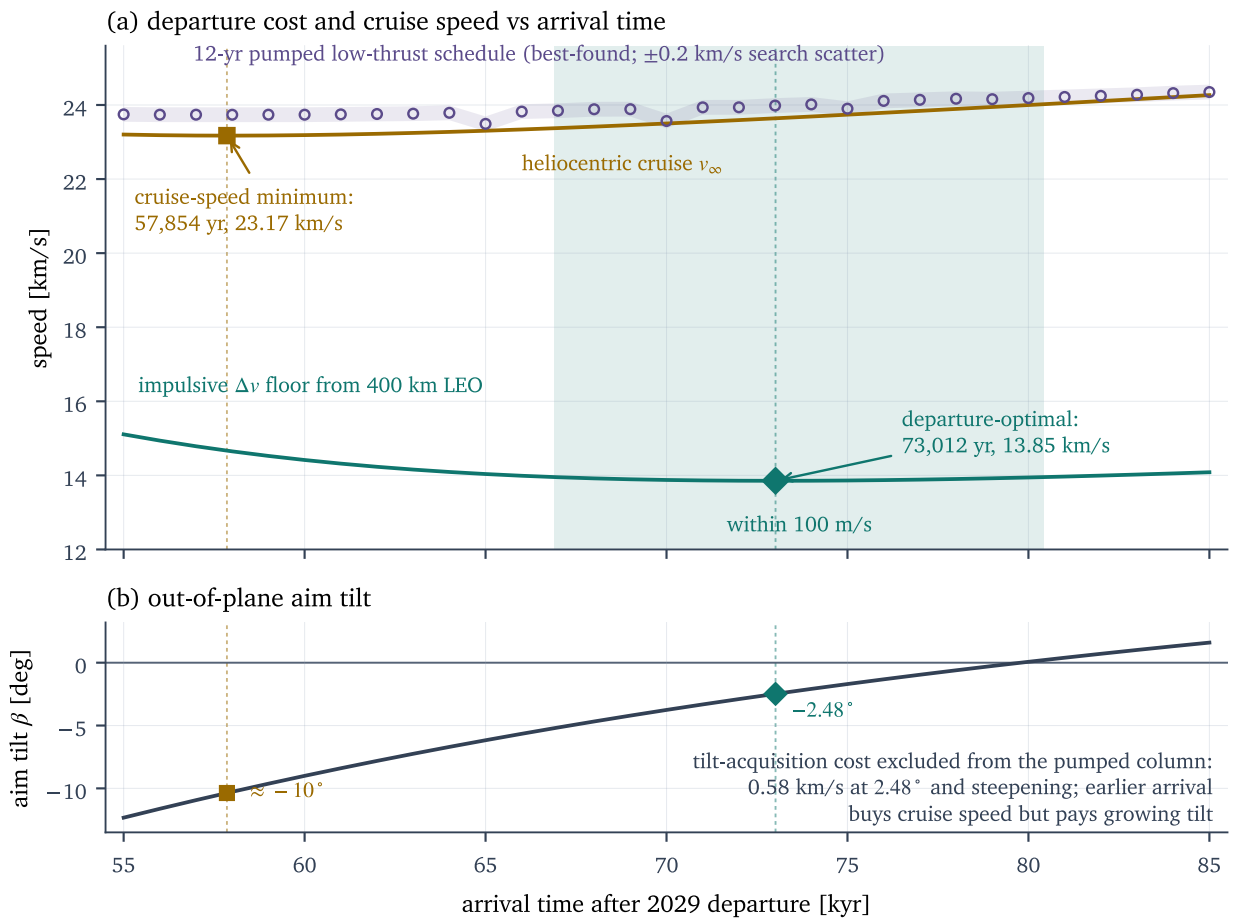


Figure 4. The arrival-epoch trade has two optima. (a) Impulsive departure floor from a 400 km low Earth orbit, required cruise v_∞, and the best 12-year pumping schedule found at each arrival epoch (open circles; each an upper bound on its optimum, with roughly $\pm 0.2\ \mathrm{km\,s^{-1}}$ scatter between search seeds). The departure-Δv optimum falls at 73,012 years and stays within $100\ \mathrm{m\,s^{-1}}$ across the shaded 66,908–80,436-year basin; the cruise-speed minimum falls at 57,854 years, $23.17\ \mathrm{km\,s^{-1}}$. (b) Aim tilt β versus arrival epoch: the tilt steepens from $-2.48°$ at the departure-Δv optimum to about $-10°$ near the cruise-speed minimum, and the cost of acquiring the tilt ($0.58\ \mathrm{km\,s^{-1}}$ at $2.48°$; excluded from the pumping-schedule column) grows with it. Earlier arrival buys cruise speed but pays growing tilt.

Table 5. Arrival-epoch trade, curated from the 31-row scan (full table in Appendix E). Cruise v_∞ and aim tilt β (negative below the ecliptic) follow from the lead relation with the 2600 AU miss tolerance; the impulsive floor is a single perigee burn from a 400 km low Earth orbit. A feasible 12-year pumping schedule reached every arrival epoch examined (Section 4).

Arrival T (yr)	Cruise v_∞ ($\mathrm{km\,s^{-1}}$)	β (deg)	Impulsive Δv ($\mathrm{km\,s^{-1}}$)	Comment
55,000	23.20	−12.33	15.11	steepest tilt examined
58,000	23.17	−10.27	14.65	at the cruise-speed minimum
65,000	23.30	−6.17	14.04	
70,000	23.50	−3.76	13.88	inside the $100\ \mathrm{m\,s^{-1}}$ basin
73,000	23.64	−2.48	13.85	departure-Δv optimum
80,000	24.00	+0.07	13.94	tilt passes through zero
85,000	24.27	+1.60	14.08	star receding

the dominant effect (avoiding the out-of-plane penalty) and misses only the small gain from intercepting earlier.

5. *The true optimum lies near 73,000 years, aimed a few degrees off plane.* **Confirmed almost exactly:** 73,012 years, $2.48°$ below the ecliptic.

2.7 Miss Tolerance and Aiming Dispersion

The mission requires only a pass within 2600 AU of the barycenter. A true intercept is not required. In energy terms this tolerance is nearly worthless: it reduces the required Earth-relative hyperbolic excess by about $83\ \mathrm{m\,s^{-1}}$, which the escape-energy term dilutes to roughly $71\ \mathrm{m\,s^{-1}}$ of impulsive Δv. The miss tolerance relaxes the *targeting* problem. The energy problem stands.

As a targeting relaxation, however, it is decisive. Hitting a 2600 AU sphere at interstellar range permits an asymptote pointing error of $0.41°$, equivalent to about $168\ \mathrm{m\,s^{-1}}$ of departure velocity dispersion. The sphere must absorb target-ephemeris error as well as execution dispersion, so the budget charges the ephemeris term first. Propagated to the 73,012-year encounter, the proper-motion uncertainty of the adopted astrometry [1] produces a cross-track dispersion of about 381 AU (1σ; some 147 AU using the cross-check orbital solution of [2]), consuming roughly 15% of the 2600 AU sphere. The radial-velocity uncertainty produces a larger along-track term: about 4600 AU (1σ; some 1500 AU with the cross-check values). An along-track error does not widen the miss distance: it shifts the encounter epoch by roughly ±900 years, and the pass still occurs, at negligible cost given the flat optimum of Section 2.4. Netting the cross-track ephemeris term against the sphere thus leaves about 85% of the miss tolerance for execution dispersion. Section 6 carries this ephemeris covariance jointly with the execution error sources through a full Monte Carlo dispersion analysis and confirms that the combined budget closes.

Within that residual tolerance, the execution-dispersion Monte Carlo of Section 6, which carries the target-ephemeris covariance alongside the thrust and attitude errors, shows the pointing tolerance is met open-loop: no trajectory correction after escape, with navigation custody held through the powered escape via star-tracker attitude knowledge and thrust telemetry. The mission therefore needs no terminal guidance, no course-correction propellant, and no late-mission contact with the vehicle; the requirement set is consistent with a fully expendable, out-of-contact cruise lasting the entire 73,000-year coast.

2.8 The 500,000-Year Horizon Is Not Binding

The mission's 500,000-year arrival ceiling exerts no pressure on the design. The optimum lies at 73,012 years, deep inside the allowed window, and the requirement *rises* toward the horizon: by 500,000 years the star recedes and the budget grows to $v_\infty = 30.61\ \mathrm{km\,s^{-1}}$ ($\Delta v = 19.93\ \mathrm{km\,s^{-1}}$). Relaxing the horizon would therefore buy nothing, and tightening it to as little as ~85,000 years would cost less than $0.1\ \mathrm{km\,s^{-1}}$. The binding constraints on this mission are propulsive and programmatic. The arrival horizon is not among them. The sections that follow take up those constraints.

3. Departure Energetics from Low Earth Orbit

The intercept geometry of the preceding section fixes the departure: the least expensive trajectory leaves Earth on a departure asymptote tilted 2.48° below the ecliptic plane, with cruise speed $v_\infty = 23.64\ \mathrm{km\,s^{-1}}$ and arrival 73,012 years after launch. Choosing the departure epoch so that the in-plane projection of the required velocity aligns with Earth's orbital motion lets the vehicle borrow Earth's 29.78 $\mathrm{km\,s^{-1}}$ of orbital speed in the ecliptic plane; only the small out-of-plane component, applied at an angle to that motion, costs propellant. The resulting Earth-relative hyperbolic excess is $v_{\infty,E} = 18.59\ \mathrm{km\,s^{-1}}$. The mission ground rules exclude gravity assists, so the baseline departure must buy that excess from a 400 km low Earth orbit on propulsion alone. The cost depends strongly on the thrust regime, and that dependence drives the entire vehicle design.

3.1 The Impulsive Floor

For a burn that is short compared with the orbital period, the Oberth effect applies: propellant spent deep in the gravity well, where the vehicle moves fastest, buys more orbital energy per unit of Δv, because the energy gained from an increment $\mathrm{d}v$ scales with the instantaneous speed at which the vehicle applies it. A single impulsive burn at perigee of a 400 km orbit therefore achieves the departure asymptote at the minimum possible cost,

$$\Delta v_{\mathrm{imp}} = \sqrt{v_{\infty,E}^2 + v_{\mathrm{esc}}^2} - v_{\mathrm{circ}}, \tag{3}$$

where $v_{\mathrm{circ}} = 7.67\ \mathrm{km\,s^{-1}}$ is the circular orbital speed at 400 km altitude and $v_{\mathrm{esc}} = \sqrt{2}\, v_{\mathrm{circ}} = 10.85\ \mathrm{km\,s^{-1}}$ is the corresponding escape speed. Evaluating Equation 3 at $v_{\infty,E} = 18.59\ \mathrm{km\,s^{-1}}$ gives

$$\Delta v_{\mathrm{imp}} = 13.85\ \mathrm{km\,s^{-1}}. \tag{4}$$

This is the floor for any high-thrust (chemical) departure without gravity assist, and it captures the full Oberth benefit. Two checks support it. An independent replication from first principles recovered the figure, together with the optimal arrival epoch, aim tilt, and Earth-relative hyperbolic excess, with a worst-case discrepancy below 0.5%. The Fermi Explorer reference analysis [14] agrees to within 0.3%: it finds 13.88 $\mathrm{km\,s^{-1}}$ at an arrival epoch of 72,800 years with a 2.4° tilt, a small offset consistent with the flatness of the optimum in the geometry analysis.

3.2 The Low-Thrust Picture

A solar-electric vehicle cannot claim the Oberth benefit at Earth. At thrust accelerations of order $10^{-4}\ \mathrm{m\,s^{-2}}$, escape from low Earth orbit proceeds as a slow spiral over many thousands of revolutions, with thrust at essentially all orbital phases. For such spirals the escape Δv from a circular orbit approaches the circular orbital speed of the starting orbit itself, the well-known penalty of low-thrust escape [8]. From 400 km that speed is 7.67 $\mathrm{km\,s^{-1}}$. We adopt an assumed budget of 7.6 $\mathrm{km\,s^{-1}}$, expended over approximately 1.5 years; the figure is likely conservative, since the escape-leg models already fold in eclipse and steering allowances (Appendix C). The spiral transits the radiation belts, and the vehicle sizing therefore includes a solar-array degradation factor of 1.25 for this phase.

The deeper consequence is that a slowly spiraling vehicle leaves Earth's sphere of influence with negligible Earth-relative hyperbolic excess. The vehicle must then accumulate the entire heliocentric Δv after escape, in interplanetary flight, where a second obstacle appears: solar-array power, and hence thrust, falls with distance from the Sun. Section 4 shows that a conventional outward spiral cannot reach the required cruise speed at all, and that a perihelion-pumping schedule can. With that heliocentric strategy, the total solar-electric departure budget is

$$\Delta v_{\mathrm{SEP}} = \underbrace{7.6}_{\text{Earth escape}} + \underbrace{[22.9,\ 24.0]}_{\text{heliocentric}} = 30.5\text{–}31.6\ \mathrm{km\,s^{-1}}. \tag{5}$$

These totals are planar. The measured out-of-plane increment for acquiring the 2.48° aim tilt adds 0.58 $\mathrm{km\,s^{-1}}$; the vehicle analysis carries it into the mass closure (Section 5). The heliocentric term is an interval whose two ends carry different pedigrees. The upper end is a measured constructive bound: an explicitly integrated, timestep-refined perihelion-pumping schedule at an initial thrust acceleration of $2.5 \times 10^{-4}\ \mathrm{m\,s^{-2}}$ reaches the required cruise speed at 23.97 $\mathrm{km\,s^{-1}}$. The lower end, 22.9 $\mathrm{km\,s^{-1}}$, is the best trajectory found by

Table 6. Departure budgets from a 400 km low Earth orbit at the optimal arrival epoch (73,012 yr), no gravity assist. The heliocentric interval is a reference interval: its upper end is a measured constructive bound, its lower end a best-found anchor (Appendix B); the closest trajectory found to the 22.26 $\mathrm{km\,s^{-1}}$ closure crossover, at four times the design thrust, remains 0.4 $\mathrm{km\,s^{-1}}$ above it.

Propulsion regime	Δv ($\mathrm{km\,s^{-1}}$)	Remarks
Impulsive (chemical)	13.85	single perigee burn; full Oberth
Solar-electric: Earth escape	7.6	spiral, ~1.5 yr; belt transit
Solar-electric: heliocentric	22.9–24.0	perihelion pumping (Section 4); reference interval
certified lower bound	16.56	throttleable class, unconstrained flight time, $r_p \geq 0.42$ AU
Solar-electric: total	30.5–31.6	reference interval

optimal-control analysis at a higher thrust acceleration ($1 \times 10^{-3}\,\mathrm{m\,s^{-2}}$), extended down to the design thrust level by a monotonicity argument valid for throttleable thrust. That result is single-method: its indirect Pontryagin solution converged, and its direct-collocation cross-check did not. It is a best-found reference anchor. It is not a certified bound. The certified lower bound is 16.56 $\mathrm{km\,s^{-1}}$, proved for the throttleable-thrust class with unconstrained flight time under the design's 0.42 AU perihelion floor (the impulsive multi-burn limit it supersedes is 16.50 $\mathrm{km\,s^{-1}}$; 15.6 $\mathrm{km\,s^{-1}}$ if the perihelion floor relaxes to 0.25 AU). Closure results later in this report quote the reference interval 22.9–24.0 $\mathrm{km\,s^{-1}}$ with one caveat: a heliocentric trajectory below the 22.26 $\mathrm{km\,s^{-1}}$ crossover, if one exists within the 12-year operational horizon, would close a 100 kg vehicle from low Earth orbit. The distances to that threshold are these: the interval's lower end, 22.9 $\mathrm{km\,s^{-1}}$, sits 0.6 $\mathrm{km\,s^{-1}}$ above it; the best trajectory found at the design thrust level costs 23.98 $\mathrm{km\,s^{-1}}$, 1.7 $\mathrm{km\,s^{-1}}$ above it; and the best found at four times the design thrust costs 22.66 $\mathrm{km\,s^{-1}}$, 0.4 $\mathrm{km\,s^{-1}}$ above it. Appendix B gives the full argument. Table 6 summarizes the departure budgets.

The factor-of-two-plus spread between the impulsive floor and the solar-electric total is the structural price of low thrust. No engineering refinement removes it: the vehicle pays once through the loss of the Oberth effect at Earth and again through the heliocentric climb. Any credible solar-electric concept must therefore carry roughly 31 $\mathrm{km\,s^{-1}}$ of total Δv. The impulsive floor is the wrong sizing basis for such a concept, and the mass-closure analysis later in this report turns on that distinction.

Both budgets already exploit the freedom to select the delivery orbit within the low-Earth class. The quoted figures assume an injection plane that contains the tilted departure asymptote, so they charge no separate plane change. The altitude lever is weak: raising the starting orbit reduces the escape cost only in proportion to the circular speed, a few tenths of a $\mathrm{km\,s^{-1}}$ across low-Earth altitudes, while slightly eroding the Oberth benefit of an impulsive burn. The delivery-orbit choice that matters lies beyond low Earth orbit altogether. Procuring a geostationary-transfer-orbit drop-off cuts the electric-propulsion escape cost from 7.6 to 4.24 $\mathrm{km\,s^{-1}}$, and Section 5 shows that this single procurement change closes the 100 kg vehicle.

4. Solar-Electric Propulsion: the Power Wall and Perihelion Pumping

4.1 Power Model

Solar-array output falls with the inverse square of heliocentric distance r, and that falloff, carried through to thrust, is what defeats the conventional escape spiral. The propulsion model used throughout this report takes

$$P(r) \;=\; P_1 \,\min\!\left[\left(\frac{r_1}{r}\right)^2,\, 4\right], \tag{6}$$

where P_1 is the array output at $r_1 = 1$ AU and the factor-of-four thermal power cap, which engages inside roughly 0.5 AU, limits harvested power. The cap is an assumption. Demonstrated practice does not include harvesting at that multiple: inner-heliosphere missions of the MESSENGER class survive 0.3–0.47 AU by off-pointing their arrays to *refuse* the elevated flux, so a cap of four leans optimistic. Section 4.6 therefore quantifies its influence on every pumping result. At fixed specific impulse and thruster efficiency, thrust is proportional

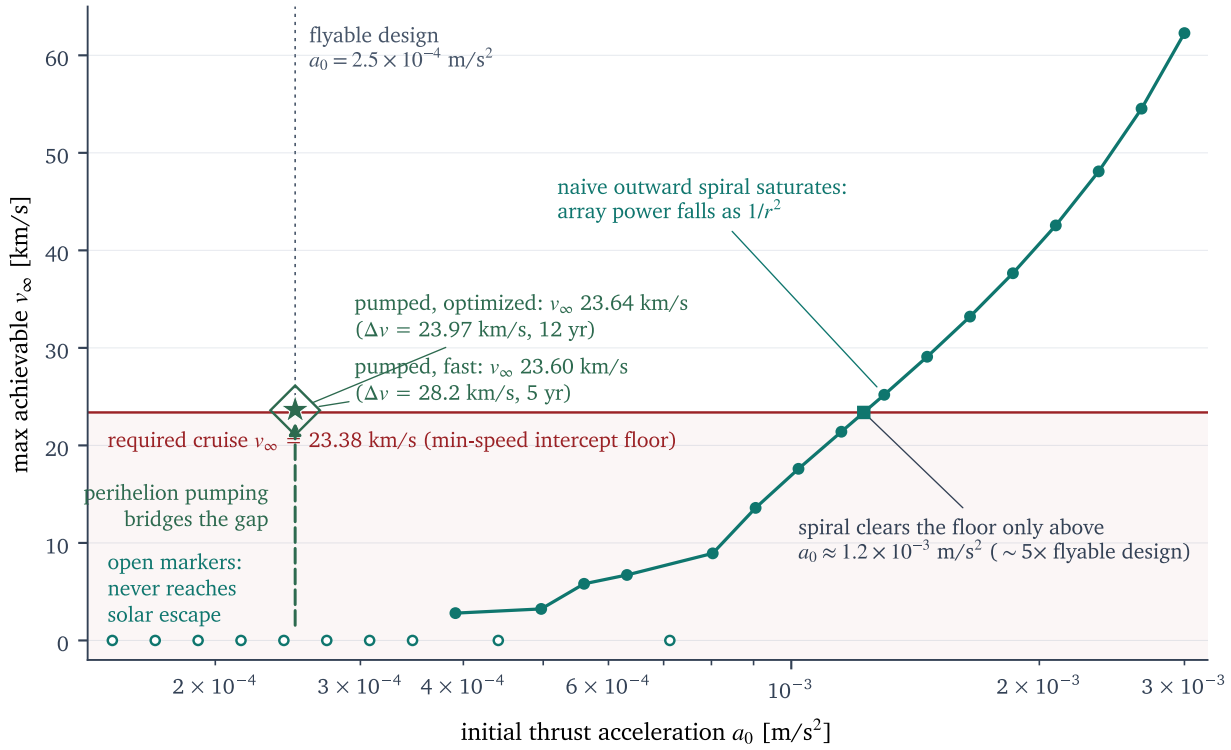


Figure 5. Achievable cruise speed versus initial thrust acceleration a_0 (logarithmic axis) for a conventional outward low-thrust spiral. The curve saturates below the 23.38 $\mathrm{km\,s^{-1}}$ cruise floor (horizontal line) at every flyable thrust level and clears it only above $a_0 \approx 1.2\times10^{-3}\,\mathrm{m\,s^{-2}}$, roughly five times the design value. The two points above the line come from perihelion-pumping schedules at $a_0 = 2.5 \times 10^{-4}\,\mathrm{m\,s^{-2}}$: the saturation bounds outward spirals; it does not bound solar-electric propulsion as such.

to power, so the available thrust acceleration inherits the same dependence on r. The single sizing parameter is a_0, the initial thrust acceleration at 1 AU and full vehicle mass.

4.2 The Power Wall: Outward-Spiral Saturation

The conventional low-thrust escape strategy thrusts continuously outward from 1 AU. Such a spiral is self-defeating: the vehicle climbs away from its own power source, so thrust decays as it recedes, and the Δv that remains deliverable arrives at ever larger radii and ever lower orbital speeds, where each increment contributes least to orbital energy. The achievable cruise speed therefore saturates as a function of a_0. The baseline analysis finds ceilings of approximately 0, 3.4, and 17.0 $\mathrm{km\,s^{-1}}$ at $a_0 = 1.5$, 5, and $10 \times 10^{-4}\,\mathrm{m\,s^{-2}}$ respectively. All three ceilings sit below the 23.38 $\mathrm{km\,s^{-1}}$ cruise floor: the minimum speed any intercept requires, set by the target system's transverse velocity. Two independent trajectory integrators reproduce the saturation curve with agreement to 2.7%; Figure 5 shows it.

On outward spirals the present analysis and the Fermi Explorer reference analysis [14] reach the same qualitative conclusion. The agreement extends to the exit from the wall: the region above $a_0 \approx 1.2 \times 10^{-3}\,\mathrm{m\,s^{-2}}$ in which Figure 5 shows outward spirals clearing the cruise floor corresponds to the pure-solar closing corner the reference analysis itself identified, above a vehicle specific power of roughly 100 W/kg, and presented as its default route. Section 4.7 compares that route with the present result. One caution applies. The reference analysis's published figure of roughly 20 $\mathrm{km\,s^{-1}}$ for a "realistic low-thrust" departure is a benchmark assumption in that analysis. It is not an output of the reference model: the reference trajectory engine itself yields 25.1 $\mathrm{km\,s^{-1}}$ for the outward spiral at the optimal epoch, and the associated vehicle sizing of roughly 500 kg

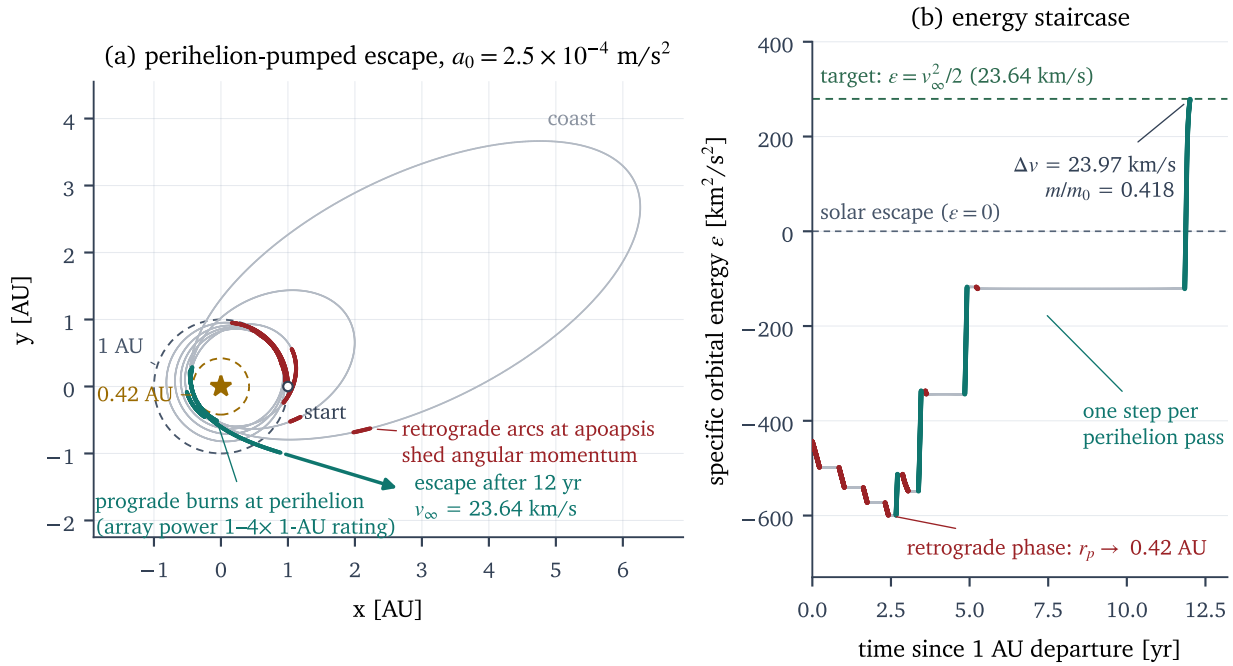


Figure 6. The optimized perihelion-pumping escape at $a_0 = 2.5 \times 10^{-4}\,\mathrm{m\,s^{-2}}$. (a) Heliocentric trajectory colored by thrust state (retrograde, coast, prograde), with the 1 AU and 0.42 AU circles dashed. Retrograde arcs near 1 AU lower perihelion; prograde arcs concentrate near perihelion, where array power is at or above its 1-AU rating. (b) Specific orbital energy versus time, colored by thrust state: each prograde perihelion pass adds one step to the energy staircase; the vehicle reaches escape energy at 12.0 yr with $\Delta v = 23.97\ \mathrm{km\,s^{-1}}$ and final mass fraction 0.418.

follows from the 20 $\mathrm{km\,s^{-1}}$ benchmark.

4.3 Perihelion Pumping

Saturation is a property of the outward-spiral trajectory class. Solar power itself imposes no such ceiling: escape requires the vehicle to leave the Sun, and nothing requires it to thrust while leaving. A perihelion-pumping schedule inverts the spiral's logic. Retrograde arcs near 1 AU shed angular momentum, lowering perihelion to 0.42 AU, just inside the radius at which the thermal power cap of Equation 6 engages, so that array power at perihelion stands at one to four times its 1-AU rating. Prograde arcs then concentrate near perihelion, where power is highest and where, by the same Oberth logic that favors the perigee burn in the impulsive case, each increment of Δv buys the most orbital energy. Successive revolutions raise aphelion while the perihelion burns repeat, until the orbit reaches escape energy. In effect, the multi-revolution structure recovers for low thrust the perihelion-burn advantage that a single fast pass cannot provide at these thrust levels.

The ingredients of this maneuver are established practice; the new element is what their combination closes. Oberth [9] identified the energetic advantage of burning deep in a gravity well, and the two-burn escape that exploits it (lower the periapsis, then burn at periapsis) is classical. Powered-flyby and solar-Oberth (perihelion-burn) escapes built on the same effect are established concepts in the interstellar-precursor literature [11], including in Fermi Explorer's own reference analysis [14]. Splitting a low-thrust maneuver into repeated apsis-centered arcs over many revolutions is likewise standard practice: SMART-1 escaped geostationary-transfer orbit on repeated perigee- and apogee-centered thrust arcs [10], and multi-revolution apsis-arc structures are routine in low-thrust trajectory optimization [7]. What the present analysis contributes is the quantified closure of this mission class by that combination (cruise speed, Δv, flight time, and mass fraction) under solar power throttled by the $1/r^2$ law of Equation 6.

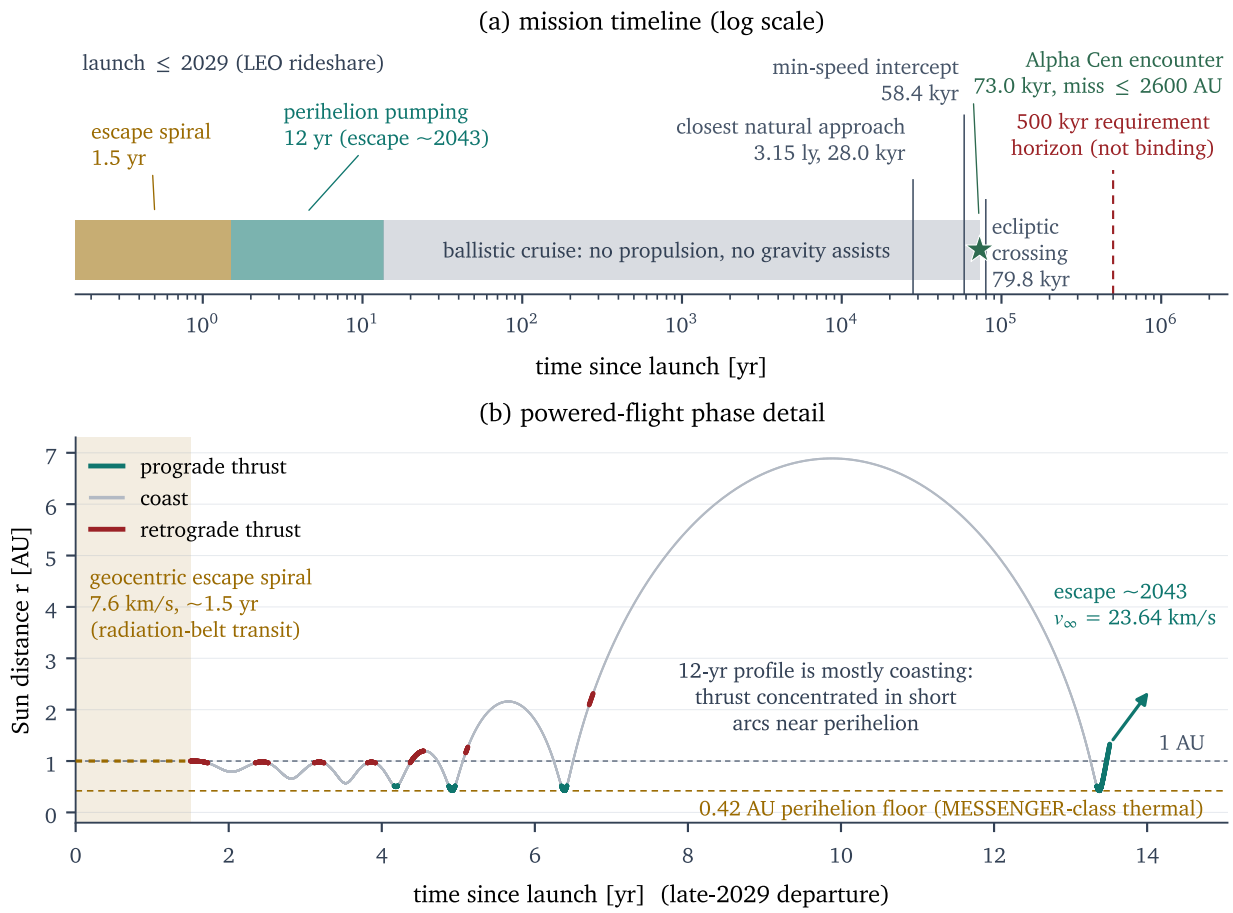


Figure 7. (a) Mission phases on a logarithmic time axis: Earth-escape spiral (~1.5 yr, 7.6 $\mathrm{km\,s^{-1}}$), heliocentric perihelion pumping (12 yr, 23.97 $\mathrm{km\,s^{-1}}$), and ballistic interstellar coast (73,000 yr) to the flyby; more than 98% of the mission is unpowered coast. (b) Powered-flight detail: heliocentric distance versus time, colored by thrust state, showing the escape spiral near 1 AU and pumping revolutions that descend to the 0.42 AU perihelion floor.

The optimized schedule at $a_0 = 2.5\times10^{-4}\,\mathrm{m\,s^{-2}}$ (Figure 6) delivers the required cruise speed of 23.64 $\mathrm{km\,s^{-1}}$ for a total Δv of 23.97 $\mathrm{km\,s^{-1}}$, over a flight time of 12.0 years of which only about 1.3 years is cumulative thrusting; the remainder is coast between thrust arcs. The final mass fraction is 0.418, consistent with the rocket equation at the 2800 s specific impulse assumed in the vehicle model, and the minimum solar distance is 0.42 AU. For a 2029 launch, escape from the solar system occurs around 2043. The arrival epoch is insensitive to the added departure time: the geometry analysis shows a departure-budget optimum flat from roughly 70 to 85 thousand years, with a spread below 0.25 $\mathrm{km\,s^{-1}}$. Figure 7 places the phases on a common time axis.

4.4 The Patience Trade

A faster pumping schedule exists: a 4.9-year variant reaches essentially the same cruise speed (23.60 $\mathrm{km\,s^{-1}}$) at a cost of 28.2 $\mathrm{km\,s^{-1}}$. Patience is therefore worth approximately 4 $\mathrm{km\,s^{-1}}$ of Δv, and the trade is decisive: the mass-closure analysis of the following section closes only with the patient 12-year schedule. Against a cruise of 73,000 years, the additional seven years of powered flight are immaterial to the mission. Their real cost is operational, in the duration of powered-flight custody, and the cost analysis prices that custody.

4.5 Bounding the Requirement

The 12-year cost is the best schedule found under an operational limit on powered-flight duration. It is not a physical floor, and the two claims separate quantitatively. The admissible class is this: any piecewise-continuous thrust program whose magnitude stays within the power-throttled envelope of Equation 6, whose heliocentric distance never falls below 0.42 AU, and which reaches the required cruise speed of 23.64 $\mathrm{km\,s^{-1}}$, with no constraint on flight time. For every trajectory in this class, the total heliocentric Δv is at least 16.56 $\mathrm{km\,s^{-1}}$. The bound is certified: a closed-form chain of analytic inequalities over the entire admissible class, owing nothing to any optimizer's output. It improves on the 16.50 $\mathrm{km\,s^{-1}}$ bi-parabolic floor, an impulsive-limit argument, by pricing the finite, power-throttled thrust directly. It is also sharp. In the adversarial numerical audit, a grazing ballistic orbit attains its tightest inequality with machine-epsilon equality, exactly where the comparison argument predicts (Appendix E.5).

The bound sits close to an achieved trajectory. With flight time unconstrained, an explicitly integrated trajectory in the same admissible class reaches the required cruise speed for 18.87 $\mathrm{km\,s^{-1}}$ after about 58 years of powered flight; the gate-passing schedules found in that search span 33–58 years at 18.9–20.5 $\mathrm{km\,s^{-1}}$. The unconstrained optimum therefore lies between 16.56 $\mathrm{km\,s^{-1}}$ (the certified lower bound) and 18.87 $\mathrm{km\,s^{-1}}$ (an integrated trajectory, so an upper bound on the optimum). The roughly 5 $\mathrm{km\,s^{-1}}$ separating that bracket from the 12-year cost is the continuation of the patience trade: program duration alone buys it back, so the 12-year figure is a schedule-driven quantity. The dynamics set no floor at that level. One consequence stands out. The 18.87 $\mathrm{km\,s^{-1}}$ trajectory lies below the 22.26 $\mathrm{km\,s^{-1}}$ crossover at which a 100 kg vehicle from low Earth orbit would close (Section 5). The 12-year schedule excludes that concept; the physics of the trajectory class does not.

On the 12-year horizon itself, the closure analysis carries the requirement as the reference interval 22.9–24.0 $\mathrm{km\,s^{-1}}$, whose two ends have the standing set out in Section 3. The upper end rests on the measured constructive schedule of Figure 6: 23.97 $\mathrm{km\,s^{-1}}$ at its production timestep, 23.98 $\mathrm{km\,s^{-1}}$ quoted conservatively across timestep refinements, rounded up to 24.0. The lower end is the best-found optimal-control anchor, computed at four times the design thrust acceleration (where the small revolution count keeps the problem numerically tractable) and carried down to the design level by monotonicity of the optimum in available thrust. The 22.9 $\mathrm{km\,s^{-1}}$ floor comes from the extremal pair itself: the 22.94 $\mathrm{km\,s^{-1}}$ extremal at a slightly higher target variant, rounded to the tenth, with the 22.66 $\mathrm{km\,s^{-1}}$ extremal at the required cruise speed lying below it. The anchor is single-method: one converged method, no independent confirmation, a reference value that bounds nothing in either direction. Its full pedigree, including the failed cross-check, appears in Section 3 and Appendix B. Figure 8 lays out the resulting two-class structure.

4.6 Sensitivity to the Thermal Power Cap

Every quantitative pumping result in this section uses the factor-of-four thermal power cap of Equation 6. Inner-heliosphere flight practice manages the 0.3–0.5 AU thermal environment by off-pointing arrays; it does not harvest the elevated flux. We therefore tested the cap directly: we re-optimized the pumping schedule at the design $a_0 = 2.5 \times 10^{-4}\,\mathrm{m\,s^{-2}}$ with the cap reduced, all other model parameters held fixed. With power capped at three times the 1-AU rating, the optimized escape requires 24.10 $\mathrm{km\,s^{-1}}$ over 19.7 years; capped at two, 25.88 $\mathrm{km\,s^{-1}}$ over 15.1 years. Both re-optimized trajectories reach escape, and both are timestep-stable (half-timestep reruns change Δv by less than 0.3%). Both, however, extend powered-flight custody to 15–20 years, beyond the 12-year horizon on which component life, burn sizing, and the operations-cost model rest. A reduced cap therefore carries a qualification and operations-cost consequence in addition to the added Δv.

The mechanism therefore survives a halving of the assumed cap at a cost of approximately 1.9 $\mathrm{km\,s^{-1}}$ of Δv and several years of flight time; it does not fail. The two cases span the assumption range examined (we did not re-optimize the sub-2× regime): cap 4 is optimistic, since it assumes the arrays harvest at four times rating in an environment MESSENGER-class practice survives by refusing that flux, while cap 2 is conservative. The mass-closure consequence falls to Section 5: the 100 kg vehicle departing from geostationary transfer orbit retains closure at the cap-2 Δv requirement, with a reduced margin, and that section traces the closing boundary.

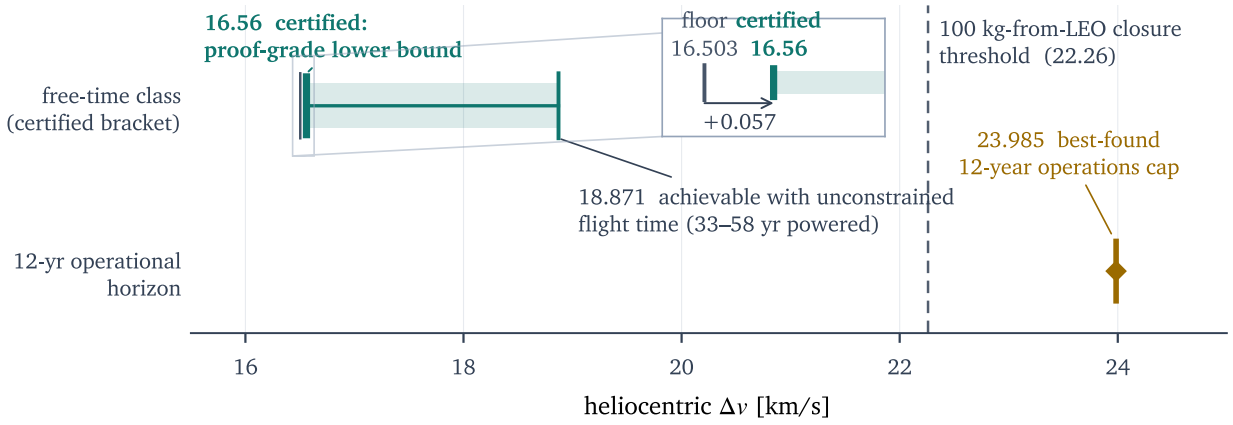


Figure 8. Two-class structure of the heliocentric Δv requirement. Upper row: with flight time unconstrained, the optimum lies between the certified lower bound of 16.56 $\mathrm{km\,s^{-1}}$ and an integrated trajectory at 18.87 $\mathrm{km\,s^{-1}}$ after about 58 years of powered flight (gate-passing schedules span 33–58 years at 18.9–20.5 $\mathrm{km\,s^{-1}}$); the 16.50 $\mathrm{km\,s^{-1}}$ bi-parabolic floor that the certified lower bound supersedes appears below the bracket. Lower row: on the 12-year operational horizon, the best schedule found costs 23.98 $\mathrm{km\,s^{-1}}$, a best-found value from a converged search; it does not bound the true optimum. The dashed line marks the 22.26 $\mathrm{km\,s^{-1}}$ crossover below which a 100 kg vehicle from low Earth orbit would close. The unconstrained-flight-time trajectory lies below that crossover, so the concept fails on schedule; the trajectory class itself permits closure.

4.7 Consequences for the Feasibility Verdict

Perihelion pumping changes the feasibility verdict for this mission class, and the change amends the Fermi Explorer reference analysis [14] at specific points. That analysis found, as this one does, that outward spirals saturate below the required cruise speed at every flyable thrust level, and it identified two ways past the wall. First, a pure-solar closing corner at high specific power: above a vehicle specific power of roughly 100 W/kg, outward-spiral solar-electric closure becomes available, and the reference analysis presented that high-specific-power route as its default. Second, a solar-Oberth architecture (lower the perihelion, then burn at perihelion), whose electric variant it rejected on the grounds that a single perihelion pass is too short for ion-class thrust to deliver the required Δv. On that basis it recommended nuclear-electric power for a near-term vehicle.

What the present result adds is the multi-revolution form of the perihelion burn, which removes the single-pass objection: the vehicle delivers over many passes the Δv that one pass cannot, with the thrust arcs re-concentrated at perihelion on every revolution. The required cruise speed then falls within reach of solar power alone at today's ~60 W/kg system-level arrays. This is a trajectory-class result; it requires no component advance, no nuclear reactor, no chemical kick stage, and no gravity assist. The amendment to the reference analysis is therefore narrow: its power-wall finding stands; its high-specific-power route remains valid but is no longer necessary; and its single-pass objection to the electric perihelion burn, correct for a single pass, does not apply to the multi-revolution schedule.

What solar-electric propulsion does demand, beyond the patience of the 12-year schedule, is the operating point of the vehicle model: a specific impulse of at least 2800 s at total efficiency near 0.55 in a 0.5–1.0 kW string, with roughly 17,000–21,000 hours of single-string firing and 64–103 kg of xenon throughput. No flown Hall thruster covers that point; flown Hall specific impulse tops out near 1,650 s at this power class [13]. Demonstrated gridded-ion hardware brackets it: NSTAR's flight-documented TH3–TH4 throttle points (0.91–1.02 kW, 2,843–2,942 s, efficiency 0.527–0.554) [13] sit on the required corner, with 30,352 hours and 235 kg of single-string ground life and 16,265 hours of single-unit flight [12], and with dual-string redundancy and the standard 1.5× throughput qualification margin available as mitigations. A demonstrated component class, gridded ion, thus meets the propulsion requirement at the price of a mission-specific qualification campaign; Section 5 carries the full life and throughput budget.

The pumping result carries independent verification of a strong form. A separate implementation, developed from scratch without access to the baseline trajectory code, constructed its own pumping schedule and reached a cruise speed of 23.60 $\mathrm{km\,s^{-1}}$ at a Δv of 28.2 $\mathrm{km\,s^{-1}}$; this independently confirms that pumping defeats the power wall at $a_0 = 2.5 \times 10^{-4}\,\mathrm{m\,s^{-2}}$. Timestep refinement of the optimized 12-year schedule changes the converged Δv by 0.06%. Appendix B details the models, the control policy, and the bracketing argument.

4.8 Limitations

The trajectory analysis carries three limitations. First, the baseline heliocentric optimization is planar; the $2.48°$ aim tilt enters through the Earth-relative hyperbolic excess. We measured the three-dimensional cost of acquiring that tilt directly. A three-dimensional trajectory optimization under the same perihelion floor, thrust envelope, and 12-year limit, constrained to place the departure asymptote within $0.04°$ of the target latitude $2.48°$ below the ecliptic, delivers the tilt for an out-of-plane increment of 0.58 $\mathrm{km\,s^{-1}}$ over the planar solution: 24.56 $\mathrm{km\,s^{-1}}$ against 23.98 $\mathrm{km\,s^{-1}}$, both quoted conservatively across successive timestep refinements (Figure 9). The measured increment falls where analytic reasoning places it: between the 22 m/s cost of steering the tilt within the escape burns at no geometric penalty and the 1.02 $\mathrm{km\,s^{-1}}$ cost of executing it as a separate plane change. The planar and three-dimensional totals are each the best solution found by a converged search; neither is a certified optimum. Their difference is therefore a measured estimate of the out-of-plane cost that bounds nothing in either direction, and Section 5 carries its consequence for mass closure. The planar treatment also leaves the early-arrival trade of Section 2.5 unsettled. Arrival epochs near 58,000 years reduce the required cruise speed to 23.17 $\mathrm{km\,s^{-1}}$ and are therefore energetically attractive, but they steepen the aim tilt toward $10°$, and we measured the tilt-acquisition cost only at $2.48°$, where it is 0.58 $\mathrm{km\,s^{-1}}$ and grows steeply with angle. The trade remains open, and we leave it to future work. Second, the repeated passes at 0.42 AU place the vehicle in a thermal environment requiring qualification comparable to inner-solar-system missions of the MESSENGER class. This is a cost and schedule item; it poses no physics risk, and the cost analysis prices it. Section 4.6 tests directly the companion assumption that the arrays harvest at up to four times their 1-AU rating in that environment. Third, the mechanism has a floor: pumping fails to reach the required cruise speed below $a_0 \approx 2.25 \times 10^{-4}\,\mathrm{m\,s^{-2}}$, so an array degradation of 10–20% over the pumping phase would erase the margin of a 100 kg-class vehicle from low Earth orbit. The mass-closure analysis treats this sensitivity, and the departure options that retire it, quantitatively.

5. Vehicle Mass Closure

The trajectory analysis fixes what the propulsion system must deliver: for a solar-electric departure from a $400\,\mathrm{km}$ low Earth orbit (LEO), a $7.6\,\mathrm{km\,s^{-1}}$ Earth-escape spiral (roughly 1.5 years, transiting the radiation belts) plus a heliocentric perihelion-pumping phase spanning the reference interval of 22.9–$24.0\,\mathrm{km\,s^{-1}}$, for a total of 30.5–$31.6\,\mathrm{km\,s^{-1}}$. The interval's two ends carry different pedigrees: the upper end is verified (independent re-integration agrees to 0.29%), the lower end best-found (Section 4.5). A vehicle of the concept class (approximately $100\,\mathrm{kg}$ at launch, predominantly xenon propellant, carrying a payload of at least $1\,\mathrm{kg}$) nearly absorbs that budget from LEO. It absorbs it with margin to spare from a geostationary transfer orbit (GTO).

5.1 Mass Model

The mass model produces the closure verdicts above: from LEO the minimum closing wet mass exceeds the concept's, and from GTO the concept closes. The model combines the rocket equation with linear subsystem scalings; Appendix C gives the full equations and the closure-boundary algebra. For a wet (launch) mass m_w and specific impulse I_{sp}, the propellant mass is $m_p = m_w\,[1 - \exp(-\Delta v_{\mathrm{tot}}/v_e)]$, where $v_e = g_0 I_{\mathrm{sp}}$ is the exhaust velocity and Δv_{tot} the total budget above. We size the thrust acceleration at $1\,\mathrm{AU}$ and wet mass, $a_0 \approx 2.5 \times 10^{-4}\,\mathrm{m/s^2}$, so that the cumulative thrusting time fits within a four-year allowance, consistent with the optimized pumping schedule, which accumulates roughly 1.3 years of thrust arcs over its twelve-year flight. The thrust $F = a_0 m_w$ then sets the $1\,\mathrm{AU}$ electrical power through the jet-power relation $P = F v_e/(2\eta)$ with an assumed thruster efficiency $\eta = 0.55$; closing designs require 0.5–$1.0\,\mathrm{kW}$ at $1\,\mathrm{AU}$. The subsystem scalings are assumed model inputs. Each subsystem scales linearly with that power and with propellant load: the solar array at $60\,\mathrm{W/kg}$ system-level specific power, penalized by a factor of 1.25 for radiation degradation during the belt transit of a LEO start; power processing at $6\,\mathrm{kg/kW}$; xenon tankage at 12% of propellant mass; and a fixed

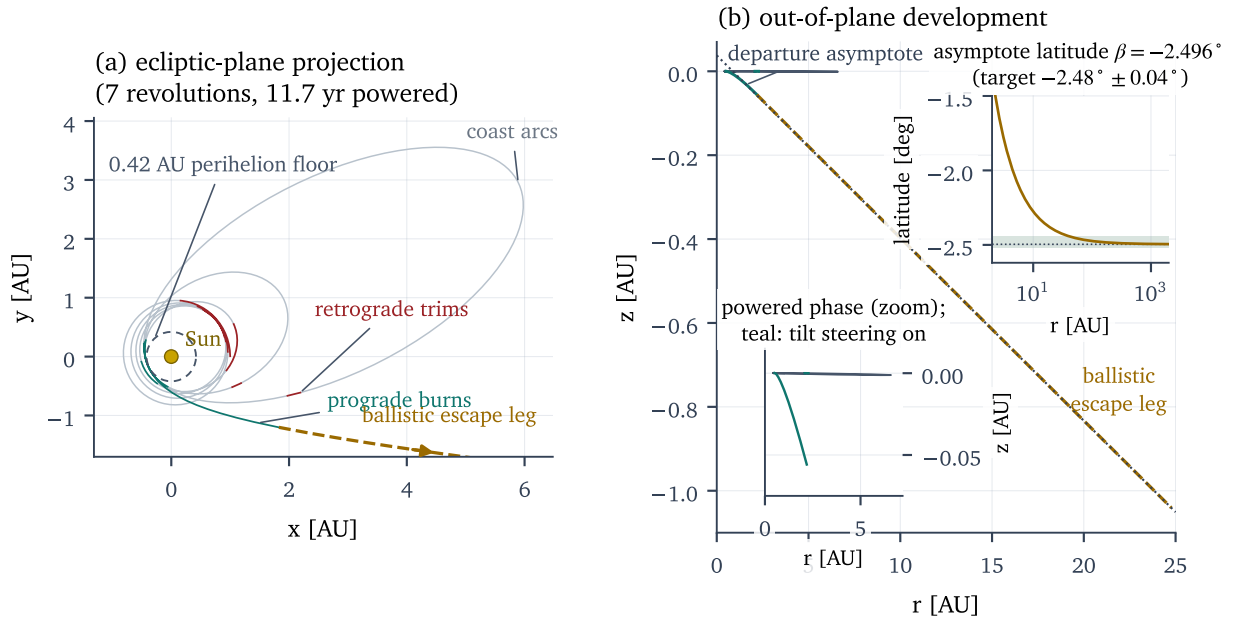


Figure 9. The three-dimensional perihelion-pumping escape at $a_0 = 2.5 \times 10^{-4}\,\mathrm{m\,s^{-2}}$, with the departure asymptote directed $2.48°$ below the ecliptic (achieved within the $0.04°$ aiming gate). Out-of-plane excursions stay below 0.06 AU and concentrate late in the escape, after the orbit is already hyperbolic. The measured cost of the tilt is 0.58 $\mathrm{km\,s^{-1}}$ over the planar schedule, for a total of 24.56 $\mathrm{km\,s^{-1}}$.

1 kg payload plus 8 kg bus. The expendable, out-of-contact posture permits a minimal antenna, whose mass is subsumed in the bus allowance. A design closes when the sum of these masses does not exceed m_w; margins below are quoted in percentage points of wet mass.

5.2 Closure Results

Figure 10 summarizes the closure results at both ends of the heliocentric reference interval. From LEO, the 100 kg concept misses closure by a measured 1.0–2.8% of wet mass everywhere in the interval: a shortfall of one to three kilograms at most. The interval's lower end is a best-found anchor; it is not a certified bound, so this verdict is conditional. The analysis cannot exclude closure of a 100 kg vehicle from LEO if some trajectory below the $\approx 22.3\,\mathrm{km\,s^{-1}}$ crossover exists within the 12-year operational horizon. The distances in hand are best stated against that crossover: the interval's lower end of $22.9\,\mathrm{km\,s^{-1}}$ sits $0.6\,\mathrm{km\,s^{-1}}$ above it; the best schedule integrated at the design thrust, $23.98\,\mathrm{km\,s^{-1}}$, sits $1.7\,\mathrm{km\,s^{-1}}$ above; and even the four-times-thrust optimal-control anchor itself, $22.66\,\mathrm{km\,s^{-1}}$ before its monotonicity extension to the design thrust, sits $0.4\,\mathrm{km\,s^{-1}}$ above. Equivalently, the minimum closing wet mass from LEO is 113.5–146 kg across the interval. Growing the vehicle to 150 kg closes the budget across the entire interval in this planar accounting, with margins of +0.2 to +2.0 points (revisited under the measured out-of-plane increment below). An independent analysis verified the closure algebra and reproduced the boundary at 113.3 kg, agreement to better than 0.2% with the design-sweep value.

Two features of the concept survive scrutiny unchanged. First, the LEO closing designs are 67–68% propellant and the GTO designs 63–64%: roughly two thirds of the vehicle, essentially all xenon. The concept's proportions are correct, and only its total is short from LEO. Second, closure is possible only with the patient pumping schedule. The twelve-year optimized schedule delivers the heliocentric phase for $23.97\,\mathrm{km\,s^{-1}}$; a fast 4.9-year variant exists but costs $28.2\,\mathrm{km\,s^{-1}}$, and no examined vehicle closes at that price. The patience trade is worth roughly $4\,\mathrm{km\,s^{-1}}$ of Δv, and against a 73,000-year cruise the additional seven years of powered flight are immaterial to the mission.

Closure also constrains the thruster. The specific-impulse requirement arises from burn-time sizing against

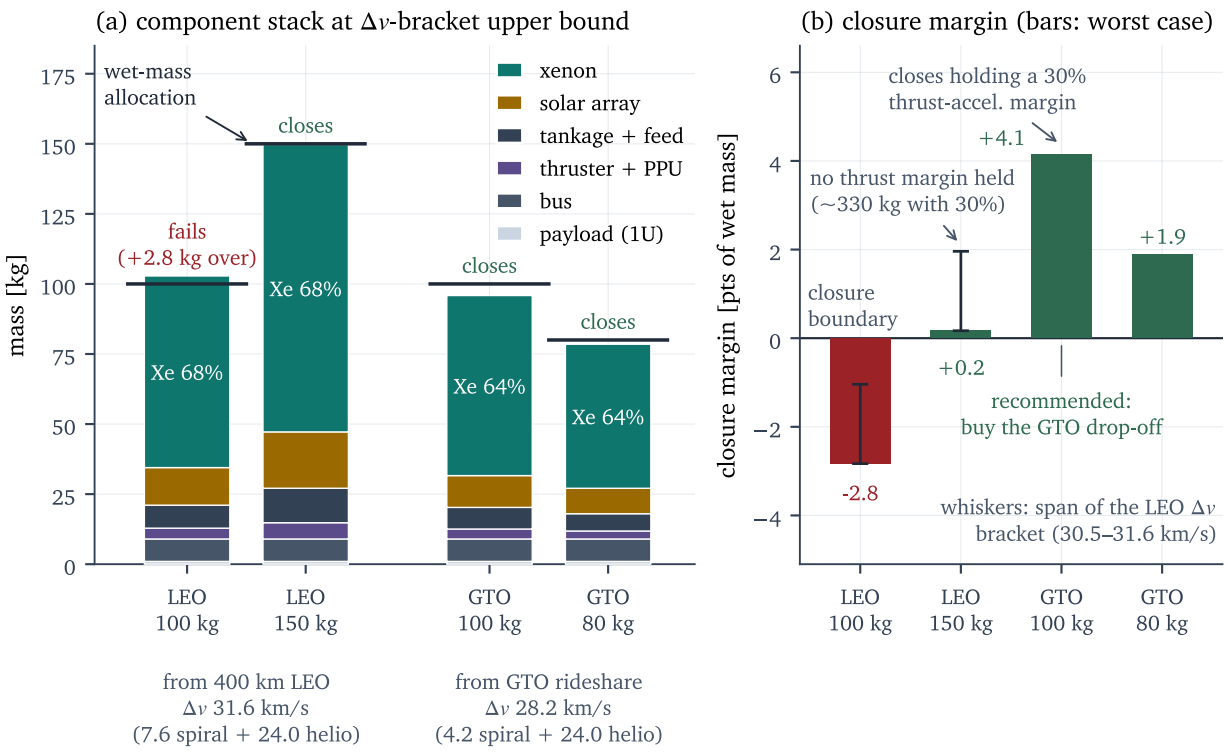


Figure 10. (a) Stacked component masses (payload, bus, thruster and power processing, tankage, solar array, xenon) at the conservative upper end of the heliocentric Δv reference interval; the horizontal cap on each bar marks the available wet mass. (b) Closure margin in percentage points of wet mass; whiskers span the reference Δv interval for the LEO cases. A 100 kg vehicle with a LEO start exceeds its cap everywhere in the interval; 150 kg closes, and both 100 kg and 80 kg close with a GTO start.

the pumping-feasibility thrust floor ($a_0 \gtrsim 2.25 \times 10^{-4}\,\mathrm{m/s^2}$). The rocket equation alone does not produce it: lowering I_{sp} raises the propellant load while the burn-time allowance holds the thrust acceleration, and with it the power chain, up. The LEO start requires a specific impulse of at least 2800 s. From GTO the requirement relaxes: a 2400 s thruster operated at the pumping-feasibility thrust floor closes 100 kg, with the closure boundary at 93.5 kg at the conservative end of the reference interval. The 2800 s figure is therefore a LEO-start requirement. It does not bind the mission as a whole. The distinction matters for component selection, because the high specific impulse pushes the operating point beyond the flown Hall-thruster envelope.

No flown Hall thruster covers the full operating point: $I_{sp} \geq 2800$ s at total efficiency $\eta \approx 0.55$ in a 0.5–1.0 kW string, with roughly 17,000–21,000 hours of cumulative single-string firing and 64–103 kg of xenon throughput. Flown Hall performance tops out near 1,650 s at this power class [13], more than 1,100 s short on specific impulse and roughly a factor of two short on demonstrated single-string hours. Demonstrated gridded-ion hardware brackets the point. The NSTAR engine's flight-documented TH3–TH4 throttle points (0.91–1.02 kW, 2,843–2,942 s, efficiency 0.527–0.554) [13] sit essentially on the required corner, and the same device demonstrated 30,352 hours and 235 kg of xenon throughput in single-string ground life testing, with 16,265 hours of single-unit flight operation [12]. Dual-string redundancy, standard practice for missions of this class, halves the per-string firing-time and throughput demands, and the demonstrated ground records already exceed the standard 1.5× throughput qualification margin applied to this mission's loads. We book one mass consequence explicitly: a gridded-ion string is heavier than the 6 kg/kW assumed for power processing in the mass model; at 10–15 kg/kW, a string adds 2–9 kg at 0.5–1.0 kW and shifts the closure boundaries by a few kilograms. The propulsion system is therefore a demonstrated component class (gridded ion) that requires a mission-specific qualification campaign.

Starting from a GTO rideshare changes the closure verdict. The departure Δv falls from 7.6 to $4.24\,\mathrm{km\,s^{-1}}$, and at the conservative (upper) end of the heliocentric reference interval an 80 kg vehicle closes with +1.9 points of margin, 100 kg with +4.1, and 150 kg with +7.1; margins are larger still at the lower end of the interval. The full closure grid appears in Appendix C.

5.3 Sensitivities

Thrust acceleration and array degradation. This is the sharpest risk in the design. The pumping strategy fails to reach the required cruise speed below $a_0 \approx 2.25\times10^{-4}\,\mathrm{m/s^2}$, so a 10–20% array degradation beyond the budgeted allowance erases the LEO-start closure margin entirely. Resizing the vehicle to carry a prudent 30% thrust-acceleration uplift (the burn-time-sized a_0 multiplied by 1.3, with thrust and the power chain rescaled accordingly) moves the closing boundaries sharply enough that the two ends of the reference interval must be quoted separately. From LEO, the minimum closing wet mass rises to 312 kg at the favorable end and 910 kg at the conservative end: three to nine times the concept. From GTO, the boundary rises to 83 kg at the favorable end and 104 kg at the conservative end: a 100 kg GTO vehicle carries the full margin at the favorable end but misses by approximately 4 kg at the conservative end, while a ~110 kg GTO vehicle carries the margin across the entire interval. The two architectures are therefore not close substitutes: the GTO start remains within a few kilograms of closure under a stress that drives the LEO start to several times its size, though at 100 kg it does not carry the full 30% margin everywhere in the interval.

Out-of-plane departure. The optimal departure asymptote is tilted 2.48° below the ecliptic. A planar treatment can only bound the cost of acquiring this tilt, between 22 m/s (if steered within the escape burns) and $1.02\,\mathrm{km\,s^{-1}}$ (worst case, executed as a separate plane change). A fully three-dimensional re-optimization of the pumping schedule measures it directly: the out-of-plane increment is $0.58\,\mathrm{km\,s^{-1}}$ (578 m/s), inside the planar band but far above its low end. The three-dimensional propagator behind that figure reproduces the planar timestep ladder bit-identically under zeroed steering, and an independent high-order re-integration of the winning trajectory reproduces its Δv to 0.29% (0.02% on the finest ladder rung) (Section E.3). Both the planar and three-dimensional figures are the best trajectories found by a heuristic search. Neither is a certified optimum, so their difference bounds the true three-dimensional cost in neither direction; it is the measured cost of the best schedule found. Propagating the increment through the closure algebra (linear interpolation of the closure scan, without a re-run of the mass model) sharpens the verdicts. The 100 kg GTO vehicle closes at both ends of the heliocentric Δv reference interval, with +5.1 points of margin at the favorable end and +3.1 at the conservative end, and it retains +2.3 points even at the band's $1.02\,\mathrm{km\,s^{-1}}$ worst case. The 150 kg LEO fallback closes at the favorable end (+1.0 points) but fails at the conservative end (−0.8 points). The bounding analysis flagged a fragility (the conservative-end LEO closure boundary moving from approximately 146 to 198 kg); the measurement now quantifies it. Any out-of-plane increment above roughly $0.10\,\mathrm{km\,s^{-1}}$ eliminates the 150 kg LEO margin at the conservative end of the interval, and the measured value is more than five times that threshold. Steering the plane change within the escape burns (the maneuver the LEO architectures relied on to stay near the 22 m/s low end) costs $0.58\,\mathrm{km\,s^{-1}}$ for the best trajectories found. On this axis the GTO start closes everywhere and the LEO fallback does not, and the measurement strengthens the recommendation of a GTO start (Section 5.4). Figure 11 summarizes the re-evaluated margins.

Thermal environment. The pumping schedule's perihelion of 0.42 AU places repeated passes inside the orbit of Mercury. The vehicle requires thermal qualification of the class demonstrated by the MESSENGER mission; this is established engineering. The cost analysis prices it as a qualification burden all the same. The pumping numbers also assume solar power harvested up to four times the 1 AU rating near perihelion (Section 4.6); with the cap halved to two, the re-optimized heliocentric requirement rises to $25.88\,\mathrm{km\,s^{-1}}$, and a deterministic re-run of the closure model at that requirement moves the GTO closure boundary to approximately 91 kg, so the 100 kg configuration still closes. The cap-reduced re-optimized schedules are, however, slower: the best schedules found take 15.1 years with the cap at two and 19.7 years at an intermediate cap of three, beyond the twelve-year operational horizon that anchors the closure claims elsewhere in this section.

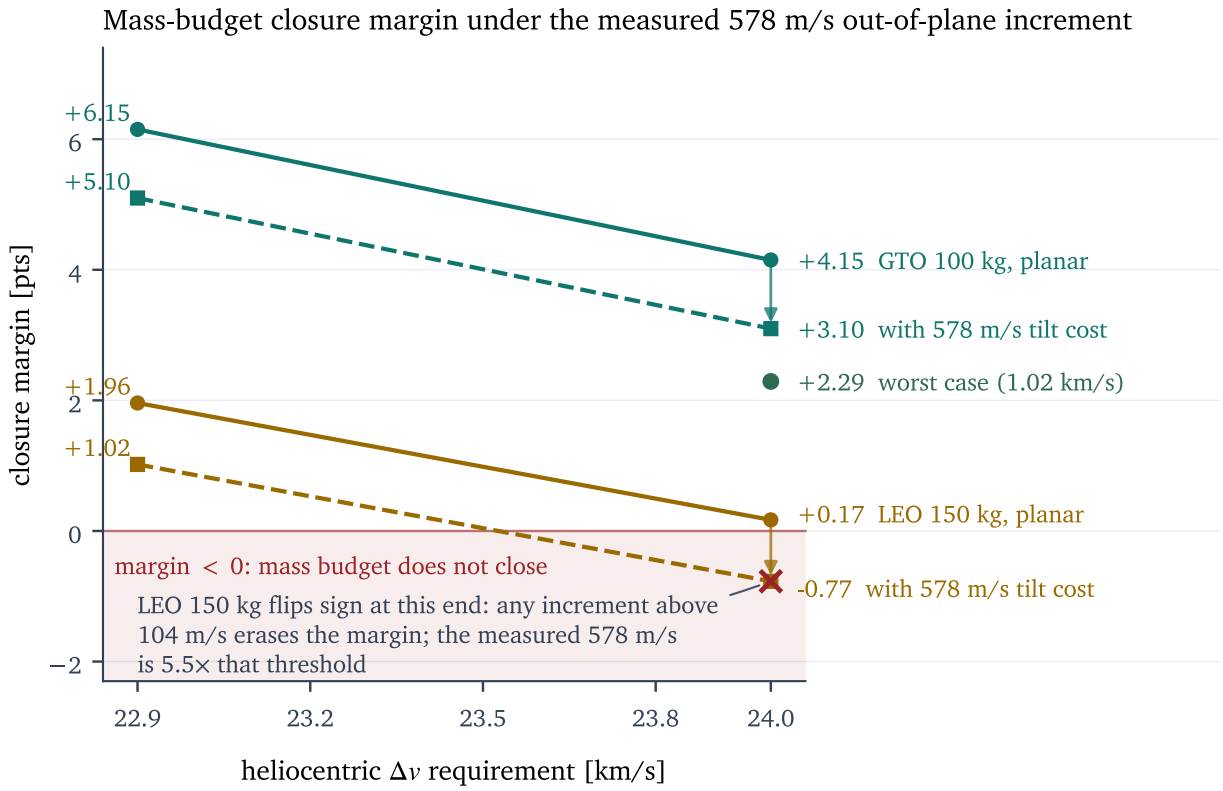


Figure 11. Closure margins re-evaluated with the measured $0.58\,\mathrm{km\,s^{-1}}$ out-of-plane increment, in percentage points of wet mass at both ends of the heliocentric Δv reference interval. The 100 kg vehicle with a GTO start closes at both ends (+5.1 and +3.1 points, and +2.3 even at the planar band's $1.02\,\mathrm{km\,s^{-1}}$ worst case); the 150 kg LEO fallback closes at the favorable end (+1.0) but fails at the conservative end (−0.8). Any increment above roughly $0.10\,\mathrm{km\,s^{-1}}$ erases the conservative-end margin of the 150 kg LEO vehicle; the measured value is more than five times that threshold.

5.4 Recommendation

The recommended change is to procure a GTO drop-off in place of the specified LEO rideshare. The case is fourfold. First, it preserves the concept: 100 kg closes with +4.1 points of margin at the conservative end of the Δv reference interval, while the LEO start misses everywhere in the interval on the trajectories found. Second, it holds under the stresses that break the LEO start: under a 30% thrust-acceleration uplift the GTO closure boundary stays within a few kilograms of the 100 kg concept (83–104 kg across the interval, versus 312–910 kg from LEO), and the GTO start absorbs the measured $0.58\,\mathrm{km\,s^{-1}}$ out-of-plane increment that defeats the 150 kg LEO fallback at the conservative end of the interval. Third, it nearly halves the departure budget (7.6 to $4.24\,\mathrm{km\,s^{-1}}$) and shortens the radiation-belt transit that drives array degradation. Fourth, GTO rideshares are a routine commercial service, so the deviation from the delivery specification is a matter of procurement. It requires no new technology. If the drop-off cannot be changed, the fallback is a 150 kg vehicle from LEO, which closes across the entire reference interval before the out-of-plane increment enters the accounting but is margin-fragile: it carries no allowance against array degradation and, with the measured $0.58\,\mathrm{km\,s^{-1}}$ out-of-plane increment charged, fails to close at the conservative end of the interval.

6. Targeting Confidence Under Execution and Ephemeris Dispersion

The aiming analysis of Section 2.7 established that the 2600 AU encounter sphere permits an open-loop asymptote pointing error of 0.41° (equivalent to approximately $168\,\mathrm{m\,s^{-1}}$ of departure velocity dispersion) and bud-

geted, term by term, a one-sigma cross-track target-ephemeris term of approximately $381\,\mathrm{AU}$ against it. With all error sources acting together, the vehicle passes inside the sphere with probability 0.982 under the stated error budget. The probability is measured: it is the passing fraction of a 20,000-draw seeded Monte Carlo simulation, run under assumed execution-error levels marked at their point of use below. A first-principles analytic error-propagation model reproduces it independently, agreement to 0.011%. It is not a flight-measured quantity; Section 6.4 quantifies how it degrades as the assumed levels grow.

6.1 Method

Each draw behind the probability above perturbs the departure state and the target's motion, then evaluates the encounter in closed form. The simulation propagates the target barycenter in uniform straight-line motion (valid here to well under the encounter-sphere scale, since the bounded Galactic-tide perturbation over the flight is about $1.2\,\mathrm{AU}$; Section 2.1), and the vehicle cruises on a straight line at its perturbed cruise velocity, so the closest approach to the moving target has an exact analytic solution per draw. The aim point is the target barycenter at the 73,012-year arrival epoch, computed from the adopted astrometry [1]; a zero-error draw reproduces that aim to below $10^{-6}\,\mathrm{AU}$ at the exact epoch, anchoring the pipeline before any error enters. At this geometry, one $\mathrm{m\,s^{-1}}$ of transverse velocity error maps to $15.4\,\mathrm{AU}$ of miss distance, so the $168\ \mathrm{m\,s^{-1}}$ allowance of Section 2.7 corresponds almost exactly to the full $2600\,\mathrm{AU}$ sphere. The two analyses are consistent by construction. The straight-line vehicle cruise meets the same standard: the Galactic-tide bound above applies to the vehicle's path as it does to the target's, and scattering by stellar encounters along the way is smaller still. Even a cumulative impulse of $1\ \mathrm{m\,s^{-1}}$, an order of magnitude beyond impulse-approximation estimates for the local stellar field, maps through the same gain to only tens of AU against the $2600\,\mathrm{AU}$ allowance.

The simulation draws four error sources independently, at the one-sigma levels of Table 7. The target-ephemeris draws perturb the parallax, both proper-motion components, and the radial velocity within the published uncertainties of the Kervella et al. astrometry [1], with the Akeson et al. solution [2] run as a self-consistent sensitivity variant (its own central values, uncertainties, and aim point). These draws use a diagonal covariance (parallax, the two proper-motion components, and the radial velocity vary independently), whereas the parameters of an orbit-corrected barycentric solution are in fact correlated; the direction of the resulting error in the success probability is not quantified here. The remaining three sources are execution errors, and their levels are assumed engineering inputs, marked as such here and in the table. The first is a $50\ \mathrm{m\,s^{-1}}$ knowledge error on the achieved departure speed. It represents the residual velocity-trim error after a telemetry-closed burn termination. It does not represent raw thruster bias: uncorrected thruster-level bias would be tens of times larger, so the assumption presumes a burn closed on tracking data. The second is a $0.10°$-per-axis pointing error on the departure asymptote, about one quarter of the total $0.41°$ open-loop allowance. The third is a $25\ \mathrm{m\,s^{-1}}$-per-axis isotropic residual representing launch-window discretization and the finite fidelity of the patched-conic departure mapping. Every run is seeded and bit-reproducible. Before the combined production run, we exercised each source alone and checked it against its closed-form transverse dispersion, which it reproduces at the percent level.

6.2 Results

With all four sources active, 19,631 of 20,000 draws pass within $2600\,\mathrm{AU}$ of the barycenter (Figure 12):

$$P(\text{closest approach} \le 2600\,\mathrm{AU}) \;=\; 0.982, \qquad \text{95\% confidence interval } [0.980,\ 0.983].$$

The median miss distance is $1076\,\mathrm{AU}$, 41% of the sphere radius; the 97.5th percentile is $2499\,\mathrm{AU}$, $101\,\mathrm{AU}$ under the requirement. The result is stable: recomputing the probability on the first 5,000 and first 10,000 draws gives 0.982 and 0.982 respectively, with the confidence-interval width halving from 5,000 to 20,000 draws exactly as the expected $1/\sqrt{n}$ scaling predicts. An independent first-principles analytic model propagates the same one-sigma levels through the closed-form encounter geometry to a two-dimensional transverse error covariance and integrates the resulting Gaussian over the $2600\,\mathrm{AU}$ disc by quadrature. It shares no code path or random draws with the simulation, and it too gives 0.982, agreement with the Monte Carlo result to 0.011%. Under the Akeson et al. variant astrometry, whose formal uncertainties are tighter, the probability rises to 0.992. The simulation also confirms the along-track argument of Section 2.7: radial errors shift the encounter epoch (by about ± 1250 years, one sigma, with all sources combined), but the pass still occurs, at negligible cost on the flat optimum of Section 2.4.

Table 7. Error sources and one-sigma levels for the dispersion Monte Carlo. The ephemeris levels are published catalog uncertainties; the three execution-error levels are engineering assumptions (plausible for a tracked, telemetry-closed departure; not flight-measured), and Section 6.4 quantifies how the verdict depends on them.

Source	One-sigma level	Basis
Target ephemeris	parallax ±0.61 mas; proper motion ±3.9 $\mathrm{mas\,yr^{-1}}$ per component; radial velocity ±0.30 $\mathrm{km\,s^{-1}}$	published uncertainties [1]; [2] as variant (±0.38 mas, ±1.5 $\mathrm{mas\,yr^{-1}}$, ±0.10 $\mathrm{km\,s^{-1}}$)
Thrust-magnitude knowledge	50 $\mathrm{m\,s^{-1}}$ on the achieved departure speed	engineering assumption (post-cutoff velocity-trim residual)
Asymptote pointing	0.10° per axis, two transverse axes	engineering assumption (attitude/thrust-vector alignment residual)
Post-departure residual velocity	25 $\mathrm{m\,s^{-1}}$ per axis, isotropic	engineering assumption (launch-window and departure-model residual)

6.3 Which Error Source Dominates

Rerunning the simulation with one source active at a time ranks the contributors by RMS transverse miss: thrust-magnitude knowledge (764 AU) leads, followed by asymptote pointing (647 AU), target ephemeris (617 AU), and the post-departure residual (547 AU) (Figure 13a). The four single-source mean-square misses sum to the full-simulation value to within 0.2%, confirming that the sources act nearly independently, and a total-effect variance decomposition preserves the same leader. No single source dominates outright (all four sit within a factor of 1.4), but the ordering carries a design lesson: the largest contributor is knowledge of the achieved departure speed. It is neither the star's ephemeris nor the pointing system. Improving thrust calibration and burn-termination velocity knowledge is therefore the highest-leverage mitigation available to the program, and it is a ground-and- telemetry investment. It requires no new flight hardware.

6.4 What the Verdict Is Conditioned On

Because three of the four error levels are engineering assumptions, the sensitivity table qualifies the headline number (Figure 13b). Doubling the assumed pointing error (to 0.20° per axis) drops the success probability from 0.982 to 0.897; doubling the thrust-magnitude knowledge error (to 100 $\mathrm{m\,s^{-1}}$) drops it to 0.858; quadrupling that error (to 200 $\mathrm{m\,s^{-1}}$) leaves 0.571, near a coin flip. Each of these excursions falls below a 0.9 design confidence threshold. The verdict therefore holds under the stated budget and fails under factor-of-two growth in the execution errors. The mission earns its 0.982 through pre-flight thrust calibration and a telemetry-closed departure. Geometric margin alone does not supply it.

6.5 A Catalog Systematic, Held Separate

The analysis deliberately excludes one effect from the probability above and reports it separately. The two published astrometric solutions for the system disagree on the barycenter's proper motion by 26.1 $\mathrm{mas\,yr^{-1}}$, formally about 6.7 times the quoted uncertainty of the adopted catalog, and on its parallax by 3.64 mas, about 6.0 times that uncertainty. The proper-motion disagreement alone would displace the cross-track aim by roughly 2550 AU at the encounter epoch; the parallax disagreement partially cancels it, and the net cross-track offset is roughly 860 AU (a first-order hand calculation gives 863 AU, agreement to 0.5%). That cancellation is a property of these two particular catalog solutions. The underlying physics does not enforce it: a re-reduction could move either term independently, so the net offset could grow as well as shrink, which strengthens the case for the observational mitigation below. This is a systematic error. Averaging the two catalogs would treat it as statistical and manufacture false confidence, so the analysis instead adopts one solution [1], runs the other [2] as a variant, and reports the offset explicitly. If the full offset were real and the adopted aim wrong, the success probability would fall to 0.962; the offset consumes only a third of the sphere, so the verdict degrades without breaking. The clean mitigation is observational: roughly seventeen years elapse between launch commitment and the final departure asymptote, and a dedicated astrometric re-reduction of the system

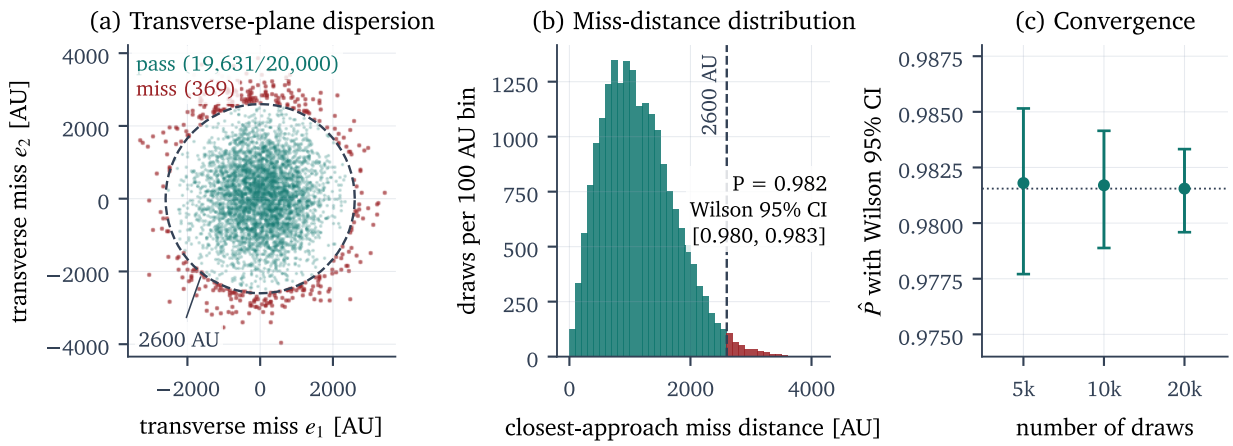


Figure 12. Encounter dispersion under the combined error budget, from the 20,000-draw seeded Monte Carlo. (a) Dispersion in the plane transverse to the approach velocity: 19,631 of 20,000 draws (teal) fall inside the 2600 AU requirement circle, 369 (red) outside. (b) Distribution of closest-approach miss distance against the requirement: the median miss is 1076 AU, the 97.5th percentile 2499 AU, and $P = 0.982$ of draws pass inside the sphere (95% confidence interval $[0.980, 0.983]$). (c) Convergence: the estimate is stable across the first 5,000, 10,000, and full 20,000 draws, with the confidence interval narrowing as $1/\sqrt{n}$. An independent analytic error-propagation model, sharing no code or random draws with the simulation, reproduces the success probability to 0.011%.

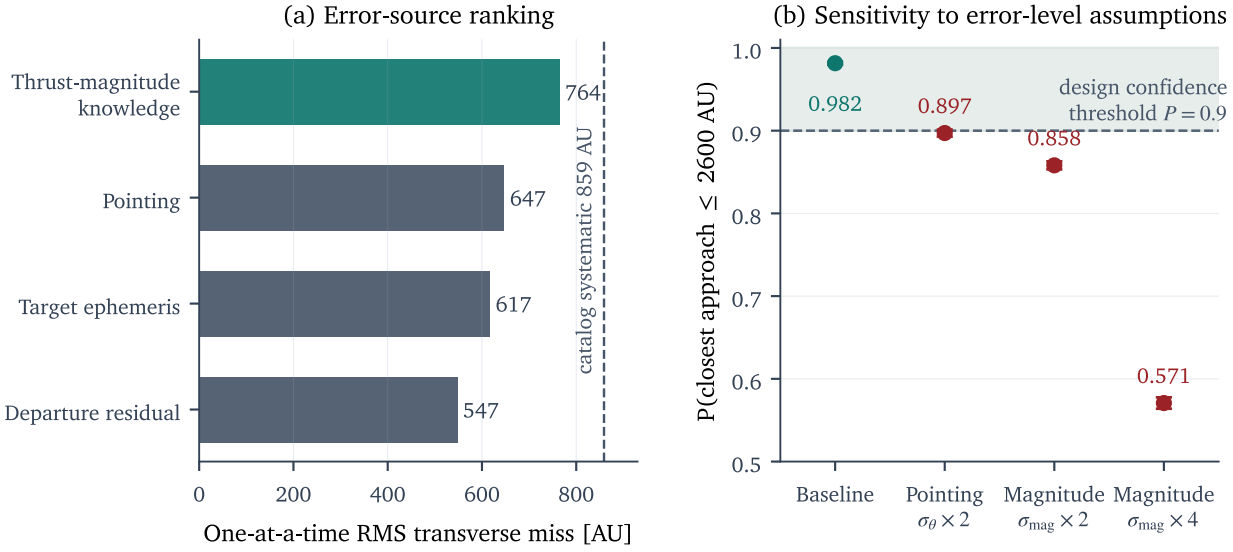


Figure 13. (a) Error-source ranking by one-at-a-time RMS transverse miss: thrust-magnitude knowledge (764 AU) leads pointing (647 AU), target ephemeris (617 AU), and the departure residual (547 AU), so thrust calibration is the highest-leverage mitigation. (b) Sensitivity of the success probability to the assumed error levels: the baseline 0.982 falls to 0.897 under a doubled pointing assumption, to 0.858 under a doubled thrust-magnitude assumption, and to 0.571 under a quadrupled one; each excursion falls below a 0.9 design confidence threshold. The verdict holds under the stated budget and fails under factor-of-two growth in either execution assumption.

over that interval would resolve the discrepancy before the program commits the aim. The analysis states that mitigation without modeling it. No number above takes credit for it.

7. Program Cost Assessment

The mission requirements set a total program cost target of $10M, allocated as $3M for launch, $3M for propulsion, and $3M for the spacecraft bus and mission operations. The three categories sum to $9M; this analysis treats the residual $1M as unallocated program reserve. Every architecture that closes the mass budget carries a measured median cost above the $10M target. The overrun from the initial $10M target cost sits entirely in the bus-and-operations category, and its origin is programmatic.

7.1 Cost Model

The cost verdict is structural. Launch and propulsion under-run their allocations by enough that a factor-of-two error in every hardware band would leave the operations-driven gap intact; the powered-flight operations line alone contributes a measured 64% of the total-cost variance (Section E.7).

A probabilistic parametric model with ten cost line items produces that verdict. Each line item carries a triangular uncertainty band (low, nominal, high) expressed in 2026 dollars per unit of its cost driver, anchored to public price lists and to published cost experience for small-spacecraft hardware, rideshare launch services, and deep-space operations. The bands are assumed inputs at scoping fidelity: the model is a class estimate. It is not a quotation. The drivers are physical quantities taken directly from the mass-closure model: launch cost scales with wet mass; the thruster and power-processing string scales with input power; the solar arrays scale with array wattage at 1 AU; propellant and tankage scale with the xenon load; and operations cost scales with the duration of powered flight. Architectures that employ perihelion pumping carry an additional thermal-qualification increment on the integration-and-test line, reflecting the high-temperature array and structure qualification required for repeated passes at 0.42 AU (Appendix D).

Monte Carlo sampling with 20,000 draws produces the total cost distribution. One modeling choice warrants an explicit caveat: the model draws the line items independently. Real programs exhibit positive correlation between line items (a schedule slip inflates operations and integration together), and positive correlation widens the total-cost distribution without changing its mean. The upper percentiles below are therefore mildly optimistic: correlated draws would raise the 90th percentile, and with it the risk of overrun. On the other side of the distribution, the $10M threshold lies deep in the left tail; widening would place slightly more probability mass below it. The independence assumption therefore understates the small probability of coming in under target. Neither effect changes the verdict: that probability is low either way. The program-margin line partially compensates by carrying program-level risk as its own band.

The physical inputs to the model rest on verified results. An independent algebraic check reproduced the mass-closure arithmetic that supplies the wet mass, propellant load, and power level; it located the closure boundary at 113.3 kg for a LEO start, within 0.2% of the design-sweep value (Section C.4). The underlying departure budget spans the 22.9–$24.0\,\mathrm{km\,s^{-1}}$ reference interval of Section 3.

7.2 Results: $16M-Class Medians

Figure 14(b) summarizes the total-cost distributions for the three architectures that close the mass budget. The LEO start at 150 kg wet mass carries a measured median (P50) cost of $16.6M, with a 10th-to-90th-percentile range of $14.2M to $19.4M. The GTO start at 100 kg (the recommended architecture) carries a P50 of $16.2M ($13.8M to $19.0M), and its 80 kg variant a P50 of $15.7M ($13.3M to $18.4M). Under conventional 2026 aerospace practice, the probability of completing any of these architectures under $10M is low.

Two features stand out. First, all three architectures land within about $1M of one another: total cost is nearly insensitive to the choice of architecture because the dominant cost lines are fixed or duration-driven. The mass-driven lines are minor. Second, the overrun does not fall where the allocation anticipated it.

7.3 Comparison with the Mission's Allocation

Figure 14(a) maps the line items onto the mission's three budget categories. The $3M/$3M/$3M split misses the actual cost structure in a favorable direction. Launch comes in at $1.0M to $1.3M and propulsion at $0.6M

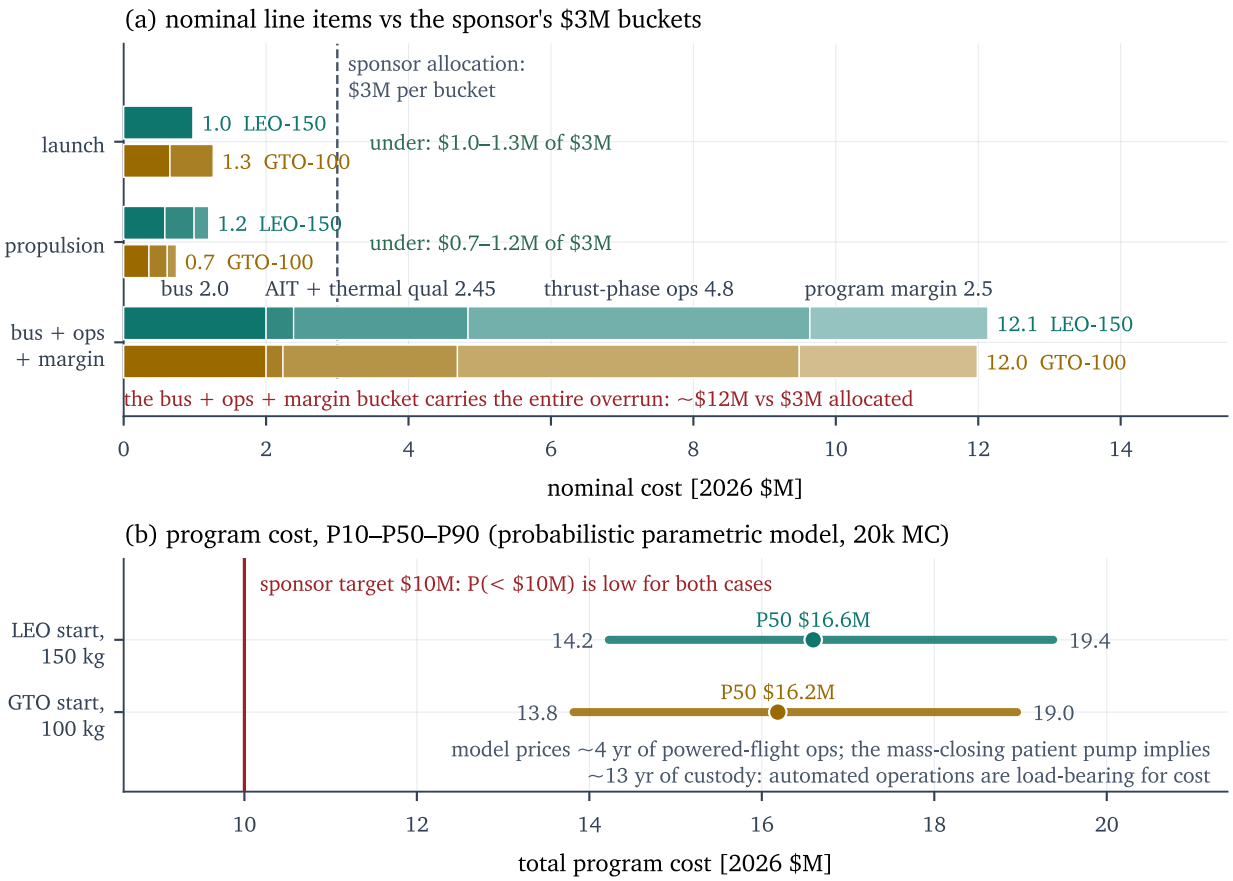


Figure 14. (a) Nominal cost by mission budget category for the two primary architectures, compared with the $3M/$3M/$3M allocation: launch and propulsion come in at roughly one third of their allocations, while the bus-and-operations category absorbs the entire overrun. (b) 10th–50th–90th-percentile total-cost ranges from the 20,000-draw Monte Carlo, compared with the $10M program target: no architecture's 10th percentile reaches the target.

to $1.2M across the closing architectures; each is roughly one third of its $3M allocation. The entire overrun from the initial $10M cost target sits in the bus-and-operations category, which also carries the program margin: approximately $12M nominal against the $3M allocated. The top cost drivers, in order, are powered-flight operations, program margin and management, and the spacecraft bus itself.

7.4 Operations Duration

In one respect the reported percentiles understate the true commitment. The model prices approximately four years of powered-flight operations. The patient pumping schedule that closes the mass budget implies roughly thirteen years of powered-flight custody: the Earth-escape spiral followed by the twelve-year pumping phase, most of it coasting. Under traditional staffed operations, the additional custody years would push the median cost higher still. Automation of the thrust phase is therefore load-bearing for program cost. It is more than a communications-link convenience. The mission concept states a tolerance for mostly-automated operations, including possible loss of contact; that tolerance is a cost-structural feature of the design.

7.5 The Path to $10M

The gap between the $16M-class medians and the $10M target is programmatic. No physical requirement drives it. The launch and propulsion categories sit at roughly one third of their $3M allocations; the roughly

$12M bus-and-operations figure prices 2026 aerospace norms: a staffed operations floor, a conventional small-sat bus, and a full environmental-qualification campaign augmented for the perihelion thermal environment. Meeting the $10M total requires compressing the conventional bus, integration-and-test, and operations practice by roughly 40 percent. That is aggressive but not implausible for a team with small-satellite heritage operating under an expendable, largely autonomous mission posture. The mission requirements describe precisely that posture. The discriminating cost issue for this program is therefore the credibility of a lean, automated operations concept. Physics sets no comparable barrier.

7.6 The $50M Tier

A higher program tier of $50M buys certainty. It buys no new physics. The most effective use of the additional budget is an escape-injection rideshare: purchasing a departure with near-zero Earth-relative hyperbolic excess from the launch provider eliminates the Earth-escape spiral entirely. The vehicle then closes at approximately 60 kg wet mass, and the purchase removes outright the three principal risk carriers of the baseline architecture: the radiation-belt transit, the array-degradation margin on the pumping phase, and the twelve-year patience requirement. The median cost of that architecture is approximately $15.6M; the purchase barely touches the $50M tier.

The additional budget does not usefully purchase alternative power or propulsion. Radioisotope generators cannot supply the kilowatt-class power the electric-propulsion system requires; a fission reactor exceeds even the $50M tier. A multi-stage chemical kick vehicle capable of the $13.85\,\mathrm{km\,s^{-1}}$ impulsive floor is both more expensive and less reliable than buying the equivalent injection energy from the launch vehicle.

Gravity assists warrant direct assessment at this tier because the $50M budget suspends the no-assist ground rule under which the rest of this report operates. At scoping fidelity, a Jupiter gravity assist supplies of order 10–15 $\mathrm{km\,s^{-1}}$ of cruise speed; the range is an assumed scoping figure. The flyby operates near Jupiter's $\sim$5.9 $\mathrm{km\,s^{-1}}$-class maximum turning benefit for slow arrivals, and the gain would substitute for most of the pumping phase. Three costs stand against it. First, the maneuver requires phase alignment of Jupiter with the departure asymptote; the alignment recurs on synodic windows, and a fixed end-2029 launch date leaves it to chance. Second, the routing adds roughly two to six years of inner- and outer-system transfer before the heliocentric cruise begins. Third, it adds deep-space navigation and operations effort of exactly the class the cost model above identifies as dominating program cost. The other assist-class option is the solar-Oberth route with a purchased kick stage examined in Fermi Explorer's own reference analysis [14]: a heat-shielded vehicle executing a chemical burn at a perihelion near ten solar radii (a thermal-protection development of the Parker Solar Probe class).

At $50M the escape-injection rideshare remains the recommended use of funds: it removes the identified risk carriers directly, with no launch-window dependence and no new development. A Jupiter assist is a launch-window-contingent opportunity worth assessing against the actual 2029 departure geometry. That window assessment is a follow-up item; this report does not perform it. The assist therefore stays off the baseline. The recommendation at the $50M tier is the same in kind as at the $10M tier: solar-electric propulsion with the escape spiral bought down or bought out, executed with margin instead of on the margin.

8. Alternative Targets

The mission requirements admit flexibility in the choice of stellar target, and the machinery developed for Alpha Centauri applies to any star with well-determined astrometry. Two of the three alternative targets offered by the Fermi Explorer team are viable, and one of them is energetically superior to the primary target. A sweep of the full close-encounter catalog (Section 8.4) found no better target; that verdict is best-found and bounds only the swept set. Every result in this section uses the method of the primary analysis: exact linear kinematics for the stellar motion, the same departure geometry from a 400 km low Earth orbit, and a miss tolerance set to 1% of each target's current distance. Astrometry for the alternative targets comes from Gaia Data Release 3 [3], except for Alpha-2 Librae, whose solution derives from the revised Hipparcos reduction [4]; the Alpha Centauri solution is that of Kervella et al. [1]. Every alternative-target result is verified: an independent implementation reproduced each from first principles, with a worst-case kinematic discrepancy below 0.5%.

Figure 15 presents the impulsive departure Δv as a function of arrival epoch for all four targets, and Table 8 summarizes the optima.

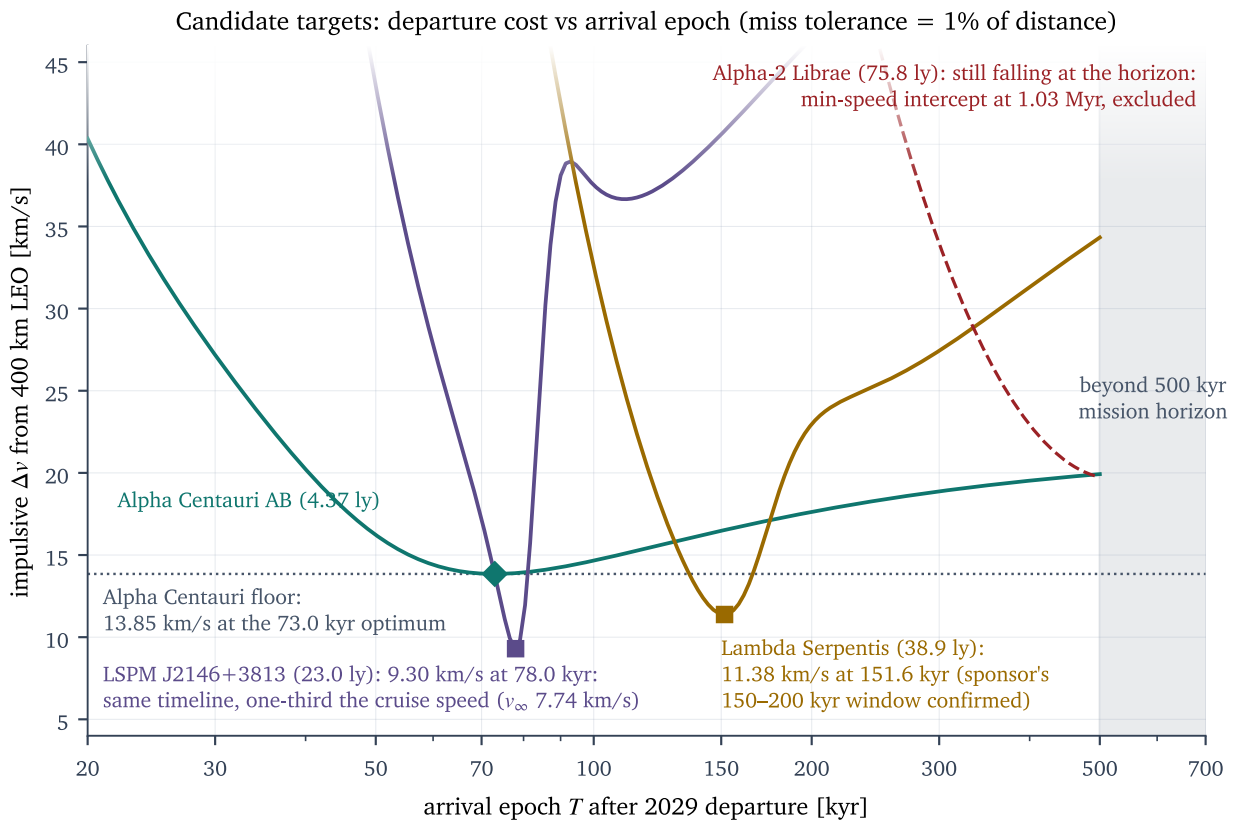


Figure 15. Minimum impulsive departure Δv from a 400 km low Earth orbit versus arrival epoch for the four candidate targets, with the minima of the three in-horizon targets marked. LSPM J2146+3813 (highlighted) achieves the lowest impulsive floor of any target examined, at essentially the same arrival epoch as Alpha Centauri; Alpha-2 Librae never approaches competitiveness inside the 500,000-year horizon.

8.1 LSPM J2146+3813: The Energetically Preferred Target

LSPM J2146+3813 is an M5V red dwarf at a present distance of 22.99 ly. Although it lies more than five times farther away than Alpha Centauri today, its space motion carries it toward the Sun: the published encounter analysis of Bailer-Jones [6], based on Gaia astrometry, gives a perihelion of 0.568 pc at +82.5 thousand years (one of the tightest known future stellar encounters). Our independent linear-kinematics computation reproduces that published perihelion to 0.2% (Section A.4).

The impulsive floor is $9.30\,\mathrm{km\,s^{-1}}$ at a 78,017-year arrival, with the departure asymptote tilted only $0.84°$ below the ecliptic. The required cruise speed is $7.74\,\mathrm{km\,s^{-1}}$: one third of the $23.64\,\mathrm{km\,s^{-1}}$ Alpha Centauri demands, on essentially the same arrival timeline (78,000 versus 73,000 years).

The reduction in cruise speed transforms the propulsion problem. The conventional outward solar-electric spiral, which saturates below the Alpha Centauri requirement at every thrust level examined (Section 4), comfortably exceeds the LSPM requirement: the spiral ceiling of $17.0\,\mathrm{km\,s^{-1}}$ at a thrust acceleration of $10{\times}10^{-4}\,\mathrm{m/s^2}$ is more than double the $7.74\,\mathrm{km\,s^{-1}}$ needed. The target requires no perihelion pumping, which removes the 0.42 AU thermal environment, the twelve-year pumping phase, and the thrust-acceleration margin sensitivity that dominate the risk posture of the Alpha Centauri design. A vehicle that closes its mass budget only narrowly against Alpha Centauri closes against this target with the spiral ceiling at more than twice the required cruise speed.

The choice before The Fermi Explorer Mission is therefore straightforward. If the objective is to reach another star's Oort cloud, LSPM J2146+3813 is the energetically preferred target: a lower impulsive floor, a far lower

Table 8. Alternative-target comparison at each target's departure-Δv optimum. All departures from a 400 km low Earth orbit, no gravity assists; miss tolerance 1% of current distance.

Target	Distance (ly)	Arrival epoch (kyr)	Impulsive floor ($\mathrm{km\,s^{-1}}$)	Published closest encounter
Alpha Centauri AB	4.37	73.0	13.85	3.15 ly at +28.0 kyr[a]
LSPM J2146+3813	22.99	78.0	9.30	0.568 pc at +82.5 kyr [6]
Lambda Serpentis	38.86	151.6	11.38	2.30 pc at +167 kyr [5]
Alpha-2 Librae	75.80	excluded[b]	19.74[b]	none

[a]Closest natural approach computed in this analysis (Section 2); the star itself passes no closer. [b]The minimum-cruise-speed intercept falls at 1.03 million years, outside the 500,000-year horizon; the quoted Δv is the best achievable within the horizon.

cruise speed, a benign trajectory with no perihelion passes, and the same arrival epoch. If the objective is Alpha Centauri specifically, the primary analysis of this report stands. The catalog sweep of Section 8.4 extends this conclusion from the offered candidates to the full encounter catalog.

8.2 Lambda Serpentis

Lambda Serpentis is a G0V star (a Sun-like star) at 38.86 ly. Its impulsive floor is $11.38\,\mathrm{km\,s^{-1}}$ at a 151,607-year arrival, which confirms the 150,000–200,000-year arrival window the Fermi Explorer team anticipated. The published encounter analysis gives a perihelion of 2.30 pc at +167 thousand years [5]; our computation agrees to 1.4% (Section A.4). Lambda Serpentis is a sound second alternative: cheaper to reach than Alpha Centauri, though at roughly double the flight time and without the exceptional encounter geometry of LSPM J2146+3813. Its solar spectral type may carry independent scientific weight in target selection.

8.3 Alpha-2 Librae: Excluded

Alpha-2 Librae, at 75.80 ly, should be removed from the candidate list. Its minimum-cruise-speed intercept falls at 1.03 million years, outside the 500,000-year mission horizon. Constrained to arrive within the horizon, it requires an impulsive departure of $19.74\,\mathrm{km\,s^{-1}}$: $5.9\,\mathrm{km\,s^{-1}}$ above Alpha Centauri's $13.85\,\mathrm{km\,s^{-1}}$ impulsive floor.

The weakest input in the target set does not overturn the exclusion. Alpha-2 Librae is a spectroscopic binary, and its systemic radial velocity is correspondingly uncertain: $-22.0 \pm 5.8\,\mathrm{km\,s^{-1}}$. Sweeping that velocity across its full $\pm 2\sigma$ range moves the minimum-speed intercept epoch; at no point within the range does the epoch enter the 500,000-year horizon. A revised systemic radial velocity outside that swept range would reopen the assessment.

8.4 Catalog Sweep: No Better Target Found

The Fermi Explorer team invited suggestions of better targets, and the candidates examined above are only the offered set. A sweep of all 61 entries of the Bailer-Jones close-encounter catalog [6] found no better target. The sweep scored every entry with the departure-budget metric of this section: minimum impulsive Δv from a 400 km low Earth orbit, arrival within the 500,000-year horizon, and the same miss rule (with the 2,600 AU allowance substituted where an encounter's own perihelion already satisfies the percentage rule). Of the 61 catalog rows, 29 are future encounters, 9 fall within the 500,000-year horizon, and 5 of those carry radial velocities free of the catalog's quality flags.

The sweep confirms LSPM J2146+3813 as the best high-confidence target. Only two catalog entries formally undercut its $9.30\,\mathrm{km\,s^{-1}}$ impulsive floor: the anonymous distant M dwarfs Gaia DR3 3984969656251772 16 (238 ly, $8.78\,\mathrm{km\,s^{-1}}$) and Gaia DR3 4155835025908320640 (311 ly, $8.82\,\mathrm{km\,s^{-1}}$). Their apparent advantage does not survive inspection. It rests on weak radial velocities, with standard errors of 8–11 $\mathrm{km\,s^{-1}}$ (as large as the floors they support), and on a near-degenerate encounter geometry: both stars pass close to the Sun themselves, so the departure budget collapses toward the metric's bare solar-escape floor while the mission value collapses with it. The result is a flyby at 150–280 $\mathrm{km\,s^{-1}}$ relative speed arriving at +330,000 to +470,000 years; LSPM J2146+3813 delivers an $83\,\mathrm{km\,s^{-1}}$ encounter at +78,000 years. That is strictly worse mission

value at nominally lower departure budget. An RV re-determination tightening the two Gaia dwarfs to about $1\,\mathrm{km\,s^{-1}}$ would reopen the comparison.

Two further results bound the search. No non-M-dwarf in the catalog approaches Lambda Serpentis's $11.38\,\mathrm{km\,s^{-1}}$: every unflagged future encounter inside the horizon is an M dwarf, so Lambda Serpentis stands as the best solar-type option available. Gliese 710, whose celebrated 0.06 pc perihelion is the tightest known future stellar encounter, lies at +1.29 million years (beyond the $500{,}000$-year horizon) and is therefore not a candidate under the mission constraints.

The sweep answers Fermi Explorer's invitation: it found no better target in the catalog. LSPM J2146+3813 is the energetically preferred target catalog-wide and within the offered set alike.

9. Conclusions and Recommendations

The analysis condenses to nine findings and six recommendations, followed by an overall assessment layered by certainty.

9.1 Findings

F1. Departure geometry and optimum. The minimum-Δv departure for Alpha Centauri falls at $T = 73{,}012$ years, requires a heliocentric cruise speed $v_\infty = 23.64\,\mathrm{km\,s^{-1}}$ aimed $2.48°$ below the ecliptic, and corresponds to an Earth-relative hyperbolic excess of $18.59\,\mathrm{km\,s^{-1}}$ and an impulsive Δv of $13.85\,\mathrm{km\,s^{-1}}$ from a 400 km low Earth orbit. A blind re-derivation of the arrival-epoch optimization, built without reference to the original solution, recovered the same optimum with agreement to 1 year and refined it to 73,012.17 years. The Δv curve is flat: within 100 m/s of the minimum over $66{,}908$–$80{,}436$ years, and still under $0.25\,\mathrm{km\,s^{-1}}$ above it out to $85{,}000$ years, leaving the arrival epoch free for launch-window phasing. The impulsive minimum is one of two distinct optima: the heliocentric cruise-speed minimum falls earlier, at $57{,}854$ years and $23.173\,\mathrm{km\,s^{-1}}$. An independent replication recomputed every number in this finding from first principles; the worst-case discrepancy is below 0.5%.

F2. The formal constraints are not binding. The $2{,}600$ AU miss tolerance is worth only about 83 m/s of Earth-relative hyperbolic excess (about 71 m/s of impulsive Δv); it relaxes targeting precision. It leaves the energy problem untouched. The $500{,}000$-year horizon is likewise not binding: the star recedes by then, and the requirement at the horizon *rises* to $v_\infty = 30.61\,\mathrm{km\,s^{-1}}$ ($\Delta v = 19.93\,\mathrm{km\,s^{-1}}$). The aiming tolerance is $0.41°$, equivalent to roughly 168 m/s of velocity dispersion. A 20,000-draw dispersion simulation with all execution and ephemeris error sources acting together confirms that this budget is achievable open-loop, with no terminal correction maneuver: the probability of passing inside the sphere is 0.982 (95% confidence interval $[0.980, 0.983]$), reproduced to 0.011% by an independent analytic error-propagation model (Section 6). The engineering error budget is an assumed input; thrust-magnitude knowledge at burn cutoff dominates it: doubling that assumption (50 m/s) drops the probability to 0.858, and doubling the assumed pointing error ($0.1°$ per axis) drops it to 0.897. Both fall below 0.9.

F3. A conventional solar-electric spiral cannot reach cruise speed. Because array power falls as the inverse square of solar distance (capped at four times the 1 AU rating inside roughly 0.5 AU by the array thermal limit), an outward low-thrust spiral saturates: the maximum achievable v_∞ is approximately 0, 3.4, and $17.0\,\mathrm{km\,s^{-1}}$ at thrust accelerations of 1.5, 5, and $10 \times 10^{-4}\,\mathrm{m/s^2}$ respectively; all fall below the $23.38\,\mathrm{km\,s^{-1}}$ cruise floor. Two independent trajectory integrators reproduce this saturation and agree to within 2.7%.

F4. Perihelion pumping defeats the power wall. Retrograde arcs near 1 AU shed angular momentum and lower perihelion to 0.42 AU; prograde arcs then concentrate near perihelion, where available power is one to four times the 1 AU rating. The optimized pumping schedule at $a_0 = 2.5 \times 10^{-4}\,\mathrm{m/s^2}$ delivers the heliocentric phase for $\Delta v = 23.98\,\mathrm{km\,s^{-1}}$ in 12.0 years of flight time (mostly coasting, with about 1.3 years of cumulative thrust) at a final mass fraction of 0.418. The total solar-electric budget from a LEO start is 30.5–$31.6\,\mathrm{km\,s^{-1}}$: a $7.6\,\mathrm{km\,s^{-1}}$ Earth-escape spiral plus a heliocentric phase quoted across the reference interval 22.9–$24.0\,\mathrm{km\,s^{-1}}$. The interval's upper end is the constructive schedule, a best-found upper bound; its lower end is a best-found optimal-control anchor carried down by monotonicity; the

certified lower bound over the full admissible class of throttleable schedules with unconstrained flight time is $16.56\,\mathrm{km\,s^{-1}}$ (Section 4.5 carries both pedigrees). The gap between that bound and the operational figure is schedule-shaped: with flight time unconstrained, an admissible trajectory reaches cruise speed for $18.87\,\mathrm{km\,s^{-1}}$ after about 58 years of powered flight (gate-passing schedules span 33–58 years at 18.9–$20.5\,\mathrm{km\,s^{-1}}$); within the 12-year operational horizon the best trajectory found is $23.98\,\mathrm{km\,s^{-1}}$. A fast 4.9-year variant costs $28.2\,\mathrm{km\,s^{-1}}$; the patience trade is worth about $4\,\mathrm{km\,s^{-1}}$, and only the patient schedule closes the mass budget. For a 2029 launch, escape occurs around 2043, with no effect on the arrival epoch.

F5. The required propulsion operating point lies within a demonstrated component class. No flown Hall thruster covers the required operating point: a specific impulse of at least $2800\,\mathrm{s}$ at total efficiency near 0.55 in a 0.5–$1.0\,\mathrm{kW}$ string, with roughly $17{,}000$–$21{,}000$ hours of single-string firing and 64–$103\,\mathrm{kg}$ of xenon throughput (flown Hall specific impulse tops out near $1{,}650\,\mathrm{s}$ at this power class [13]). Demonstrated gridded-ion hardware brackets it: NSTAR's flight-documented TH3–TH4 throttle points (0.91–$1.02\,\mathrm{kW}$, $2{,}843$–$2{,}942\,\mathrm{s}$, efficiency 0.527–0.554), with $30{,}352$ hours and $235\,\mathrm{kg}$ of single-string ground life and $16{,}265$ hours of single-unit flight [12]. Dual-string redundancy and the standard $1.5\times$ throughput qualification margin are the available mitigations. Closure therefore rests on a demonstrated component class (gridded ion) and requires a mission-specific qualification campaign. The $2800\,\mathrm{s}$ requirement is specific to the LEO start; from a GTO start a $2400\,\mathrm{s}$ thruster closes at the pumping-feasibility thrust floor (boundary near $93.5\,\mathrm{kg}$; Section 5).

F6. Mass closure is achievable; no trajectory found closes $100\,\mathrm{kg}$ from a LEO start. From a LEO start, the minimum closing wet mass is 113.5–$146\,\mathrm{kg}$ across the 22.9–$24.0\,\mathrm{km\,s^{-1}}$ reference interval; the $100\,\mathrm{kg}$ concept misses closure by 1.0–2.8% of wet mass for every trajectory found, while $150\,\mathrm{kg}$ closes throughout in the planar accounting. Closure of the concept cannot be excluded outright: it would follow from a heliocentric trajectory below the ~$22.3\,\mathrm{km\,s^{-1}}$ crossover within the 12-year operational horizon, and no trajectory found at the design thrust level comes within $0.6\,\mathrm{km\,s^{-1}}$ of that crossover. From a GTO start, where the departure budget falls from 7.6 to $4.24\,\mathrm{km\,s^{-1}}$, an $80\,\mathrm{kg}$ vehicle closes with $+1.9$ points of margin, $100\,\mathrm{kg}$ with $+4.1$, and $150\,\mathrm{kg}$ with $+7.1$. Under a 30% thrust-acceleration margin, the GTO start retains $100\,\mathrm{kg}$ closure only at the favorable end of the interval (boundary $83.4\,\mathrm{kg}$ at $22.9\,\mathrm{km\,s^{-1}}$ versus $103.7\,\mathrm{kg}$ at $24.0\,\mathrm{km\,s^{-1}}$; approximately $110\,\mathrm{kg}$ carries the margin throughout), while the same margin drives the LEO-start boundary to 312–$910\,\mathrm{kg}$.

The measured out-of-plane increment is $0.58\,\mathrm{km\,s^{-1}}$: the difference between a fully three-dimensional re-optimization of the pumping schedule and the planar total. It sits inside the planar analysis's $22\,\mathrm{m/s}$-to-$1.02\,\mathrm{km\,s^{-1}}$ band and far above the band's low end. Under that increment, the GTO start closes at both ends of the reference interval ($+5.1$ and $+3.1$ points of margin, and $+2.3$ points even at the band's worst case) while the $150\,\mathrm{kg}$ LEO fallback fails at the conservative end (-0.8 points). The LEO architectures rely on steering the plane change during powered flight to stay near the band's low end; for the best trajectories found that maneuver costs $0.58\,\mathrm{km\,s^{-1}}$, against the roughly $0.10\,\mathrm{km\,s^{-1}}$ that margin can absorb. The closing designs are 63–68% propellant, essentially all xenon: the concept's proportions are correct. An independent algebraic recomputation verified the closure arithmetic; it reproduced the closure boundary at $113.3\,\mathrm{kg}$, within 0.2%.

F7. The Fermi Explorer reference analysis is confirmed on geometry and amended on propulsion. Our independent computation agrees with the reference analysis [14] on the departure geometry to within 0.3% (their optimum: $13.88\,\mathrm{km\,s^{-1}}$ at 72,800 years, tilt $2.4°$). Their published figure of roughly $20\,\mathrm{km\,s^{-1}}$ for a realistic low-thrust departure is a benchmark assumption in their analysis; their own trajectory engine yields $25.1\,\mathrm{km\,s^{-1}}$ for the outward spiral. Multi-revolution perihelion pumping amends two of their conclusions, the verdict that pure solar-electric propulsion does not close at today's array specific power and the rejection of a single-pass electric perihelion burn, and their remaining conclusions stand (Section 4).

F8. The modeled median program cost is $15.7M–$16.6M under conventional aerospace practices, higher than the initial $10M cost target. The probability of meeting the initial $10M target is low under 2026 aerospace norms. No physical constraint sets the gap. The allocation weights the categories

differently from the modeled cost structure, in a favorable direction: launch ($1.0M–$1.3M) and propulsion ($0.6M–$1.2M) each come in at roughly one third of their $3M allocations, and the entire overrun sits in the bus, operations, and program-margin category (roughly $12M nominal against $3M allocated). The cost model prices about four years of powered-flight operations, whereas the mass-closing schedule implies roughly thirteen years of custody. Automation is therefore essential to the cost estimate itself, beyond its communications rationale. At a $50M tier, an escape-injection rideshare closes the vehicle at approximately $60\,\mathrm{kg}$ and removes the belt-transit, array-degradation, and patience risks outright, at a median cost near $15.6M; radioisotope generators cannot supply the required kilowatt-class power, and a fission reactor exceeds even that tier.

F9. An energetically superior target exists. LSPM J2146+3813, an M5V dwarf with a published closest approach of $0.568\,\mathrm{pc}$ at $+82.5$ thousand years [6], requires an impulsive floor of $9.30\,\mathrm{km\,s^{-1}}$ (versus 13.85) and a cruise speed of $7.74\,\mathrm{km\,s^{-1}}$ (versus 23.64) at an arrival epoch comparable to Alpha Centauri's. The target needs no perihelion pumping: the conventional spiral ceiling exceeds the requirement more than twofold. Lambda Serpentis is a sound second alternative ($11.38\,\mathrm{km\,s^{-1}}$ at $151{,}607$ years) and confirms the Fermi Explorer team's anticipated window. Alpha-2 Librae is excluded: its minimum-cruise-speed intercept stays outside the mission horizon across the full $\pm 2\sigma$ range of its systemic radial velocity. A sweep of the Bailer-Jones encounter catalog found no higher-confidence target inside the $500{,}000$-year horizon and no Sun-like challenger to Lambda Serpentis (Section 8).

9.2 Recommendations

R1. Procure a GTO drop-off with a 100–110 kg vehicle. This preserves the concept vehicle with $+4.1$ points of closure margin at the conservative end of the reference interval. It also survives the two stress cases that break the LEO start: a 30% thrust-acceleration margin holds at $100\,\mathrm{kg}$ at the favorable end of the interval and at approximately $110\,\mathrm{kg}$ throughout (against 312–910 kg from a LEO start), and under the measured $0.58\,\mathrm{km\,s^{-1}}$ out-of-plane increment the GTO start closes at both ends of the interval while the $150\,\mathrm{kg}$ LEO fallback fails at the conservative end. If the delivery orbit cannot be changed, grow the vehicle to $150\,\mathrm{kg}$. That variant closes from a LEO start across the reference interval in the planar accounting but is margin-fragile: it carries no allowance against array degradation and, once the measured $0.58\,\mathrm{km\,s^{-1}}$ out-of-plane increment is charged, fails at the conservative end of the interval even with the plane change steered during powered flight.

R2. Select a gridded-ion propulsion string and start its qualification campaign at once. For the LEO start, closure requires a specific impulse of at least $2800\,\mathrm{s}$; from the recommended GTO start, a $2400\,\mathrm{s}$ thruster closes at the pumping-feasibility thrust floor (boundary near $93.5\,\mathrm{kg}$), so the $2800\,\mathrm{s}$ requirement is specific to the LEO contingency. No flown Hall thruster covers the operating point (flown Hall tops out near $1{,}650\,\mathrm{s}$ at this power class); demonstrated gridded-ion hardware brackets it, at NSTAR's TH3–TH4 throttle points, with $30{,}352$ hours and $235\,\mathrm{kg}$ of single-string ground life. Thruster selection should therefore target the demonstrated gridded-ion class, with dual-string redundancy and the standard $1.5\times$ throughput qualification margin. Treat the mission-specific qualification campaign as the pacing schedule item for an end-2029 launch.

R3. Adopt an automation-first operations architecture. The mass-closing trajectory implies roughly thirteen years of powered-flight custody, while the cost estimate prices four; staffed operations over the full duration would push the median cost higher still. Autonomy during the escape spiral and pumping phase is essential to both the mass budget and the cost estimate. The mission's stated tolerance for mostly-automated operations, including possible loss of contact, should enter the design as a structural feature. Holding it in reserve forfeits the cost case.

R4. Scope thermal qualification to the perihelion environment early. The pumping schedule executes repeated passes at $0.42\,\mathrm{AU}$, inside the orbit of Mercury, and requires thermal qualification of the class demonstrated by the MESSENGER mission. This is established engineering but a real cost and schedule item, and it should enter the qualification program from the start; together with the propulsion qualification campaign (R2), it is one of the two items on which the end-2029 launch date is conditional.

R5. Weigh LSPM J2146+3813 as the energetically dominant alternative. If the mission objective is

a passage through another star's Oort cloud, this target offers a lower impulsive floor, one third the cruise speed, no perihelion passes, and the same arrival epoch; a vehicle that is marginal against Alpha Centauri closes against it, with the conventional spiral ceiling more than twice the requirement. If the objective is Alpha Centauri itself, the comparison is moot.

R6. Carry five items into the next analysis phase. First, thrust-magnitude calibration. The dispersion analysis identifies thrust-magnitude knowledge at burn cutoff as the dominant targeting error: at the assumed 50 m/s residual the probability of passing inside the encounter sphere is 0.982, doubling that assumption drops it to 0.858, and quadrupling it drops it to 0.571. Telemetry-closed thrust calibration and cutoff accuracy must therefore enter the explicit propulsion requirements and the qualification campaign (R2). Second, an ephemeris re-reduction. The two published Alpha Centauri astrometric solutions disagree by a systematic offset worth roughly 860 AU of cross-track aim at the encounter epoch. That offset is held separate from the headline probability; averaging it in would understate the systematic. Roughly seventeen years of pre-departure astrometry are available to resolve it before the asymptote is committed. Third, a detailed bus design: the bus and its operations concept carry the entire cost overrun, while launch and propulsion under-run their allocations. Fourth, a full schedule analysis: the end-2029 launch verdict is conditional on the propulsion and thermal qualification campaigns, and this report constructed no integrated schedule. Fifth, the early-arrival trade. A survey of arrival epochs from $55{,}000$ to $85{,}000$ years finds a feasible pumping schedule at every epoch examined, and the cruise-speed minimum at $57{,}854$ years points to arrivals some fifteen thousand years earlier than the departure optimum at little added cruise-speed cost. The departure asymptote, however, steepens to roughly 10° below the ecliptic at $58{,}000$ years, while this report measured the out-of-plane increment only at the optimum's 2.48° tilt ($0.58\,\mathrm{km\,s^{-1}}$); the increment grows steeply with angle. Pricing the tilt at early-arrival geometries is the natural next computation.

9.3 Overall Assessment

The assessment resolves into four layers, in decreasing order of certainty.

Trajectory-class feasibility: established. A solar-electric vehicle can reach the required cruise speed with solar power alone via multi-revolution perihelion pumping. This rests on a constructive trajectory, explicitly integrated and timestep-refined, that delivers $24.0\,\mathrm{km\,s^{-1}}$ at the design thrust acceleration; the verification pass reconstructed the mechanism independently. The physics that appeared to forbid the outcome, the inverse-square decay of solar power, constrains only outward spirals; it does not constrain trajectories that do their burning at or inside 1 AU.

Quantitative closure of the 100 kg concept from a LEO start: not achieved within the operational horizon. No trajectory found in this analysis closes 100 kg from the specified LEO start within the twelve-year powered-flight horizon that the component-life and burn-sizing assumptions price; the shortfall is 1.0–2.8% of wet mass across the 22.9–$24.0\,\mathrm{km\,s^{-1}}$ reference interval. Closure would require a heliocentric trajectory below the $\sim$22.3 $\mathrm{km\,s^{-1}}$ crossover, and the verdict is horizon-scoped. With flight time unconstrained, an admissible trajectory below the crossover exists: $18.87\,\mathrm{km\,s^{-1}}$ after about 58 years of powered flight, far beyond the twelve-year operational horizon. No certified lower bound above the crossover is therefore possible once flight time is freed, while the certified lower bound over the full class stands at $16.56\,\mathrm{km\,s^{-1}}$. Within the twelve-year horizon, every trajectory found sits above the crossover: the lower end of the reference interval, $22.9\,\mathrm{km\,s^{-1}}$, by $0.6\,\mathrm{km\,s^{-1}}$; the best trajectory found at the design thrust, $23.98\,\mathrm{km\,s^{-1}}$, by $1.7\,\mathrm{km\,s^{-1}}$; and the best-found trajectory at four times the design thrust, $22.66\,\mathrm{km\,s^{-1}}$, by $0.4\,\mathrm{km\,s^{-1}}$.

Specification compliance: conditional. The mission closes at 100 kg from a GTO start (a delivery-orbit deviation from the LEO ground rule) or at approximately 150 kg from a LEO start; the second option is margin-fragile, and the measured $0.58\,\mathrm{km\,s^{-1}}$ out-of-plane increment defeats it at the conservative end of the reference interval. The launch-before-end-2029 objective is schedule-conditional on the propulsion qualification campaign within the demonstrated gridded-ion component class and on MESSENGER-class thermal qualification; this report performed no full schedule analysis.

Cost: the modeled median program cost is $15.7M–$16.6M under conventional aerospace practices, higher than the initial $10M cost target. Launch and propulsion each come in under their allocations, and the entire gap concentrates in the bus, operations, and margin category. The gap is programmatic; no physical constraint sets it. The path to the target runs through lean automated operations and compressed integration and test, a

posture the mission's own ground rules invite.

The program can fly this trajectory class. The discriminating questions are whether it can qualify the propulsion string in time and whether it can build an operations concept lean enough to afford the years the flight takes.

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A. Astrometric Inputs and Intercept Kinematics

This appendix documents the astrometric inputs for all four candidate targets, the coordinate transformation applied to them, and the closed-form kinematic results used throughout Section 2. The closed forms serve two purposes: they make the geometry transparent, and they provide exact internal checks on the numerical sweeps. Independently written code rederived every quantity in this appendix from first principles; the worst disagreement across all kinematic quantities was below 0.5%.

A.1 Astrometric Inputs and Provenance

For Alpha Centauri the orbital motion of the A and B components contaminates single-star astrometric solutions, so we adopt the orbit-corrected barycentric solution of Kervella et al. [1], cross-checked against the independent orbital study of [2]: distance $4.365\,\mathrm{ly}$ and space velocity $32.38\ \mathrm{km\,s^{-1}}$, resolved into a radial component $v_r = -22.40\ \mathrm{km\,s^{-1}}$ (approaching) and a transverse component $v_t = 23.38\ \mathrm{km\,s^{-1}}$. The two single stars, LSPM J2146+3813 and Lambda Serpentis, use Gaia DR3 astrometry [3] directly. Alpha-2 Librae is too bright and too disturbed (it is itself a spectroscopic binary) for a clean Gaia solution; we use the Hipparcos re-reduction of van Leeuwen [4]. Its systemic radial velocity, $-22.0 \pm 5.8\ \mathrm{km\,s^{-1}}$, is by far the weakest input in Table 9; Section A.4 examines whether that uncertainty could alter the star's exclusion.

Table 9. Adopted astrometry for the four candidate targets. Section A.2 describes the transformation of distances and space motions to the heliocentric ecliptic frame. The published-encounter column lists the closest-approach solutions from the stellar-encounter literature, used as an external cross-check in Section A.4.

Target	Type	d (ly)	Astrometric source	Published encounter
Alpha Centauri AB	binary barycenter	4.365	Kervella et al. [1]	none
LSPM J2146+3813	M5V	22.99	Gaia DR3 [3]	0.568 pc at +82.5 kyr [6]
Lambda Serpentis	G0V	38.86	Gaia DR3 [3]	2.30 pc at +167 kyr [5]
Alpha-2 Librae	spectroscopic binary	75.80	Hipparcos [4]	none

A.2 Transformation to the Heliocentric Ecliptic Frame

All intercept computations use Cartesian coordinates in the heliocentric ecliptic frame, with x toward the vernal equinox and z toward the north ecliptic pole. Each catalog provides, in the ICRS equatorial frame, the right ascension α, declination δ, parallax ϖ, proper-motion components $\mu_{\alpha*} \equiv \mu_\alpha \cos\delta$ and μ_δ, and radial velocity v_r. The transformation proceeds in four steps.

1. *Distance.* $d = 1/\varpi$, with ϖ in arcseconds and d in parsecs. At the parallax precision of the sources in Table 9 the simple inversion is appropriate; no distance prior is needed.
2. *Position.* The unit line-of-sight vector in equatorial coordinates is $\hat{\mathrm{n}} = (\cos\delta\cos\alpha,\ \cos\delta\sin\alpha,\ \sin\delta)$, and the position vector is $\mathrm{r}_0 = d\,\hat{\mathrm{n}}$.
3. *Velocity.* With $\hat{\mathrm{e}}_\alpha$ and $\hat{\mathrm{e}}_\delta$ the local east and north unit vectors on the sky,

$$\mathrm{v}_\star \;=\; v_r\,\hat{\mathrm{n}} \;+\; \kappa\, d\,(\mu_{\alpha*}\,\hat{\mathrm{e}}_\alpha + \mu_\delta\,\hat{\mathrm{e}}_\delta)\,, \qquad \kappa = 4.740\ \mathrm{km\,s^{-1}}\ \text{per}\ (\mathrm{arcsec\,yr^{-1}\,pc}), \tag{7}$$

   where κ is the standard conversion between proper motion at distance d and transverse velocity.
4. *Rotation.* A rotation about the common x-axis through the obliquity of the ecliptic, $\varepsilon \simeq 23.44°$, carries both vectors from the equatorial to the ecliptic frame.

The model then propagates the star as $\mathrm{r}(t) = \mathrm{r}_0 + \mathrm{v}_\star t$. It neglects perspective acceleration and Galactic tidal acceleration. Section A.4 validates both omissions against the published encounter literature, and an internal check that integrates the target's motion through the Galactic potential leaves the intercept epoch essentially unchanged.

A.3 Closed-Form Intercept Kinematics

Lead-the-target relation and the minimum-speed identity. A vehicle cruising in a straight line at velocity V meets the star at time T if $\mathrm{V}T = \mathrm{r}_0 + \mathrm{v}_\star T$, i.e. $\mathrm{V}(T) = \mathrm{r}_0/T + \mathrm{v}_\star$. Writing $\mathrm{v}_\star = v_r\,\hat{\mathrm{n}} + v_t\,\hat{\mathrm{t}}$, with $\hat{\mathrm{t}}$ the unit vector transverse to the line of sight in the plane of the star's motion,

$$v_\infty^2(T) \;=\; \left(\frac{d}{T} + v_r\right)^2 + v_t^2. \tag{8}$$

For an approaching star ($v_r < 0$) the radial term vanishes at

$$T^* \;=\; \frac{d}{|v_r|}, \qquad v_\infty(T^*) \;=\; v_t, \tag{9}$$

so the minimum-speed intercept occurs when the star's own approach exactly closes the line-of-sight distance, and the minimum cruise speed equals the star's transverse velocity. The identity is exact under linear motion. It is not an approximation. For Alpha Centauri, Equation 9 with the adopted astrometry gives $T^* = 58{,}422$ years and $v_\infty = 23.38\ \mathrm{km\,s^{-1}}$, reproducing the numerical sweep exactly; the identity thus serves as a built-in consistency check on the sweep machinery.

Ecliptic-plane crossing. In ecliptic coordinates the star's out-of-plane position is $z(t) = z_0 + v_z t$, so the plane crossing (when it exists, i.e. when z_0 and v_z have opposite signs) occurs at

$$T_\times \;=\; -\frac{z_0}{v_z}. \tag{10}$$

For Alpha Centauri, which lies below the ecliptic and moves north, Equation 10 gives $T_\times = 79{,}786$ years, at which epoch the star is $6.425\,\mathrm{ly}$ from the Sun.

Closest natural approach. Minimizing $s(t) = |\mathrm{r}_0 + \mathrm{v}_\star t|$ gives the epoch and distance of the star's closest approach to the Sun:

$$t_{\mathrm{ca}} \;=\; -\frac{\mathrm{r}_0 \cdot \mathrm{v}_\star}{|\mathrm{v}_\star|^2} \;=\; \frac{d\,|v_r|}{|\mathrm{v}_\star|^2}, \qquad s_{\mathrm{ca}} \;=\; \frac{|\mathrm{r}_0 \times \mathrm{v}_\star|}{|\mathrm{v}_\star|} \;=\; \frac{d\,v_t}{|\mathrm{v}_\star|}. \tag{11}$$

For Alpha Centauri, $s_{\mathrm{ca}} = 4.365 \times 23.38/32.38 = 3.152\,\mathrm{ly}$ at $t_{\mathrm{ca}} = 27{,}955$ years.

Earth-relative hyperbolic excess. The model treats the departure as a patched conic. To leave the solar system with cruise speed v_∞, the heliocentric speed required at Earth's distance is

$$v_{\mathrm{dep}} \;=\; \sqrt{v_\infty^2 + v_{\mathrm{esc},\odot}^2}, \qquad v_{\mathrm{esc},\odot}(1\,\mathrm{AU}) = 42.12\ \mathrm{km\,s^{-1}}. \tag{12}$$

When the departure asymptote is tilted an angle β out of the ecliptic with its in-plane projection aligned with Earth's orbital velocity $v_\oplus = 29.78\ \mathrm{km\,s^{-1}}$ (the alignment is set by choosing the departure epoch within the year), the law of cosines gives the required Earth-relative hyperbolic excess:

$$v_{\infty,E}^2 \;=\; v_{\mathrm{dep}}^2 + v_\oplus^2 - 2\,v_{\mathrm{dep}}\,v_\oplus\cos\beta. \tag{13}$$

At the departure optimum ($v_\infty = 23.64\ \mathrm{km\,s^{-1}}$, $\beta = 2.48°$), Equation 12 gives $v_{\mathrm{dep}} = 48.30\ \mathrm{km\,s^{-1}}$ and Equation 13 gives $v_{\infty,E} = 18.59\ \mathrm{km\,s^{-1}}$, matching the values quoted in Section 2.4. One patched-conic approximation deserves note: Equation 13 assigns the full asymptote tilt β to the heliocentric velocity at 1 AU, whereas in-plane turning along the departure hyperbola allows the departure-velocity tilt to be smaller for the same asymptote aim, so the roughly $73\ \mathrm{m\,s^{-1}}$ that the tilt adds to $v_{\infty,E}$ is an estimate from the conservative side. The effect is below $0.1\ \mathrm{km\,s^{-1}}$ and sits within the replication tolerance quoted below.

Impulsive departure from low Earth orbit. A single tangential burn at perigee of a circular 400 km orbit, with circular speed $v_{\mathrm{circ}} = 7.67\ \mathrm{km\,s^{-1}}$ and local escape speed $v_{\mathrm{esc}} = 10.85\ \mathrm{km\,s^{-1}}$, captures the full Oberth benefit and requires

$$\Delta v = \sqrt{v_{\infty,E}^2 + v_{\mathrm{esc}}^2} - v_{\mathrm{circ}} = \sqrt{18.59^2 + 10.85^2} - 7.67 = 13.85\ \mathrm{km\,s^{-1}} \tag{14}$$

at the optimum. This is the impulsive floor of Table 4. Independent replication confirmed each quantity defining the optimum (T, v_∞, β, $v_{\infty,E}$, and Δv) to better than 0.5%. The optimum agrees with the Fermi Explorer reference analysis [14] ($13.88\ \mathrm{km\,s^{-1}}$ at 72,800 years, 2.4° tilt) to within 0.3%.

Miss tolerance and aiming dispersion. Relaxing the aim condition from true intercept to a pass within $2600\ \mathrm{AU}$ of the barycenter reduces the required Earth-relative hyperbolic excess by approximately $83\ \mathrm{m\,s^{-1}}$; after dilution by the escape-energy term in Equation 14 this is worth only about $71\ \mathrm{m\,s^{-1}}$ of impulsive Δv. The same sphere admits an asymptote pointing error of 0.41°, equivalent to roughly $168\ \mathrm{m\,s^{-1}}$ of departure velocity dispersion; this tolerance is the basis for the open-loop guidance conclusion of Section 2.7.

A.4 Consistency with Published Encounter Catalogs

The stellar-encounter literature provides an external check on the linear-motion model: it derives closest-approach solutions by integrating stellar orbits through a Galactic potential with full propagation of astrometric uncertainty. Two of our targets have published solutions. For LSPM J2146+3813, Bailer-Jones [6] finds a perihelion of $0.568\,\mathrm{pc}$ at $+82.5\,\mathrm{kyr}$, one of the tightest known future stellar encounters; our linear-motion perihelion agrees to 0.2%. For Lambda Serpentis, the published solution [5] gives $2.30\,\mathrm{pc}$ at $+167\,\mathrm{kyr}$; ours agrees to 1.4%. Agreement at the percent level with orbit-integrated solutions confirms that straight-line kinematics is adequate at these ranges over 10^5-year horizons, and validates the neglect of perspective and Galactic accelerations in Section A.2.

For Alpha-2 Librae no comparably tight published solution exists, and its systemic radial velocity is uncertain at the $\pm 5.8\ \mathrm{km\,s^{-1}}$ level. The exclusion of this star from the target set survives that uncertainty: across the $\pm 2\sigma$ range of the radial velocity, the minimum-speed intercept epoch of Equation 9 stays outside the 500,000-year mission horizon (the nominal value is 1.03 Myr), and forcing an arrival inside the horizon requires an impulsive departure of $19.74\ \mathrm{km\,s^{-1}}$, worse than Alpha Centauri itself. The exclusion therefore does not depend on the quality of the weakest astrometric input.

The two published-encounter cross-checks above concern the targets carried forward into the analysis. Section E.7 documents the systematic sweep of the full 61-row encounter catalog that produced the candidate list, including its filters, exclusions, and ranking metric.

B. Trajectory Models and Verification

This appendix documents the trajectory models behind the departure and heliocentric results of Section 3 and Section 4. It covers the control policy and optimization of the perihelion-pumping escape; the bounding analysis for the heliocentric Δv requirement, comprising a certified lower bound, constructive upper bounds at two flight-time horizons, and a best-found reference anchor; and the independent verification applied to each headline number.

B.1 Equations of Motion and Thrust Law

The model treats the heliocentric phase as planar two-body motion about a point-mass Sun with a power-throttled, steerable thrust:

$$\ddot{\mathrm{r}} = -\mu_\odot \frac{\mathrm{r}}{r^3} + \frac{F(r)}{m}\,\hat{\mathrm{u}}, \tag{15}$$

$$F(r) = F_1 \,\min\left[\left(\frac{r_1}{r}\right)^2, 4\right], \tag{16}$$

$$\dot{m} = -\frac{F(r)}{g_0\, I_{\mathrm{sp}}}, \tag{17}$$

where r is the heliocentric position with magnitude r, $\mu_\odot$ is the solar gravitational parameter, m is the vehicle mass, F_1 is the thrust at $r_1 = 1$ AU, g_0 is standard gravity, and I_{sp} is the specific impulse. The control policy described below commands the thrust direction $\hat{\mathrm{u}}$ to one of three states along the velocity vector: prograde, retrograde, or off (coast). This three-state law at full envelope thrust is the policy class of the constructive schedule; the bounding analysis later in this appendix embeds it in a broader throttleable class and defines the heliocentric minimum over that class. The factor-of-four cap in Equation 16 models the thermal power cap on the array; it engages inside roughly 0.5 AU. The characteristic parameter is the initial 1-AU thrust acceleration $a_0 \equiv F_1/m_0$, with m_0 the initial mass; because thrust at fixed power is independent of mass, the instantaneous acceleration grows as m_0/m while propellant depletes, and the integrated results include this effect.

B.2 Earth-Escape Spiral

The model does not integrate the geocentric escape phase revolution by revolution. It budgets the phase by the standard low-thrust result that continuous-thrust escape from a circular orbit costs approximately the circular orbital speed of the starting orbit [8]. From a 400 km orbit ($v_{\mathrm{circ}} = 7.67\ \mathrm{km\,s^{-1}}$) the model carries a departure budget of 7.6 $\mathrm{km\,s^{-1}}$, expended over approximately 1.5 years. The spiral transits the radiation belts, and the array sizing applies a degradation factor of 1.25 for this phase. A vehicle escaping by slow spiral leaves Earth's sphere of influence with negligible Earth-relative hyperbolic excess, so the vehicle then builds the full heliocentric requirement under Equation 15–Equation 17.

B.3 Perihelion-Pumping Control Policy and Optimization

A four-parameter feedback policy operating on the osculating orbit generates the pumping escape:

(i) the target perihelion radius, to which retrograde thrusting drives the orbit (the optimizer selects approximately 0.42 AU, just inside the radius of the thermal power cap);

(ii) the fraction of each revolution, centered on aphelion, over which retrograde thrust lowers the perihelion;

(iii) the heliocentric radius bounding the prograde burn window, so that accelerating arcs fire only near perihelion where power is at or above the 1-AU rating;

(iv) the specific-orbital-energy threshold at which pumping terminates and the vehicle thrusts continuously to escape.

We optimized the four parameters numerically, with a coarse parameter sweep followed by local refinement, taking as objective the total Δv to reach the required cruise speed of 23.64 $\mathrm{km\,s^{-1}}$ at $a_0 = 2.5 \times 10^{-4}\ \mathrm{m\,s^{-2}}$. Concretely, the search drew 400 seeded uniform-random candidates over the parameter box and polished

the five best by Nelder–Mead simplex descent; an independently seeded re-optimization with a larger budget reproduced the same schedule and carried 23.98 $\mathrm{km\,s^{-1}}$ from its most conservative timestep rung. A fixed-step kick-drift-kick scheme integrates the trajectories with the thrust policy frozen across each step, a step law of $\max(600\ \mathrm{s},\ 0.002 \times \text{local circular period})$ scaled by the refinement factor, and a rocket-equation mass update on thrusting steps; on every accepted schedule the integrated Δv matches the rocket-equation value implied by the final mass to $10^{-13}\ \mathrm{km\,s^{-1}}$. Seeds, acceptance gates, and both optimization records appear in Section E.2. The best schedule (Figure 6) delivers the cruise speed at a Δv of 23.97 $\mathrm{km\,s^{-1}}$ over 12.0 years of flight time, of which only about 1.3 years is cumulative thrusting (0.9 years retrograde, 0.4 prograde), with a minimum solar distance of 0.42 AU. The final mass fraction of 0.418 agrees with the rocket equation evaluated at $I_{\mathrm{sp}} = 2800$ s, an internal consistency check on the mass integration. Timestep refinement established convergence, with one caveat to disclose: at one-half the production step the integration initially terminated at the simulation-horizon cap short of cruise speed. With the horizon extended, the half-step run reaches cruise speed at a Δv of 23.969 $\mathrm{km\,s^{-1}}$ and the quarter-step run at 23.985 $\mathrm{km\,s^{-1}}$: agreement within 0.07%, with the residual refinement drift covered by carrying the upper bound at 24.0 $\mathrm{km\,s^{-1}}$.

B.4 Bounding the Heliocentric Requirement

Let $\Delta v^*(a_0)$ denote the minimum heliocentric Δv that reaches the required cruise speed (heliocentric hyperbolic excess of 23.64 $\mathrm{km\,s^{-1}}$) from the post-escape state at 1 AU under Equation 15–Equation 17 with initial 1-AU thrust acceleration a_0. The minimum is taken over the throttleable admissible class: piecewise-continuous thrust programs with free in-plane direction, magnitude commandable anywhere in the power envelope, $0 \le F \le F_1 \min[(r_1/r)^2,\ 4]$, and heliocentric distance $r(t) \ge 0.42$ AU, the minimum solar distance carried by the thermal design, which the constructive schedule respects ($r_{\min} = 0.423$ AU). Throttleability is physically justified: a flight electric-propulsion string throttles continuously through its power processing unit. The three-state bang-off policy class of the preceding subsection is a subset of this class, so every constructive result above is admissible a fortiori. The class definition places no constraint on flight time; where a result below instead refers to the 12-year operational horizon carried by the mission design, that restriction is stated explicitly. The report carries four quantities for $\Delta v^*(2.5 \times 10^{-4}\ \mathrm{m\,s^{-2}})$, of different evidentiary standing: a certified lower bound, an impulsive floor retained as a consistency reference, constructive upper bounds at two flight-time horizons, and a best-found reference anchor.

Monotonicity on the throttleable class. Δv^* is non-increasing in a_0. Under a larger acceleration bound $a_0' > a_0$, any admissible throttleable program at bound a_0 remains admissible: its commanded thrust magnitude at every instant lies inside the enlarged envelope and is realized by scaling the throttle setting by a_0/a_0' relative to full thrust. The trajectory, the mass history, and the expended Δv are unchanged, so the feasible set only grows with a_0 and the minimum cannot increase. The argument requires throttleability: a bang-off class restricted to full envelope thrust does not contain the rescaled program. That requirement is why the definition above names the admissible class explicitly.

Certified lower bound. Every trajectory in the admissible class expends a heliocentric Δv of at least 16.56 $\mathrm{km\,s^{-1}}$:

$$\Delta v^*(2.5 \times 10^{-4}\ \mathrm{m\,s^{-2}}) \ \ge\ 16.56\ \mathrm{km\,s^{-1}}. \tag{18}$$

This is a proof-grade bound on the true class optimum. It is not the cost of any particular trajectory. A six-step bounding argument, in which every step is a closed-form inequality, establishes it: (i) the trajectory is truncated at first attainment of the target energy, and propellant accounting converts the mass history into an explicit ceiling on achievable thrust acceleration as a function of the total Δv expended; (ii) the final escape leg makes exactly one transit of the region near the perihelion floor, because the available thrust falls an order of magnitude short of what lingering there would require; (iii) the duration of that single transit is bounded by a comparison argument, capping both the Δv deliverable and the orbital energy gained inside the transit window; (iv) every remaining increment of orbital energy must then be purchased at a price no better than the reciprocal of the fastest admissible speed near the perihelion floor, and integrating that price over the required energy gain bounds the cost of the escape leg; (v) the initial segment, ending when the orbit first reaches escape energy, is bounded below by $(\sqrt{2}-1)$ times Earth's orbital speed, about 12.3 $\mathrm{km\,s^{-1}}$, by a separate self-contained argument; (vi) the pieces close through a self-consistency (fixed-point) check on the assumed total: if the total Δv were below 16.56 $\mathrm{km\,s^{-1}}$, the thrust ceiling of step (i) would be tight enough that steps (ii)–(v) force a total of at least 16.5659 $\mathrm{km\,s^{-1}}$, a contradiction. Because the admissible class carries no flight-time constraint, the

bound holds a fortiori for every time-restricted subclass, including the 12-year operational horizon; like the class itself, it is a statement about the planar model of Equation 15. An adversarial audit checked the argument step by step; a numerical harness additionally tested every inequality in the chain against three independently integrated trajectories (21 falsification checks, zero violations), and a separate recomputation of the assembled constants from first principles matched the derivation to better than 10^{-5} relative. Section E.5 documents the complete chain and the numerical harness applied to it.

Impulsive floor (consistency reference). Removing the thrust-magnitude bound altogether ($a_0 \to \infty$) only enlarges the feasible set, so the impulsive multi-burn minimum over the same class also bounds Δv^* from below; the $r \geq 0.42$ AU constraint of that class is what makes the floor nontrivial (without any perihelion bound the impulsive floor decays to the 1-AU escape cost, about 12.3 $\mathrm{km\,s^{-1}}$). That minimum is the bi-parabolic transfer: an escape burn at 1 AU costing $v_{\mathrm{esc}} - v_{\mathrm{circ}} \approx 12.3\ \mathrm{km\,s^{-1}}$, a vanishing redirection burn at great distance, and a periapsis burn at 0.42 AU costing $\sqrt{v_\infty^2 + v_{\mathrm{esc}}^2(r_p)} - v_{\mathrm{esc}}(r_p) \approx 4.2\ \mathrm{km\,s^{-1}}$ at $v_\infty = 23.64\ \mathrm{km\,s^{-1}}$. The floor is 16.5 $\mathrm{km\,s^{-1}}$ (15.6 $\mathrm{km\,s^{-1}}$ if the perihelion bound relaxes to 0.25 AU). The certified lower bound of Equation 18 exceeds this floor by 0.06 $\mathrm{km\,s^{-1}}$, as any sound bound must.

Constructive upper bounds at two horizons. Two explicitly integrated admissible trajectories bracket the optimum from above, at different flight-time horizons. At the 12-year operational horizon (the component-life and burn-sizing basis of the mission design) the optimized pumping schedule of the preceding subsection reaches the cruise speed at 23.97 $\mathrm{km\,s^{-1}}$, carried as 24.0 $\mathrm{km\,s^{-1}}$; an independent re-optimization at the same horizon reproduced 23.98 $\mathrm{km\,s^{-1}}$. With the flight-time restriction removed, an admissible schedule 5.1 $\mathrm{km\,s^{-1}}$ cheaper exists: a long-horizon trajectory of the same pumping family, integrated at four timestep resolutions and quoted at the most conservative value across them, reaches the same cruise speed at 18.87 $\mathrm{km\,s^{-1}}$ with about 58 years of flight time (the family of such schedules found spans 33–58-year flight times). Both figures are the best schedules found. Neither is a proven optimum. Combined with Equation 18, the unconstrained-flight-time optimum is bracketed as

$$16.56\ \mathrm{km\,s^{-1}} \ \leq\ \Delta v^*(2.5 \times 10^{-4}) \ \leq\ 18.87\ \mathrm{km\,s^{-1}}, \tag{19}$$

while the 12-year-constrained optimum is known only to lie between 16.56 and 24.0 $\mathrm{km\,s^{-1}}$: the certified lower bound does not exploit the time limit, and how much of the 18.87-to-24.0 gap is recoverable within 12 years is open.

Best-found reference anchor. At a higher thrust acceleration the optimal-control problem is numerically tractable, and we solved the following problem as a reference point: planar two-body dynamics per Equation 15 with the power-throttled thrust envelope per Equation 16, initial 1-AU thrust acceleration $a_0' = 1 \times 10^{-3}\ \mathrm{m\,s^{-2}}$ (four times the design value), departure from the post-escape heliocentric state at 1 AU with negligible Earth-relative hyperbolic excess, terminal condition equal to the required cruise speed, flight time fixed at 0.82 years (matching the elapsed time of a pumping heuristic at the same target; with flight time left free this problem degenerates toward the impulsive limit), and expended Δv as the objective. An indirect method based on Pontryagin's minimum principle (costate integration with shooting on the boundary conditions) converges cleanly to an extremal expending 22.66 $\mathrm{km\,s^{-1}}$ at the 23.64 $\mathrm{km\,s^{-1}}$ cruise speed; the same solver bit-for-bit reproduces the previously computed 22.94 $\mathrm{km\,s^{-1}}$ extremal at a 24.14 $\mathrm{km\,s^{-1}}$ target variant. The solve imposes no perihelion-floor path constraint, but the extremal never descends below 0.62 AU, so the 0.42 AU floor is inactive. We attempted a direct-collocation cross-check of this extremal; it did not converge. Attempts at 24, 48, and 60 mesh segments either stalled at their iteration limits near the initializing heuristic, some 22% above the indirect value, or failed re-integration checks. The 22.66 $\mathrm{km\,s^{-1}}$ value therefore carries single-method provenance and the label best-found: an extremal satisfying first-order optimality conditions whose global optimality is unproven. It serves as an upper reference on the optimum at the higher thrust level. It bounds nothing at the design thrust level. No recommendation rests on it; it enters only as the favorable end of the reference interval, and discarding it moves every verdict toward the 24.0 $\mathrm{km\,s^{-1}}$ end (verified by independent re-integration to 0.29%), where the recommended architecture still closes.

What the report's conclusions rest on. The certified statement is the lower bound of Equation 18, paired with the demonstrated upper bounds of 18.87 $\mathrm{km\,s^{-1}}$ (unconstrained flight time) and 24.0 $\mathrm{km\,s^{-1}}$ (12-year horizon). By the closure algebra of Appendix C, a 100 kg vehicle on the LEO start closes if a heliocentric trajectory below the ≈22.3 $\mathrm{km\,s^{-1}}$ crossover (22.26 $\mathrm{km\,s^{-1}}$) exists. Unconstrained-flight-time solutions cross that threshold: the demonstrated 18.87 $\mathrm{km\,s^{-1}}$ schedule sits 3.4 $\mathrm{km\,s^{-1}}$ below it, at the price of 33–58 years of

flight time. No known 12-year solution crosses it: the best 12-year schedule sits 1.7 $\mathrm{km\,s^{-1}}$ above the crossover, and nothing between 16.56 and 24.0 $\mathrm{km\,s^{-1}}$ is excluded for the 12-year class. The closure statements in the body of the report are therefore scoped to the 12-year operational horizon, with the unconstrained-flight-time result stated alongside them as the demonstrated long-duration alternative.

B.5 Independent Verification

An independent replication reproduced each headline trajectory result from first principles, without access to the baseline implementation and with the explicit objective of refuting the claimed value; the figures quoted below come from that replication.

- *Departure kinematics.* The replication reproduced the optimal arrival epoch, aim tilt, Earth-relative hyperbolic excess, and the 13.85 $\mathrm{km\,s^{-1}}$ impulsive floor, with a worst-case discrepancy below 0.5%. The same quantities agree with the Fermi Explorer reference analysis [14] to within 0.3%.
- *Outward-spiral saturation.* Two independently written trajectory integrators reproduced the power-wall ceiling of Figure 5, agreeing to 2.7%.
- *Perihelion pumping.* A from-scratch implementation reconstructed the mechanism independently: it designed its own pumping schedule and reached a cruise speed of 23.60 $\mathrm{km\,s^{-1}}$ at a Δv of 28.2 $\mathrm{km\,s^{-1}}$ over 4.9 years (the fast variant discussed in Section 4), confirming both the breaking of the power wall at $a_0 = 2.5 \times 10^{-4}\,\mathrm{m\,s^{-2}}$ and the cost of the impatient schedule. The optimized 12-year schedule reaches the same cruise speed at 23.97 $\mathrm{km\,s^{-1}}$.
- *Heliocentric lower bound.* Beyond the adversarial step-by-step audit of the derivation, the numerical harness tested every inequality in the six-step chain against three independently integrated trajectories (21 falsification checks in all, with zero violations), and a separate harness recomputed the constants entering the final fixed-point evaluation from first principles, agreeing with the derivation to better than 10^{-5} relative.

B.6 A Note on Direct Collocation

We attempted direct transcription of the low-thrust escape at the design thrust acceleration (Hermite–Simpson collocation over the full heliocentric phase); it did not converge. The failure mode is structural. It is not incidental. The optimal solution alternates coast arcs lasting months with perihelion passes whose dynamical timescale is days: a mesh fine enough to resolve the fast passes, repeated across many revolutions, grows to an intractable problem size, while a coarser mesh aliases exactly the burn arcs on which the solution depends. This difficulty of direct methods in many-revolution low-thrust problems is well documented [7]. Direct collocation also failed at the four-times-higher anchor thrust level of the bounding analysis above, where the revolution count is small: repeated attempts across mesh resolutions stalled near their initializing heuristic and did not descend to the extremal found by the indirect method, which is why the anchor value carries single-method provenance. No conclusion in this report depends on direct collocation: the heliocentric budget rests on the certified lower bound and the explicitly integrated constructive schedules, and the anchor is carried as a best-found reference with its caveat attached.

C. Mass-Closure Model

This appendix documents the vehicle mass-closure model summarized in Section 5: the governing equations and their coefficients, the algebra behind the sharp threshold in closing wet mass, the full closure grid, and the independent verification of the arithmetic.

C.1 Model Equations

Let m_w denote the wet (launch) mass, I_{sp} the thruster specific impulse, and $v_e = g_0 I_{\mathrm{sp}}$ the exhaust velocity, with g_0 the standard gravitational acceleration. For $I_{\mathrm{sp}} = 2800\,\mathrm{s}$, $v_e = 27.46\,\mathrm{km\,s^{-1}}$.

Δv budget. The total budget is the sum of a departure leg and a heliocentric leg,

$$\Delta v_{\mathrm{tot}} = \Delta v_{\mathrm{dep}} + \Delta v_{\mathrm{hel}}, \qquad \Delta v_{\mathrm{hel}} \in [22.9,\ 24.0]\,\mathrm{km\,s^{-1}}, \tag{20}$$

where the heliocentric interval is the perihelion-pumping reference interval of the trajectory analysis (Section 3): its upper end a verified constructive bound (independent re-integration agrees to 0.29%), its lower end a best-found reference anchor. For a LEO start, $\Delta v_{\mathrm{dep}} = 7.6\,\mathrm{km\,s^{-1}}$: a slow low-thrust escape spiral from a circular orbit costs approximately the initial circular speed [8], here evaluated with eclipse and steering effects included. That the budget sits marginally below the ideal circular speed of $7.67\,\mathrm{km\,s^{-1}}$ is not a contradiction: eclipse outages lengthen the spiral in time without adding Δv, steering losses do add Δv, and the offsetting credit comes from the near-escape endgame, where the orbit is far from circular and the remaining cost falls below what the circular-speed rule assigns it. For a GTO start, the departure leg follows from a standard $250 \times 35{,}786\,\mathrm{km}$ geostationary transfer orbit, whose semi-major axis is $a = 24{,}396\,\mathrm{km}$. The model takes the low-thrust escape budget, computed from orbital energy, as the local circular speed at that semi-major axis, with a 5% multiplicative allowance covering eclipse outages and steering losses:

$$\Delta v_{\mathrm{dep}} \;=\; 1.05\,\sqrt{\mu_\oplus / a} \;=\; 1.05 \times 4.04\,\mathrm{km\,s^{-1}} \;=\; 4.24\,\mathrm{km\,s^{-1}}, \tag{21}$$

with $\mu_\oplus$ the Earth gravitational parameter. Two features of this model deserve note, since the GTO start carries the report's principal closure recommendation. First, Eq. (21) applies a circular-orbit escape result to an orbit of eccentricity 0.73; low-thrust escapes from geostationary transfer orbit flown as repeated apsis-centered thrust arcs (the operational practice SMART-1 established from this same orbit [10]) generally do better than the circular-orbit estimate, so the model is likely conservative here. Second, this leg is an assumed model input; no explicit integration backs it. The closure algebra therefore carries a $\pm 10\%$ sensitivity on it, which also partially covers the circular-orbit approximation: raising the leg to $4.66\,\mathrm{km\,s^{-1}}$ lowers the GTO-100 margins of Table 10 by 0.6–0.7 percentage points (about 1.5 points per $\mathrm{km\,s^{-1}}$ of departure-leg growth), so the GTO-100 case still closes at both ends of the reference interval. The interval ends correspond to total budgets of 30.5 and $31.6\,\mathrm{km\,s^{-1}}$ from LEO, and 27.14 and $28.24\,\mathrm{km\,s^{-1}}$ from GTO.

Propellant. The rocket equation gives the propellant fraction and mass,

$$\zeta \;=\; \frac{m_p}{m_w} \;=\; 1 - \exp\!\left(-\frac{\Delta v_{\mathrm{tot}}}{v_e}\right), \qquad m_p = \zeta\, m_w. \tag{22}$$

Phase-split burn-time sizing. The thrust acceleration a_0 (defined at wet mass and at the $1\,\mathrm{AU}$ power rating) is a design variable bounded below by an operations constraint on cumulative thrusting time. Splitting the budget into its two phases, the propellant fractions of the initial wet mass expended in each phase are

$$\zeta_{\mathrm{dep}} = 1 - e^{-\Delta v_{\mathrm{dep}}/v_e}, \qquad \zeta_{\mathrm{hel}} = \left(1 - \zeta_{\mathrm{dep}}\right)\left(1 - e^{-\Delta v_{\mathrm{hel}}/v_e}\right), \tag{23}$$

and the cumulative burn time is modeled as

$$t_b \;=\; \left(K_{\mathrm{dep}}\,\zeta_{\mathrm{dep}} + K_{\mathrm{hel}}\,\zeta_{\mathrm{hel}}\right)\frac{v_e}{a_0}, \tag{24}$$

where the ideal constant-thrust burn time for a phase is $\zeta\, v_e/a_0$ and the dimensionless factors K correct it for the realities of each phase: $K_{\rm dep} = 1.54$ for the LEO spiral (the reciprocal of a 0.65 eclipse duty cycle) and 1.2 for the GTO spiral; $K_{\rm hel} = 1.84$ for the pumped heliocentric phase, calibrated against the integrated pumping schedules, whose burn arcs are localized near perihelion where available power meets or exceeds the $1\,$AU rating. The model sets a_0 to the smallest value for which $t_b \le 4\,$yr; the resulting $a_0 \approx 2.5\times10^{-4}\,\mathrm{m/s^2}$ is consistent with the optimized pumping schedule, which accumulates roughly 1.3 years of thrust within its twelve-year flight time. Because Eq. (24) involves only mass *fractions*, the sized a_0 is independent of m_w, a property the closure algebra below exploits.

Power and subsystem masses. Thrust and power at the $1\,$AU rating follow from

$$F = a_0\, m_w, \qquad P = \frac{F\, v_e}{2\eta}, \tag{25}$$

with thruster total efficiency $\eta = 0.55$. The subsystem masses are

$$m_{\rm arr} = f_{\rm rad}\,\frac{P}{\alpha}, \qquad m_{\rm pp} = \kappa\, P, \qquad m_{\rm tank} = f_t\, m_p, \tag{26}$$

where $\alpha = 60\,$W/kg is the system-level array specific power, $f_{\rm rad}$ the radiation-degradation factor on the array (1.25 for the LEO start with its long belt transit; 1.15 for the shorter GTO transit), $\kappa = 6\,$kg/kW the power-processing specific mass, and $f_t = 0.12$ the xenon tankage fraction. The fixed mass is $m_{\rm fix} = 9\,$kg ($1\,$kg payload plus $8\,$kg bus). The design closes when the required mass fits under the wet mass; the margin fraction is

$$f_m \;=\; \frac{m_w - m_{\rm req}}{m_w}, \qquad m_{\rm req} = m_{\rm fix} + m_p + m_{\rm tank} + m_{\rm arr} + m_{\rm pp}, \tag{27}$$

with closure defined by $f_m \ge 0$. The model sizes thrust and power at the $1\,$AU rating. The variation of available power along the trajectory is the province of the trajectory analysis. The mass model does not track it.

C.2 Closure-Boundary Algebra

Every term in $m_{\rm req}$ except the fixed $9\,$kg scales linearly with wet mass: m_p and $m_{\rm tank}$ through the propellant fraction ζ, and $m_{\rm arr}$ and $m_{\rm pp}$ through $P = a_0 m_w v_e/(2\eta)$ with a_0 independent of m_w. The requirement therefore collapses to

$$m_{\rm req} = m_{\rm fix} + \varphi\, m_w, \qquad \varphi \;=\; \zeta\,(1+f_t) \;+\; \frac{a_0 v_e}{2\eta}\left(\frac{f_{\rm rad}}{\alpha} + \kappa\right), \tag{28}$$

where φ is the *variable mass fraction*: the kilograms of propellant, tankage, array, and power processing required per kilogram of wet mass. The margin becomes $f_m = 1 - \varphi - m_{\rm fix}/m_w$, so closure requires

$$\varphi < 1 \qquad \text{and} \qquad m_w \;\ge\; \frac{m_{\rm fix}}{1-\varphi}. \tag{29}$$

The boundary is sharp because each additional kilogram of wet mass buys only $(1-\varphi)$ kilograms of net margin. For the LEO start, $1-\varphi$ is 0.06–0.08, so the $9\,$kg of fixed mass forces the wet mass above roughly 110–$150\,$kg before the margin turns positive. And if $\varphi \ge 1$, which occurs when the thrust acceleration roughly doubles (the power chain in Eq. (28) is proportional to a_0), no wet mass closes at any size.

Worked examples at the interval ends make the boundary concrete. For the LEO start ($I_{\rm sp} = 2800\,$s, $v_e = 27.46\,\mathrm{km\,s^{-1}}$): at $\Delta v_{\rm tot} = 30.5\,\mathrm{km\,s^{-1}}$, Eq. (22) gives $\zeta = 0.671$, so the propellant-plus-tankage term is $\zeta(1+f_t) = 0.751$; the burn-time-sized a_0 makes the power chain $a_0 v_e/(2\eta) \approx 6.3\,$W per kilogram of wet mass, and the combined array-plus-processing specific mass is $f_{\rm rad}/\alpha + \kappa = 0.0268\,$kg/W, contributing 0.169; hence $\varphi = 0.921$ and $m_w \ge 9/(1-\varphi) = 113.3\,$kg. At $\Delta v_{\rm tot} = 31.6\,\mathrm{km\,s^{-1}}$, $\zeta = 0.684$, the power chain contributes 0.173, $\varphi = 0.938$, and the boundary rises to $145.6\,$kg. These reproduce the 113.5–$146\,$kg range quoted in Section 5, the small differences reflecting the finite step of the design sweep. The same algebra for the GTO start gives $\varphi \approx 0.85$–0.87 and a closure boundary near 59–$68\,$kg, below the smallest case examined ($80\,$kg); hence every GTO entry in the grid closes.

C.3 Closure Grid

Table 10 gives the full closure grid: both departure modes, wet masses of 80–150 kg, and both ends of the heliocentric Δv reference interval. Margins are in percentage points of wet mass (Eq. (27)); a design closes when its margin is non-negative. The grid exhibits the structure derived above: on the LEO start the 100 kg vehicle misses everywhere in the interval (by 1.0–2.8 points) and 150 kg closes everywhere; on the GTO start every case closes, with margins that survive the conservative end of the interval.

Table 10. Mass-closure grid. m_p is propellant mass and P the required electrical power at 1 AU; the margin is the closure margin f_m in percentage points of wet mass, with negative values indicating failure to close. Common parameters: $I_{\mathrm{sp}} = 2800$ s, $\eta = 0.55$, $\alpha = 60$ W/kg, $\kappa = 6$ kg/kW, $f_t = 0.12$, 1 kg payload plus 8 kg bus. Departure legs: LEO 7.6 km s^{-1}, GTO 4.24 km s^{-1}.

		$\Delta v_{\mathrm{hel}} = 22.9$ km s^{-1}			$\Delta v_{\mathrm{hel}} = 24.0$ km s^{-1}		
Departure	m_w (kg)	m_p (kg)	P (kW)	margin	m_p (kg)	P (kW)	margin
LEO	80	53.7	0.50	−3.29	54.7	0.52	−5.08
LEO	100	67.1	0.63	−1.04	68.4	0.64	−2.83
LEO	120	80.5	0.76	+0.46	82.0	0.77	−1.33
LEO	150	100.6	0.95	+1.96	102.5	0.97	+0.17
GTO	80	50.2	0.46	+3.90	51.4	0.47	+1.90
GTO	100	62.8	0.58	+6.15	64.2	0.59	+4.15
GTO	120	75.3	0.69	+7.65	77.1	0.71	+5.65
GTO	150	94.2	0.87	+9.15	96.4	0.89	+7.15

C.4 Independent Verification

A separate verification analysis replicated the closure arithmetic independently, from the model statement alone. The replication reproduced the closure boundary algebraically at 113.3 kg, in agreement with Eq. (29) above and within 0.2% of the design-sweep value, and confirmed the two pivotal grid entries at the lower end of the interval: a 100 kg vehicle on the LEO start misses closure by approximately one kilogram, while 120 kg closes. The same check also exercised the no-close condition of Eq. (29): sizing the thrust acceleration at roughly twice the nominal value drives the variable mass fraction φ above unity, so that no wet mass of any size closes.

The closure results also inherit the asymmetric pedigree of the heliocentric interval, stated at Eq. (20); the full pedigree, including the certified 16.56 km s^{-1} lower bound and the distance of every known trajectory from the 22.26 km s^{-1} closure crossover, is set out in Section 3 and Appendix B. Closure margins quoted across the interval are therefore conditional at its best-found lower edge. A separate analysis reconstructed the pumping mechanism independently; the closure conclusions of Section 5 rest on no single computation.

D. Cost Model Detail

This appendix documents the program cost model summarized in Section 7: the line-item values for the two primary architectures, the definitions of the cost drivers, the thermal-qualification increment for perihelion passes, the statistical-independence assumption and its directional effect, and what the estimate deliberately excludes.

D.1 Line-Item Values

Table 11 lists the nominal line-item values, in millions of 2026 dollars, for the two primary mass-closing architectures: the LEO start at 150 kg wet mass and the recommended GTO start at 100 kg. The model carries ten triangular line items, and the table groups the rows under the mission's three budget categories.

Table 11. Nominal cost line items (millions of 2026 dollars) for the two primary architectures, grouped by mission budget category. Columns may not sum exactly to the subtotals and totals because of rounding.

Line item	Cost driver	LEO start, 150 kg	GTO start, 100 kg
Launch			
Rideshare launch	wet mass	0.98	0.65
Escape-injection premium	wet mass	0	0.61
Subtotal (allocation $3.0M)		*0.98*	*1.26*
Propulsion			
Thruster and power-processing string	input power	0.58	0.36
Xenon propellant, loaded	propellant mass	0.41	0.26
Tankage and feed system	propellant mass	0.21	0.13
Subtotal (allocation $3.0M)		*1.19*	*0.74*
Bus and mission operations			
Spacecraft bus	fixed	2.00	2.00
Solar arrays	array wattage	0.39	0.24
Integration, test, and qualification	fixed	2.45	2.45
Powered-flight operations	burn duration	4.80	4.80
Program margin and management	fixed	2.50	2.50
Subtotal (allocation $3.0M)		*12.14*	*11.99*
Total, nominal		14.31	13.99
Total, median (P50), 20,000-draw Monte Carlo		16.60	16.19

The Monte Carlo median exceeds the sum of nominal values because the triangular uncertainty bands are right-skewed: for most hardware and services line items the credible upside in price is larger than the credible downside, so the median of the summed distribution lies above the sum of the modes.

The bands are class estimates at scoping fidelity. They are not a vendor-quoted budget. The rideshare line brackets the publicly published price schedules of commercial rideshare providers. The escape-injection premium scales from the same public pricing by the energetic fraction of the escape burden the launch provider assumes. The bus, integration, and operations bands reflect the range observed across recent small-spacecraft missions of comparable class; they are not quotations for this vehicle. Read the $10M verdict of Section 7 at that fidelity: it locates the cost class of the mission and where its spending concentrates. It is not a budget commitment.

D.2 Cost-Driver Definitions

The model prices each line item per unit of a physical driver taken from the mass-closure model output for the costed architecture. The drivers are:

- **Fixed.** Independent of vehicle scale: the spacecraft bus, the integration-test-and-qualification campaign,

and the program margin and management line.

- **Wet mass.** Sets the rideshare launch line and the escape-injection premium. The premium depends on the departure orbit: zero for a LEO start, a factor of 0.45 for a GTO start (roughly the energetic halfway point to escape), and the full amount when the launch vehicle provides escape injection directly.
- **Input power.** The model prices the thruster and power-processing string per kilowatt of electric-propulsion input power at 1 AU.
- **Array wattage.** The model prices the solar arrays per watt of array output at 1 AU.
- **Propellant mass.** The model prices the loaded xenon and the tankage and feed system each per kilogram of xenon carried.
- **Powered-flight duration.** The model prices operations per year of powered flight, with a floor of one year. As noted in Section 7.4, the model prices approximately four years of powered-flight operations. The patient pumping schedule that closes the mass budget implies roughly thirteen years of custody. Under staffed-operations assumptions the operations line is therefore a lower bound.

Because these drivers come directly from the mass-closure model, the cost estimate inherits that model's verification status: an independent algebraic check replicated the closure arithmetic supplying the wet mass, propellant load, and power level, with agreement to 0.2% (Section C.4).

D.3 Perihelion Thermal-Qualification Increment

Architectures that employ perihelion pumping make repeated solar passes down to 0.42 AU. Their integration-test-and-qualification line therefore carries a multiplier of 1.4 relative to a benign-cruise mission, representing a high-temperature qualification campaign for the arrays and structure of the kind flown on previous inner-heliosphere missions. The values tabulated in Table 11 include this multiplier. The increment follows the heliocentric trajectory. It does not follow the departure mode: an escape-injection architecture that still pumps at perihelion retains it in full.

D.4 Independence Assumption

The Monte Carlo total is the sum of statistically independent triangular draws, one per line item; Section E.7 (Table 13) records the (low, nominal, high) parameters of each triangle, the sampling seed, and the closure scan supplying the cost drivers. Independence has two consequences. First, it makes the variance decomposition exact (the variance of the total is exactly the sum of the line-item variances), which is what makes the ranking of cost drivers reported in Section 7.3 well defined. Second, its neglect of correlation acts differently on the two tails. Real programs exhibit positive correlation between line items: a schedule slip inflates operations and integration together, and a propellant-market shock co-moves with launch demand. Positive correlation widens the total-cost distribution about an unchanged mean, thickening both tails at once. The widening raises the 90th percentile, so the independent model is mildly optimistic about overrun risk. The widening also places more probability mass in the low tail, so independence understates the already-small probability of remaining under $10M; on that axis the estimate is conservative. Read the reported upper percentiles with the first bias in mind. The program-margin line partially compensates by carrying program-level, correlated risk as a band of its own.

D.5 Scope Exclusions

The estimate prices the flight segment, its launch, and its powered-flight operations. It excludes development of the 1 kg payload itself, which the model prices as furnished; monitoring or custody of the vehicle after the end of powered flight, including any attempt to maintain contact during the interstellar coast; ground-segment capital beyond the tracking and operations services carried in the operations rate; launch insurance; and technology development beyond the first-unit qualification increments already carried in the propulsion and integration lines. None of these exclusions alters the structural conclusion of Section 7: the cost problem concentrates in the bus, operations, and margin category, and its resolution is programmatic. It is not physical.

E. Computational Campaign and Verification Record

This appendix records the computations behind every headline number in this report: search algorithms, sample counts, random seeds, timestep-refinement ladders, acceptance gates, and runtimes, together with the independent check applied to each result and the negative results retained alongside the positive ones. This report states the inputs, seeds, integrator settings, and acceptance gates from which an independent implementation can reproduce every headline number. This appendix maps each computation to the independent means that checked it.

E.1 Campaign Structure and Reproducibility Conventions

The analysis ran as a series of self-contained computational campaigns; each campaign records its inputs, seeds, and results in a single machine-readable artifact. Four conventions hold uniformly across the campaigns.

- *Pinned environment.* Every campaign ran in one pinned Python environment, fixed by a locked dependency manifest (Python 3.14.3, NumPy 2.5.0, SciPy 1.18.0, macOS on arm64), and each artifact embeds its environment block.
- *Named seeds.* Every stochastic computation uses an explicitly seeded generator (NumPy PCG64); the artifact records the seed next to the result it produced. Deterministic computations record that they drew no random numbers. Reruns, where performed, reproduce artifacts byte-for-byte.
- *Code digests.* Artifacts record SHA-256 digests of the scripts that produced them, binding each result to the exact code version.
- *Guarded execution.* Long-running solves ran under wall-clock guards with the guard limits recorded; runtimes serve as provenance only, and no result depends on them. Several producers refuse to write output at all if any of their internal self-checks fails.

Table 12 maps each headline number to its producing campaign and the independent check applied to it. The verification is layered: every load-bearing quantity except one carries a check from a method that shares no code with its producer. The exception is the planar twelve-year pumping schedule: its checks (timestep ladder, rocket-equation fingerprint, independent re-optimization) all run through the same fixed-step integrator. Its independent-code coverage arrives indirectly, through the three-dimensional propagator, which reproduces the planar ladder bit-for-bit under zeroed steering and whose own winning trajectory a separately implemented high-order method re-integrated (Section E.3). That chain is indirect; this record flags it as such and keeps it out of the blanket claim.

Two further verification layers sit above the per-campaign gates. First, a separate verification pass re-derived every headline quantity with independently written implementations that work from the cited astrometry and the stated models; the pass never calls the originating code. The kinematic quantities carried a 0.5% agreement gate, and the worst observed discrepancy falls below it; the sections that follow report case-by-case tolerances elsewhere. Second, an automated regression suite of more than 220 checks pins the assembled results: it recomputes the headline numbers and the intermediate quantities feeding them and fails on any value drift. The suite ran green against every result quoted in this report.

E.2 Perihelion-Pumping Schedule Search

Two independently seeded searches optimized the four-parameter pumping policy of Appendix B. Both use the same trajectory integrator: a fixed-step kick-drift-kick scheme with the thrust policy frozen across each step, an adaptive step law $\Delta t = \max(600\ \mathrm{s},\ 0.002\,P_{\mathrm{circ}}(r)) \times s$, where $P_{\mathrm{circ}}(r)$ is the local circular-orbit period and s the refinement scale, and a rocket-equation mass update on every thrusting step.

The original search drew 400 uniform-random candidates over the parameter box from a generator seeded at 11, then polished the five best by Nelder–Mead simplex descent; the best polished schedule reaches the cruise speed at 23.966 $\mathrm{km\,s^{-1}}$ in 11.90 years. Timestep refinement exposed the seam disclosed in Appendix B: at half step the run initially terminated at the simulation-horizon cap short of cruise speed, and completing it at an extended horizon gives 23.969 $\mathrm{km\,s^{-1}}$, with the quarter-step run at 23.985 $\mathrm{km\,s^{-1}}$.

Table 12. Headline results, producing campaigns, and independent checks. Each campaign recorded its seeds and acceptance record in its machine-readable artifact.

Result	Producing campaign	Independent check
Departure optimum (73,012 yr; $v_\infty = 23.64\ \mathrm{km\,s^{-1}}$; 2.48° tilt; 13.85 $\mathrm{km\,s^{-1}}$ floor)	Deterministic departure-geometry scan	Closed-form identities (Appendix A); from-scratch replication, $< 0.5\%$
12-yr pumping schedule, 24.0 $\mathrm{km\,s^{-1}}$	Seeded schedule search and re-optimization (Section E.2)	Timestep ladder; rocket-equation fingerprint; independent re-optimization
Free-flight-time schedule, 18.87 $\mathrm{km\,s^{-1}}$	Six-cell long-horizon sweep (Section E.3)	Four-resolution timestep ladder
Certified lower bound, 16.56 $\mathrm{km\,s^{-1}}$	Six-lemma bounding chain (Section E.5)	Adversarial proof audit; 21-check numerical harness (Section E.5)
Reference anchor, 22.66 $\mathrm{km\,s^{-1}}$	Indirect optimal-control solve (Section E.4)	Direct-collocation cross-check attempted; unconverged (single-method, so labeled)
Out-of-plane increment, 578 m/s	Three-dimensional trajectory sweep (Section E.3)	Bit-level planar-embedding regression; independent re-integration
Intercept probability, 0.982	Seeded dispersion Monte Carlo (Section E.6)	Closed-form analytic propagation, 0.011% agreement
Mass-closure boundaries and grid	Sixteen-point closure scan (Section E.7)	Algebraic replication (Appendix C)
Catalog sweep and target ranking	Catalog encounter sweep (Section E.7)	Published-encounter solutions (Section A.4)

The re-optimization repeated the search from an independent seed (20260716) with the perihelion floor raised from the search-phase 0.25 AU to the 0.42 AU class floor: 802 coarse evaluations over the box $r_p \in [0.42, 0.6]$ AU, aphelion fraction $\in [0.55, 0.98]$, burn radius $\in [0.45, 1.0]$ AU, termination energy $\in [-220, -20]\ \mathrm{km^2\,s^{-2}}$, followed by Nelder–Mead polish of the eight best candidates (400 function evaluations each). The search then integrated each surviving candidate on a three-rung timestep ladder ($s = 1, 0.5, 0.25$) and accepted it only if it passed five gates: (i) the integrated Δv matches the rocket-equation value implied by the final mass to within $10^{-13}\ \mathrm{km\,s^{-1}}$ (a mass-bookkeeping fingerprint; worst case 6×10^{-14}); (ii) the ladder spread is within 1% (measured 0.077%); (iii) the minimum solar distance respects the 0.42 AU floor on every rung; (iv) the commanded thrust never exceeds the power envelope, checked pointwise after the fact; and (v) the artifact records the seeds and integrator metadata in full. The winner is the schedule from the original search: the re-optimization found nothing better from a fresh seed. The carried value is the most conservative rung of its ladder, 23.9845 $\mathrm{km\,s^{-1}}$, quoted as 24.0. The full search took approximately 4 seconds of wall time. The policy class limits the 12-year value. Search effort does not limit it.

E.3 Long-Horizon and Three-Dimensional Trajectory Sweeps

Long-horizon probe. The unconstrained-flight-time upper bound of Section B.4 comes from a six-cell sweep over horizons of 60 and 150 years crossed with three values of an aphelion-switch radius (5, 10, 20 AU) governing a fifth policy parameter that widens the prograde-burn window once the orbit grows large. Each cell drew 400 seeded coarse candidates (seed 20260715) and polished the best by Nelder–Mead. The sweep ran under a pre-registered stop rule: halt when a schedule passes all acceptance gates (cruise speed reached, perihelion floor respected, timestep ladder within 1% with an additional eighth-step corroboration rung, envelope compliance, and the rocket-equation fingerprint) at a Δv below the 22.26 $\mathrm{km\,s^{-1}}$ closure crossover. The rule fired at 18.871 $\mathrm{km\,s^{-1}}$ (ladder maximum) in the 60-year cell with a 10 AU switch radius, on a trajectory using about 58 years of flight; gate-passing schedules across the cells span 33–58-year flight times at 18.9–20.5 $\mathrm{km\,s^{-1}}$. One cell (20 AU switch, 150-year horizon) produced a nominally cheaper schedule at 18.68 $\mathrm{km\,s^{-1}}$ that failed the extended ladder gate: it is timestep-fragile at its horizon cap. The sweep excluded that schedule from quotation and recorded the exclusion in the artifact.

Three-dimensional sweep. A three-dimensional extension of the same propagator measured the out-of-plane cost of the $2.48°$ departure tilt; in this extension the steering direction acquires an out-of-plane component commanded by up to four additional parameters. Five sweep cells ran, each with its own seed (20260715 through 20260719), 600 coarse candidates, and eight polished finalists: tilt applied within the terminal escape burn only; tilt applied during the retrograde arcs beyond an activation radius; both combined; a variant with the pump parameters frozen at the planar optimum and only the steering free; and a control cell targeting zero tilt, which recovered the planar 23.966 $\mathrm{km\,s^{-1}}$ result exactly. The direction requirement is a hard gate of $0.04°$ on the achieved asymptote latitude against the $-2.48°$ target: one tenth of the $0.41°$ open-loop aiming budget, so the trajectory gate does not consume the dispersion allowance of Section 6. During the coarse search the gate relaxes to a penalty of 0.413 $\mathrm{km\,s^{-1}}$ per degree of direction error, which keeps the search from starving.

Before the sweep, the 3-D propagator itself passed a four-part self-test battery, recorded alongside the sweep results: (i) a planar-embedding regression, in which the 3-D code with the steering zeroed must reproduce the planar ladder (23.9661, 23.9835, 23.9845 $\mathrm{km\,s^{-1}}$) bit-identically, with the out-of-plane coordinate bitwise zero at every step; (ii) a cross-method asymptote check, comparing the departure direction constructed analytically from the osculating orbit at thrust cutoff against an independent high-order integration (SciPy DOP853, relative tolerance 10^{-10}, absolute 10^{-12}-scaled) coasting to 100 AU, with agreement to 7×10^{-12} radians; (iii) validation of the closed-form out-of-plane gain factor against finite-difference velocity-kick experiments in three orbital regimes; and (iv) measurement of the radial distribution of retrograde thrust in the planar optimum, which showed all retrograde Δv inside 2.3 AU and none beyond 4 AU, fixing the activation-radius search box.

The winning cell is the combined parameterization: carried 3-D value 24.5627 $\mathrm{km\,s^{-1}}$ (ladder maximum over the same three-rung refinement and gate set as the planar search), against the planar carried value of 23.9845 $\mathrm{km\,s^{-1}}$. The increment is 578.19 m/s, inside the $[22\ \mathrm{m/s},\ 1.02\ \mathrm{km\,s^{-1}}]$ band that the closed-form bracket of Section 4 predicts. Both endpoints are best-found constructive schedules, so the increment is a difference of upper bounds. It does not bound the true optimal increment. The retrograde-arc-only cell produced no candidate passing the full gate set; the artifact records that negative. An independent re-integration then checked the winner with a method sharing nothing with the fixed-step propagator: SciPy DOP853 at relative tolerance 10^{-10}, with the control law evaluated continuously at every derivative call. The fixed-step code freezes the law across each step. The re-integration reproduced the Δv to within 0.3% of every ladder rung (0.02% of the finest) and the asymptote latitude to $0.013°$.

Finally, the margin slopes of the Appendix C scan propagate the increment through the mass-closure grid: the recommended 100 kg GTO architecture keeps margins of $+5.1$ and $+3.1$ points at the two ends of the heliocentric reference interval, while the 150 kg LEO architecture moves to $+1.0$ and -0.8 points. The conservative end flips sign: that case tolerates only about 104 m/s of growth. This is the basis for the statement in Section 5 that the three-dimensional cost leaves the GTO recommendation intact while removing the conservative-end margin of the LEO alternative. The full 3-D campaign, all five cells and ladders, took 103 seconds of wall time.

E.4 Optimal-Control Solves and the Direct-Collocation Record

Indirect solves. An indirect method produced the reference-anchor extremals of Section B.4: it integrates the costate equations of Pontryagin's minimum principle with SciPy's DOP853 at relative tolerance 10^{-8} and absolute tolerance 10^{-10}, smooths the bang-bang thrust switching by a continuation schedule that sharpens the switching function in three stages ($\varepsilon = 0.3, 0.1, 0.03$), and closes the boundary conditions by shooting from five random multistarts (seed 20260702). Convergence requires the shooting residual below 10^{-6}; the accepted solutions close it to 4.4×10^{-13}. The 23.64 $\mathrm{km\,s^{-1}}$-target solve converges in 0.4 s to 22.660 $\mathrm{km\,s^{-1}}$ (minimum solar distance 0.62 AU, so the perihelion floor is inactive). The same solver reproduces the previously computed 22.944 $\mathrm{km\,s^{-1}}$ extremal at the 24.14 $\mathrm{km\,s^{-1}}$ target variant bit-for-bit; that reproduction checks environment drift only. It is not an independent verification.

Direct-collocation attempts. Four attempts at confirming the anchor extremal by direct transcription all failed; the record stays because the failures are informative. The first three (uniform-mesh Hermite–Simpson at 24, 48, and 60 segments) stalled at their iteration limits near the initializing heuristic or failed re-integration, as summarized in Appendix B. The fourth was the strongest attempt the problem structure permits: a 100-segment Hermite–Simpson transcription on a nonuniform mesh with nodes concentrated where the dynamics are fast (node density proportional to a blend of $r^{-3/2}$-warped and uniform time, giving a five-to-one ratio

of largest to smallest segment), warm-started from the indirect extremal itself, with the steering and throttle histories seeded from the costate solution. The nonlinear program converged by every internal measure (successful termination in 148 iterations, collocation defects of 4.4×10^{-10}) and claimed 21.62 $\mathrm{km\,s^{-1}}$, below the indirect value. Re-integrating the claimed control history through the dynamics, however, returns 21.18 $\mathrm{km\,s^{-1}}$ of actual expenditure, a 2.1% discrepancy exceeding the 2% acceptance gate: the transcription exploited discretization slack. It found no better trajectory. The record lists the attempt as a rejected solution, and the anchor retains its single-method label.

Abandoned design-thrust solves. Optimal control at the design thrust acceleration itself ($a_0 = 2.5 \times 10^{-4}\,\mathrm{m\,s^{-2}}$, where a solution would sharpen the closure verdict directly) ended as a negative result, closed out under a documented disposition procedure. Two solver jobs, a segment-grid direct solve and a free-horizon indirect solve, ran to approximately 20.4 and 20.1 CPU-hours respectively (essentially full CPU over roughly 26 hours of wall time each) without completing a single case. The close-out procedure froze the environment, confirmed the process identities, captured stack samples (both processes were live inside the integrator's derivative evaluations; neither was hung), hashed every log before and after interruption, and adjudicated all recovered output against the certified interval: no converged admissible trajectory existed in any log. The procedure records one recovered line, a 120-segment direct attempt reporting 12.9 $\mathrm{km\,s^{-1}}$ with a failed convergence flag, as inadmissible: it lies below the certified 16.56 $\mathrm{km\,s^{-1}}$ bound, so it cannot correspond to any real trajectory. The roughly 40 CPU-hours of abandoned solves are the strongest available evidence that the many-revolution problem at design thrust is out of reach for these methods. The heliocentric budget therefore rests on the certified bound and the constructive schedules.

E.5 Lower-Bound Certificate and Numerical Falsification Harness

The certified 16.56 $\mathrm{km\,s^{-1}}$ bound of Section B.4 is a chain of six closed-form lemmas, machine-checked end to end. Section B.4 states the six steps; this record adds the constants and inequality surfaces that the numerical harness attacks. The single-transit lemma shows that every radial-velocity zero on the escape leg is a strict upward crossing; hovering near the perihelion floor would require some thirteen times the available thrust. The transit-duration lemma takes its comparison at a shell radius of 0.46 AU (slightly above the 0.42 AU floor, where the bound is sharpest), capping both the Δv deliverable and the energy gained inside the window. The initial-segment lemma runs through a verification function whose gradient inequality reduces to a two-variable polynomial bound. The chain closes by self-consistency: evaluated at an assumed total of 16.56 $\mathrm{km\,s^{-1}}$, the assembled bound returns 16.5659 $\mathrm{km\,s^{-1}}$, a contradiction for any cheaper trajectory. The closure is insensitive to the assumed total: the assembled bound varies only between 16.566 and 16.564 $\mathrm{km\,s^{-1}}$ as the assumption ranges from 16.503 (the impulsive floor) to 17 $\mathrm{km\,s^{-1}}$. Two simpler relaxation strategies preceded the chain and stand as recorded negative results: a scalar energy relaxation certifies only 11.39 $\mathrm{km\,s^{-1}}$ and a two-variable box relaxation at most 13.31 $\mathrm{km\,s^{-1}}$, both below the impulsive floor and therefore useless. The main chain salvages their pointwise inequalities as components.

Because the bound is a theorem-grade claim, its audit was adversarial: a dedicated review pass, conducted separately from the derivation, attempted to defeat each lemma in turn, probing its case splits, boundary values, and sign conventions for counterexamples, and the audit accepted the chain only after every such attempt failed. A dedicated falsification harness then attacked the chain numerically. The harness evaluated seven inequality targets drawn from the lemmas (turning-point sign structure, the radial-velocity lower bound, the transit-window duration cap, the in-window energy cap, the pointwise shell-speed bounds, and the polynomial gradient bound with its boundary values) on three trajectories of different character: the optimized pumping schedule, the anchor extremal, and a freshly integrated ballistic orbit grazing the shell (DOP853 at relative tolerance 10^{-12}, energy conserved to 6×10^{-15} over the arc). None of the 21 checks failed. The harness records checks whose case-split premises do not apply to a given trajectory as vacuous; it does not count them as passes. The tightest margin, -9×10^{-15} on the radial-velocity bound along the ballistic trajectory, is machine-epsilon-scale: the ballistic trajectory attains the inequality with equality, exactly as the comparison argument predicts. Equality there confirms the bound is sharp; a slack bound would leave a finite margin. The harness additionally checked the polynomial gradient bound in a cancellation-free factored form on a 400×400 grid and against a 1,000-point random polynomial-identity cloud (seed 20260715), and it recomputed the eight assembled constants of the fixed-point evaluation from first principles, matching the derivation to 5.4×10^{-6}

relative.

E.6 Aiming-Dispersion Monte Carlo

The intercept-probability analysis of Section 6 is a 20,000-draw Monte Carlo: a dedicated closed-form dispersion-propagation module, driven by a thin campaign script, perturbs four error sources on each draw and propagates the miss at closest approach in closed form, with no numerical integration per draw. The four sources are target ephemeris (catalog parallax, proper-motion, and radial-velocity covariances), pointing (two transverse rotation angles), velocity magnitude, and the injection residual; their dispersion levels are assumed engineering inputs, marked at their point of use in Section 6. Every run in the campaign carries its own named seed, fourteen in all (42–55): the headline run (seed 42), the alternate-catalog variant (43), four one-at-a-time source isolations (44–47), four leave-one-out complements (48–51), three assumption-sensitivity doublings (52–54), and the cross-catalog systematic diagnostic (55). The producer is deterministic (reruns are byte-identical) and refuses to write its artifact if any of its 29 internal self-checks fails.

The headline result is $P = 0.98155$ (19,631 of 20,000 draws inside the 2,600 AU miss-tolerance sphere), with a Wilson 95% interval of $[0.97959,\ 0.98332]$ and an exact Clopper–Pearson interval agreeing to 1.7×10^{-4}. Prefix subsets of the single seed-42 stream establish convergence: the 5,000-, 10,000-, and 20,000-draw intervals are mutually contained, and the interval width contracts by a factor of 0.502 from 5,000 to 20,000 draws, against the 0.5 that $1/\sqrt{n}$ scaling predicts. Ranked by one-at-a-time root-mean-square miss, the sources contribute 764 AU (velocity magnitude), 647 AU (pointing), 617 AU (ephemeris), and 547 AU (injection residual); a leave-one-out total-effect ranking promotes ephemeris above pointing once the ranking includes interactions with the geometry, and the variance budget closes to 0.2% (the interaction residual between the summed one-at-a-time second moments and the full-run second moment), confirming the sources act nearly independently. A genuinely independent method verified the headline probability: a first-principles linear-Gaussian propagation of the transverse covariance through the closed-form gains, with the disc probability evaluated by 400-node Gauss–Legendre quadrature, sharing no code or random numbers with the Monte Carlo. It gives $P = 0.98166$, a relative difference of 1.1×10^{-4} against an acceptance gate of 2×10^{-2}. Substituting the alternate astrometric catalog self-consistently (centers, uncertainties, and aim point together; seed 43) gives $P = 0.9919$. The deliberate cross-catalog experiment (aiming with one catalog while drawing truth from the other) produces the ≈ 859 AU cross-track systematic offset discussed in Section 6.5, with $P = 0.9623$; a first-order analytic prediction of the offset agrees to 0.5%. The assumption-sensitivity rows are part of the record: doubling the pointing dispersion lowers P to 0.897, doubling the magnitude dispersion to 0.858, and quadrupling it to 0.571, which is why Section 6 flags the velocity-trim requirement as the binding engineering assumption. The artifact embeds the SHA-256 digests of the producing code and the pinned-environment block.

E.7 Catalog Sweep and Cost Monte Carlo

Catalog sweep. The alternative-target scan of Section 8 rests on a full sweep of the Bailer-Jones stellar-encounter catalog [6]. The sweep screened all 61 rows of the published close-encounter table (retrieved from the VizieR service on 2026-07-14, with Gaia DR3 astrometry for the survivors fetched the same day) for future encounters with perihelion inside 1 pc arriving within the 500,000-year horizon and free of the catalog's dubious-astrometry and binarity flags. The filter removes 32 past encounters, 20 whose perihelia lie beyond the horizon, and four flagged rows, each exclusion recorded with its reason; all four flagged stars carry spurious radial velocities of hundreds of $\mathrm{km\,s^{-1}}$. The sweep then ranked the five surviving candidates (Proxima Centauri among them) by minimum impulsive departure from low Earth orbit, alongside reference solutions for Alpha Centauri and Lambda Serpentis computed identically. The metric has a floor of 8.757 $\mathrm{km\,s^{-1}}$ (bare solar escape from a 400 km orbit with full Earth-velocity alignment), which no target can undercut. Two anonymous distant M dwarfs formally beat LSPM J2146+3813: 8.78 and 8.82 against 9.30 $\mathrm{km\,s^{-1}}$. Both sit barely above the floor, carry weak radial velocities (± 8–$11\ \mathrm{km\,s^{-1}}$), and present near-degenerate miss geometry, which is why Section 8 carries LSPM J2146+3813 as the practical minimum-energy target and Lambda Serpentis (11.38 $\mathrm{km\,s^{-1}}$) as the best Sun-like alternative. The sweep confirms every unflagged surviving candidate is an M dwarf.

Cost Monte Carlo. The cost distributions of Section 7 are sums of ten independent triangular draws, one per line item, sampled 20,000 times from a generator seeded at 7 in a single vectorized call of the cost model.

Table 13. Triangular (low, nominal, high) parameters of the cost Monte Carlo, in 2026 dollars per driver unit. The escape-injection premium scales by departure mode (0 from LEO, 0.45 from GTO, 1.0 for direct escape injection); the integration-and-qualification line carries the 1.4 perihelion thermal-qualification multiplier for architectures that pump at perihelion (Appendix D). The model prices operations per year of powered flight with a one-year floor.

Line item	Unit	Low	Nominal	High
Spacecraft bus	fixed	$1.2M	$2.0M	$4.0M
Thruster and power-processing string	per kW	$0.35M	$0.60M	$1.2M
Solar arrays	per W	$250	$400	$800
Xenon propellant, loaded	per kg	$2,500	$4,000	$12,000
Tankage and feed system	per kg Xe	$1,000	$2,000	$4,000
Rideshare launch	per kg	$5,000	$6,500	$9,000
Escape-injection premium	per kg	$8,000	$13,500	$25,000
Integration, test, and qualification	fixed	$1.0M	$1.75M	$3.0M
Powered-flight operations	per year	$0.6M	$1.2M	$2.5M
Program margin and management	fixed	$1.0M	$2.5M	$5.0M

Table 13 gives the per-unit (low, nominal, high) parameters behind the nominal values of Table 11; Appendix D grades the basis for the bands. Percentiles at this sample count are stable to well under the $0.1M quotation grain. The costed architectures come from the sixteen-point closure scan: both departure modes, four wet masses, and both ends of the heliocentric reference interval (the same scan that populates the closure grid of Appendix C), with the full cost model attached to each closing case of interest. The variance decomposition, exact under independence, attributes about 64% of the cost variance to the powered-flight operations line, which is what makes the structural conclusion of Section 7 insensitive to the hardware bands.

E.8 Arrival-Epoch Re-derivation and Trade Table

As a final check on the departure geometry, a blind re-derivation repeated the arrival-epoch optimization: a separate implementation, written without access to the originating code or to any of its results, received only the cited astrometry and the stated intercept convention and scanned arrival epochs at one-year resolution. It converged on the same optimum, agreeing to one year of arrival epoch: the minimum of the impulsive floor falls at 73,012 years (refined to 73,012.17), with the 13.85 $\mathrm{km\,s^{-1}}$ floor value, the 23.64 $\mathrm{km\,s^{-1}}$ cruise speed, and the $-2.48°$ aim tilt of the headline geometry. The same scan locates a distinct minimum of the heliocentric cruise speed itself: 23.17 $\mathrm{km\,s^{-1}}$ at an arrival epoch of 57,854 years. The two optima differ because the departure budget is Earth-relative: at earlier arrivals the required asymptote points farther below the ecliptic and aligns less well with Earth's orbital velocity, so the impulsive floor rises even as the cruise speed falls. The impulsive floor is flat near its optimum: it stays within 100 m/s of its minimum for arrivals between 66,908 and 80,436 years.

Table 14 extends the re-derivation into a full trade table over arrival epochs from 55,000 to 85,000 years in 1,000-year steps. For each row it lists the required heliocentric cruise speed, the aim tilt, the impulsive floor from a 400 km orbit, and the heliocentric Δv of a twelve-year pumping schedule re-optimized for that row's cruise-speed target. Each row runs from its own recorded seed, with the same 800-candidate search, timestep ladder, and five acceptance gates as the planar schedule search of Section E.2, under the design vehicle's thrust, power-envelope, and 0.42 AU perihelion-floor constraints. Every row closes: a gate-passing twelve-year schedule exists across the entire window.

Three caveats govern the pumped column. First, each entry is the best schedule found for that row: a constructive upper bound on the true optimum of the policy class. It is not that optimum. Second, the entries carry roughly ± 0.2 $\mathrm{km\,s^{-1}}$ of search scatter from their per-row seeds, and the scatter is comparable to the inter-row differences it must resolve: the 65,000-, 70,000-, and 75,000-year rows each found a better schedule than both of their neighbors (the best, 23.49 $\mathrm{km\,s^{-1}}$ at 65,000 years, is likely available at nearby epochs as well). Third, the column is planar and excludes the cost of acquiring the out-of-plane aim tilt, measured at 0.58 $\mathrm{km\,s^{-1}}$ for the $2.48°$ tilt of the departure optimum and growing steeply with angle; the tilt steepens past $-10°$ by 58,000

years. The apparent attraction of earlier arrivals (a lower cruise speed at 57,854 years than at the departure optimum) is therefore promising but unsettled: the planar pumped column understates the cost of exactly those rows where the tilt is largest, and settling the trade requires a three-dimensional re-optimization across the window.

Table 14. Arrival-epoch trade, 55,000–85,000 years. Every epoch in the window closes with a twelve-year pumped schedule. The pumped column is the best schedule found per row (an upper bound on the optimum, with $\approx \pm 0.2\ \mathrm{km\,s^{-1}}$ seed scatter); it is planar and excludes the tilt-acquisition cost, which is 0.58 $\mathrm{km\,s^{-1}}$ at $2.48°$ and grows steeply with tilt.

Arrival epoch (yr)	v_∞ ($\mathrm{km\,s^{-1}}$)	Tilt β (deg)	Impulsive floor ($\mathrm{km\,s^{-1}}$)	Pumped 12-yr Δv ($\mathrm{km\,s^{-1}}$)
55,000	23.202	−12.33	15.108	23.75
56,000	23.185	−11.62	14.939	23.74
57,000	23.175	−10.94	14.786	23.74
58,000	23.173	−10.27	14.649	23.74
59,000	23.177	−9.63	14.525	23.74
60,000	23.187	−9.00	14.415	23.74
61,000	23.202	−8.40	14.318	23.75
62,000	23.222	−7.81	14.232	23.76
63,000	23.246	−7.25	14.157	23.77
64,000	23.273	−6.70	14.092	23.79
65,000	23.305	−6.17	14.036	23.49
66,000	23.339	−5.66	13.990	23.82
67,000	23.376	−5.16	13.951	23.84
68,000	23.415	−4.68	13.919	23.88
69,000	23.457	−4.21	13.894	23.89
70,000	23.500	−3.76	13.876	23.57
71,000	23.545	−3.32	13.863	23.94
72,000	23.592	−2.90	13.856	23.94
73,000	23.640	−2.48	13.854	23.98
74,000	23.689	−2.09	13.856	24.02
75,000	23.739	−1.70	13.862	23.90
76,000	23.790	−1.32	13.872	24.11
77,000	23.841	−0.96	13.886	24.14
78,000	23.893	−0.61	13.902	24.17
79,000	23.946	−0.26	13.922	24.16
80,000	23.999	+0.07	13.944	24.19
81,000	24.052	+0.39	13.968	24.22
82,000	24.105	+0.71	13.994	24.25
83,000	24.159	+1.01	14.023	24.28
84,000	24.212	+1.31	14.053	24.32
85,000	24.266	+1.60	14.085	24.35

TECHNICAL MEMORANDUM AUGUST 2026

The Perihelion-Pumped Arrival-Epoch Optimum: Independent Verification and Crossing-Epoch Reconciliation

Technical memorandum for The Fermi Explorer Mission

Prepared by	Physical Superintelligence PBC
Prepared for	The Fermi Explorer Mission, a 501(c)(3) nonprofit organization
Date	August 2026
Scope	Adjudication of the perihelion-pumped arrival-epoch optimum and the ecliptic-crossing epoch against the Fermi team's independent analysis, with the six-item measurement program that stress-tested both
Status	Final

1. Executive summary

BOTTOM LINES UP FRONT (force-ranked)

1. The Fermi team's 77,500 yr perihelion-pumped arrival optimum is correct twice over: our independent model bottoms at 77,499 yr (verified), and the adjudicated empirical basin puts both optima 16 $\mathrm{m\,s^{-1}}$ above its measured bottom, 24.156 $\mathrm{km\,s^{-1}}$ at 79,000 yr, flat to 25 $\mathrm{m\,s^{-1}}$ over 77,500 to 79,000 yr with the bottom unresolvable beyond ±1,000 yr (measured; Section 3).
2. The 73,012 yr impulsive-departure optimum also stands at 13.854 $\mathrm{km\,s^{-1}}$ (verified; blind re-derivation), and the two optima price different vehicles, so both are right (Section 2).
3. The 534 yr crossing-epoch gap is closed: their 79,252 yr reproduces exactly from their inputs, and the gap is +533.5 yr of inputs, distance +528.8 yr of it, plus −0.01 yr of conventions (verified; Figure 4).
4. Their input state mixes catalogs, the Akeson 2021 distance with Kervella-era proper motions, and the catalogs disagree at 5.1 sigma in parallax, so the mix matters (measured; Table 5).
5. The carried radial velocity is spectroscopic in both catalogs, the kinematic value is -22.332 ± 0.005 $\mathrm{km\,s^{-1}}$ against the carried −22.400, the correction moves the crossing 299 to 332 yr later, and the four catalog-convention scenarios span 79,467 to 80,118 yr on their state clocks (verified convention; measured shifts; Section 4.5).
6. The crossing design epochs cost 22.1 $\mathrm{m\,s^{-1}}$ (Akeson working state) and 27.1 $\mathrm{m\,s^{-1}}$ (Kervella) above the measured basin bottom, 0.11% of total, so the measurement endorses the Fermi design default (measured; Section 3.3).
7. End to end the crossings hold: arrival probability 0.965 (Kervella) and 0.984 (Akeson working state; 0.9858/0.9866 on the Akeson J2019.5 catalog state) against the 0.982 baseline, GTO-100 closes at both with +3.8 pts, LEO-150 still does not close, and the measured open-loop powered-phase residual is 99.6 $\mathrm{m\,s^{-1}}$ against the assumed 25 $\mathrm{m\,s^{-1}}$ per axis (simulation and measured; Section 5).
8. The joint end-to-end probability at the Kervella crossing is 0.964 at assumed execution levels and 0.606 under the measured residual, and radial velocity drives 99.1% of arrival-epoch variance (simulation; Section 6).
9. The recommended public statement is pinned to the Akeson 2021 state at its native epoch J2019.5 in the kinematic frame, with T counted from the 2029.0 departure: $T_{\mathrm{cross}} = 79{,}756$ yr after departure, crossing date AD 81,785, program band ±400–800 yr; the observed-γ variant gives 79,457.5 yr, AD 81,486.5 (measured; Section 7).

This memorandum closes out the two questions raised by the Fermi Explorer team's independent perihelion-pumped (PP) arrival-epoch analysis, whether their 77,500 yr propellant optimum is correct and why their ecliptic-crossing epoch (79,252 yr) differs from ours (79,785.8 yr) by 534 yr, and it reports the six-item measurement program that stress-tested the answers: an empirical basin sweep with outlier adjudication, a radial-velocity convention audit, design-epoch dispersion and closure, powered-phase execution residuals, linearity bounds, and a joint end-to-end Monte Carlo. All questions are settled above. Every number in this memorandum traces to a machine artifact or to the Fermi team's cited files, and every claim carries its evidence grade in the sentence that states it. Neither implementation was tuned toward the other at any point. An independent adversarial verification pass re-ran every analysis script to completion, recomputed the headline values through different call paths, and confirmed every verdict (Appendix A).

Two subsidiary findings complete the picture. The 0.04 $\mathrm{km\,s^{-1}}$ v_∞ offset between the two epoch tables has the same single cause as the crossing gap: the raw unshaved v_∞ at 73,000 yr is 23.7145 $\mathrm{km\,s^{-1}}$ on their inputs and 23.8087 $\mathrm{km\,s^{-1}}$ on ours, convention-independent to 0.0001 $\mathrm{km\,s^{-1}}$, and distance drives the difference (measured; Section 4.7). And the Fermi team's own error budget anticipated the main result: their error-budget table lists "Astrometry inputs (sub-0.5% differences move the crossing itself ~500 yr)", which is exactly the effect we measure. Their numerical discipline, their dt/8 convergence checks, and their custody-gated acceptance criteria all held up under our audit (Section 3).

2. The two-optima structure: why both numbers are correct

The arrival epoch T fixes the required departure asymptote: the heliocentric cruise speed $v_\infty(T)$ and its ecliptic latitude $\beta(T)$, both set by the Alpha Centauri AB barycentric astrometry with the 2,600 AU miss allowance credited. Three different objectives place their minima at three different epochs.

- **Cruise speed.** $v_\infty(T)$ alone bottoms at 57,854 yr at 23.173 $\mathrm{km\,s^{-1}}$ (measured; 1 yr grid over 50,000 to 100,000 yr). Over the 70,000 to 82,000 yr window relevant here, $v_\infty(T)$ rises monotonically (measured on a 49-point grid).
- **Impulsive departure.** The single-impulse Oberth departure from a 400 km LEO pays $\sqrt{v_{\mathrm{esc}}^2 + v_\infty^2} - v_{\mathrm{circ}}$, which trades the rising $v_\infty(T)$ against the shrinking geometric tilt. Its minimum is 73,012.17 yr at 13.8539 $\mathrm{km\,s^{-1}}$ (verified; blind re-derivation agreed with the heritage value 73,012.04 yr to 0.2 yr).
- **Perihelion-pumped campaign.** The PP vehicle acquires tilt by steering thrust out of plane during the pump campaign. That cost is a separate additive term that falls quadratically to zero where the asymptote crosses the ecliptic (measured; decay exponent 1.92, Section 3). Adding it to the shaved $v_\infty(T)$ moves the minimum to 77,499 yr in our model (measured) and 77,500 yr in their direct simulation (their table; verified against our model to 1 yr), and the measured basin of Section 3.3 places both inside its flat bottom.

The 73,012 yr and 77,500 yr numbers therefore answer different questions. The impulsive optimum prices a chemical departure stage. The PP optimum prices the pumped vehicle's own propellant. The Fermi team's document scopes their analysis to the PP vehicle explicitly, and within that scope their epoch table is correct (verified; Section 3). Their design default is the ecliptic crossing itself, 33 $\mathrm{m\,s^{-1}}$ above their own basin bottom and 22 to 27 $\mathrm{m\,s^{-1}}$ above our measured basin bottom (measured; Section 3.3), chosen because it is the one epoch fixed by geometry alone. That design logic survives our adjudication and our measurement program: the crossing epoch is invariant under every pricing-model revision, its residual uncertainty is astrometric (Section 4), and its end-to-end dispersion and vehicle closure hold at the crossing (Section 5).

3. Independent verification of the PP optimum

3.1 Measured tilt-acquisition costs

We measured the 3-D tilt-acquisition cost directly with the certified propagator at four arrival epochs, holding everything else fixed. Each row targets the exact kinematic aim (v_∞, β) of its epoch, runs the certified 3-D pump search under the heritage acceptance gates plus a direction gate $|\beta_{\mathrm{achieved}} - \beta_{\mathrm{target}}| \le 0.04°$, and subtracts a matched planar pump run at the same v_∞ target. The carried value is the maximum of the gate-passing

Table 1. Measured 3-D tilt-acquisition cost versus asymptote ecliptic latitude. Carried 3-D and matched-planar totals are best-found values under identical gates and seeds (seed base 20260803); the increment column is their difference, computed before rounding. The zero-tilt row is the planar-collapse control: the 3-D search at $\beta_{\mathrm{target}} = 0$ lands 0.015 $\mathrm{km\,s^{-1}}$ above its own planar embedding, which bounds the method's residual.

Arrival T (yr)	$\lvert\beta_{\mathrm{target}}\rvert$ (deg)	Carried 3-D ($\mathrm{km\,s^{-1}}$)	Matched planar ($\mathrm{km\,s^{-1}}$)	Tilt cost ($\mathrm{km\,s^{-1}}$)
73,012	2.48	24.563	24.006	0.556
75,000	1.70	24.345	24.052	0.293
77,500	0.78	24.173	24.111	0.062
79,786	0.00	24.200	24.200	0.000

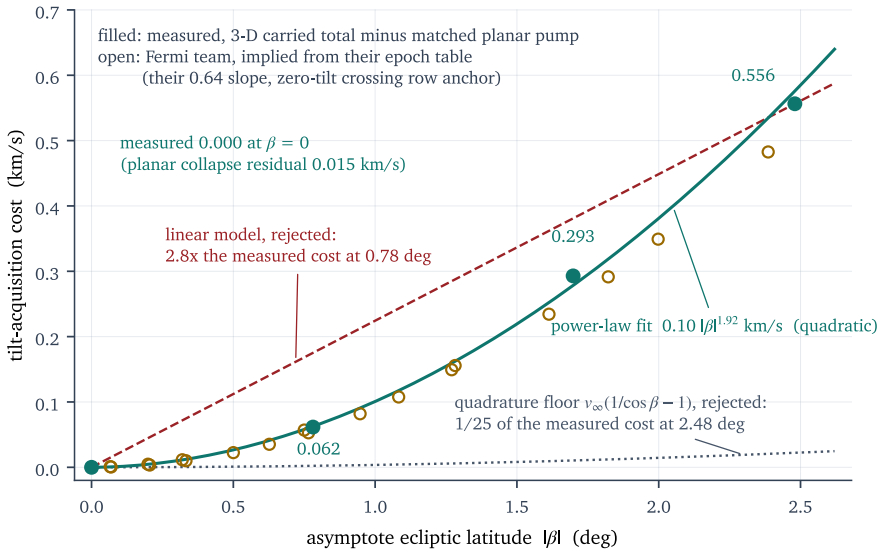


Figure 1. Tilt-acquisition cost versus asymptote ecliptic latitude. Filled points: our measured 3-D carried totals minus the matched planar pump (Table 1). Open points: the Fermi team's implied per-epoch tilt costs from their published table. The quadratic fit $0.10\,|\beta|^{1.92}\,\mathrm{km\,s^{-1}}$ (measured) passes through both data sets. The linear model and the quadrature floor, both fixed a priori, are rejected at ratios 2.8 and 25 (measured).

$\times 1.0/\times 0.5/\times 0.25$ time-step ladder. The 3-D totals are best-found values from a seeded stochastic search with per-row scatter of $\pm 0.2\,\mathrm{km\,s^{-1}}$; the differences in Table 1 are the quantity the model uses.

A power-law fit to the measured increments gives $0.10\,|\beta|^{1.92}\,\mathrm{km\,s^{-1}}$ with β in degrees (measured; amplitude 0.1007, exponent 1.92, quadratic within the scatter). Two alternative models fixed a priori are rejected by the same data: a linear model anchored at the 2.48° point overprices the 0.78° epoch by a factor of 2.8, and the geometric quadrature floor $v_\infty(1/\cos\beta - 1)$ underprices the 2.48° point by a factor of 25 (measured; Figure 1). The quadratic law was asserted from the single heritage anchor before the multi-epoch measurement and was not adjusted afterward.

3.2 The epoch curve and the 1 yr agreement

Our tilt-inclusive model is the shaved kinematic $v_\infty(T)$ plus the quadratic anchor term $0.58\,(\beta(T)/2.48°)^2\,\mathrm{km\,s^{-1}}$, with the $0.58\,\mathrm{km\,s^{-1}}$ anchor taken from the heritage 3-D measurement before this analysis began. Its minimum is 77,499 yr (measured). The Fermi team's direct simulation, an unrelated implementation with its own thermal power curve, steering law, and step control, bottoms at 77,500 yr on a 500 yr grid with a basin flat to $7\,\mathrm{m\,s^{-1}}$

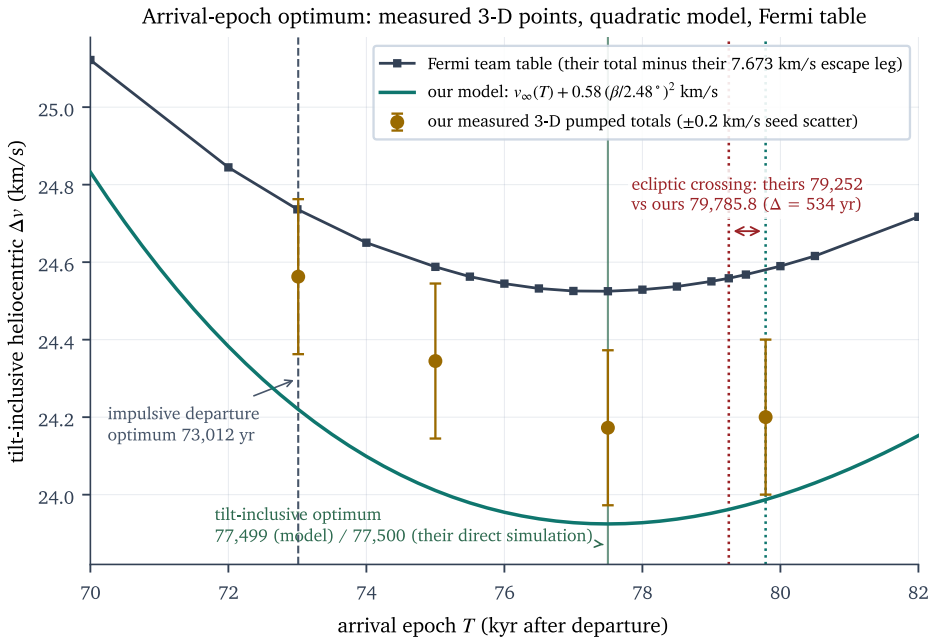


Figure 2. Arrival-epoch curves. Dark curve: the Fermi team's published tilt-inclusive totals with their 7.673 $\mathrm{km\,s^{-1}}$ escape leg removed. Teal curve: our model, $v_\infty^{\mathrm{shaved}}(T) + 0.58\,(\beta/2.48°)^2\,\mathrm{km\,s^{-1}}$ (measured components). Points with bars: our measured 3-D pumped totals with the ±0.2 $\mathrm{km\,s^{-1}}$ best-found seed scatter. The two curves sit at different absolute levels because their direct simulation prices the full campaign under its own conventions while our model prices kinematics plus the tilt anchor; the adjudicated quantity is each curve's basin location, which the level offset does not move (their record states the convention deltas are epoch-smooth). The two minima, 77,499 yr (ours, measured) and 77,500 yr (theirs), agree to 1 yr, and both sit 16 $\mathrm{m\,s^{-1}}$ above the measured basin bottom of Figure 3. Vertical lines mark the impulsive optimum at 73,012 yr and the two crossing epochs whose 534 yr gap Section 4 decomposes; Figure 4 resolves that gap bar by bar.

over 76,500 to 78,000 yr (their table; verified against our model to 1 yr). Figure 2 overlays the two curves and our four measured 3-D points. No parameter in either implementation was tuned toward the other. Both minima are model results; the direct 15-epoch measurement of Section 3.3 locates the physical basin around them. Their own cross-checks strengthen the agreement. Their independent 3-D re-integration of the tilt cost, written from a docstring specification with its own stepping, gives 610 $\mathrm{m\,s^{-1}}$ at the 2.48° point against their engine's 606 $\mathrm{m\,s^{-1}}$ and our measured 578 $\mathrm{m\,s^{-1}}$: three implementations inside a 5.5% spread (verified from their audit record and ours). Their step-size discipline is equally concrete: a dt/8 re-integration of three checked rows moves the totals by at most 4.1 $\mathrm{m\,s^{-1}}$ (75,000 yr: −3.2; 77,500 yr: −3.5; crossing: −4.1 $\mathrm{m\,s^{-1}}$) against their 40 $\mathrm{m\,s^{-1}}$ audit gate (their record; verified against their published cross-check list). Their per-row acceptance discipline is explicit as well: every epoch row must reach the target v_∞, land the asymptote latitude within 0.06° of $-\beta$, and keep stellar custody at or below 15 yr (their record). Their adversarial audit history is documented and credible: they found and corrected a one-sided miss-allowance convention that overpriced tilted epochs by up to 100 $\mathrm{m\,s^{-1}}$, and they retracted an earlier corroboration claim that rested on a seed-scatter outlier in our planar table.

3.3 The measured basin: swept, adjudicated, flat

The model agreement above is necessary evidence; the definitive evidence is a direct empirical re-trace of the tilt-inclusive 3-D pumped basin. The sweep covers epochs 74,000 to 81,000 yr in 500 yr steps with five seeds per epoch (seed base 20260804, explicit per-run offsets) and imports the bounded 3-D search recipe verbatim

Table 2. Basin-sweep anchor gates (verified; from the completed sweep artifact, re-checked by the adjudication run). The collapse gate is 0.05 $\mathrm{km\,s^{-1}}$, declared in-script against the prior measured 0.0149 $\mathrm{km\,s^{-1}}$; it is a declared loose gate with its prior value referenced.

Anchor	Quantity	Value ($\mathrm{km\,s^{-1}}$)	Result
A	Heritage planar carried Δv	23.984510138158	pass (bit-level)
B	Heritage certified 3-D carried Δv	24.562696	pass (bit-level)
C	77,500 yr winner re-ladder	24.172672236	pass (exact vs prior artifact)
D	Crossing-winner planar collapse	+0.014893	pass (gate 0.05, declared)

from the certified heritage code. All four anchor gates pass (Table 2). The sweep completed in 1,715 s against its 2,100 s budget. All 75 cells are feasible at 5/5 seeds (measured).

Two cells show the single-seed outlier signature seen before in this project: seed 20260918 at 79,500 yr found a 23.830 $\mathrm{km\,s^{-1}}$ schedule 0.33 $\mathrm{km\,s^{-1}}$ below both neighbors, and seed 20260938 at 80,500 yr found 24.114 $\mathrm{km\,s^{-1}}$, 0.07 to 0.12 $\mathrm{km\,s^{-1}}$ below both neighbors, each from a remote corner of the search box. An adjudication run re-ran the 79,500 cell with five fresh seeds and the 80,500 cell with three, spot re-ran 78,500 and 79,000, and re-laddered both outlier winners on an extended dt ladder down to ×0.0625. The verdict is clean (measured). Zero of eight fresh seeds reproduce either mode; all eight converge to the family values, 24.186090 and 24.228154 $\mathrm{km\,s^{-1}}$. Both spot re-runs reproduce their sweep values exactly. Both outlier winners pass all gates on the extended ladder; the 79,500 winner drifts +12.5 $\mathrm{m\,s^{-1}}$ at the finer rungs and the 80,500 winner drifts 0.0 $\mathrm{m\,s^{-1}}$. Each outlier run is therefore a valid upper bound at its own epoch, found by one seed in ten (79,500) or one in eight (80,500), and both are excluded from the envelope because a mode that rare distorts cross-epoch comparison under a fixed search budget. The exclusion is conservative for the design conclusion: a rare deeper schedule family near 79,500 to 80,500 yr moves the basin bottom toward the crossing epochs.

The adjudicated envelope (Figure 3) bottoms at 24.156 $\mathrm{km\,s^{-1}}$ at 79,000 yr with zero within-cell seed scatter at all 15 cells (measured; every value a best-found upper bound). Flatness: 78,500 to 79,000 yr lie within 10 $\mathrm{m\,s^{-1}}$ of the bottom, 77,500 to 79,000 within 25 $\mathrm{m\,s^{-1}}$, and 77,000 to 80,000 within 50 $\mathrm{m\,s^{-1}}$. The basin is too flat to localize its bottom beyond ±1,000 yr: the 25 $\mathrm{m\,s^{-1}}$ band spans 1,500 yr, and the quadratic fits put the vertex at 78,747 yr (five cells around the bottom) or 78,589 yr (full window) with rms residuals of 7.7 and 7.3 $\mathrm{m\,s^{-1}}$. That flatness is the finding. The 77,499 yr model fit and Fermi's 77,500 yr direct optimum each sit 16.3 $\mathrm{m\,s^{-1}}$ above the measured bottom, inside the 25 $\mathrm{m\,s^{-1}}$ band, so the verified 1 yr model agreement and the measured basin tell one consistent story; Fermi's own table holds 76,500 to 78,000 yr within 7 $\mathrm{m\,s^{-1}}$ of their bottom, which matches a basin this flat. The crossing design epochs cost 22.1 $\mathrm{m\,s^{-1}}$ (Akeson-based working state, Table 5, 79,372 yr) and 27.1 $\mathrm{m\,s^{-1}}$ (Kervella, 79,786 yr) above the measured bottom, both inside the 50 $\mathrm{m\,s^{-1}}$ band. Choosing the crossing as the design epoch gives up under 30 $\mathrm{m\,s^{-1}}$ of Δv, 0.11% of the 24.16 $\mathrm{km\,s^{-1}}$ total, so the crossing design epoch costs almost nothing and the measurement strengthens the recommendation of Section 7. The joint Monte Carlo of Section 6 carries a surrogate-roughness term $\sigma_{\mathrm{surr}} = 0.0091\,\mathrm{km\,s^{-1}}$; the measured quadratic-detrend rms of the adjudicated envelope is 0.0073 $\mathrm{km\,s^{-1}}$, so the surrogate over-covers the measured roughness and stands as conservative (measured).

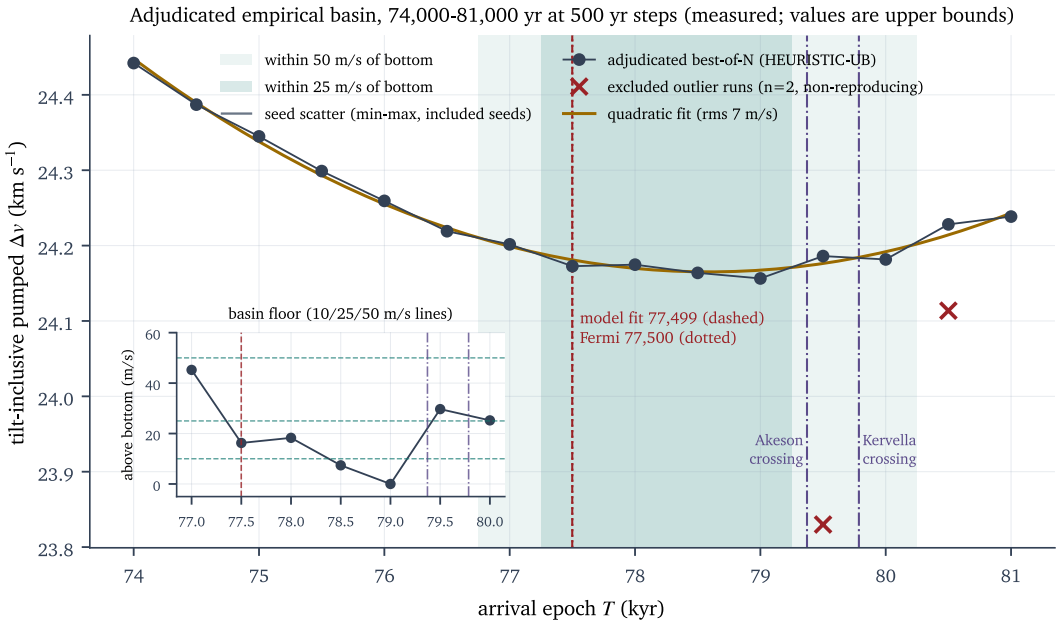


Figure 3. Adjudicated empirical basin (measured; every point a best-of-N upper-bound envelope value). Points: adjudicated best per epoch with min-max seed whiskers (zero spread at every cell); red crosses: the two excluded single-seed outlier runs, with zero of eight fresh seeds reproducing them; amber curve: full-window quadratic fit (rms 7 $\mathrm{m\,s^{-1}}$); shaded spans: epochs within 25 and 50 $\mathrm{m\,s^{-1}}$ of the bottom; dashed and dotted red lines: the model fit 77,499 and Fermi 77,500 yr, each 16 $\mathrm{m\,s^{-1}}$ above the measured bottom; dash-dot purple lines: the Akeson working-state (79,372) and Kervella (79,786) crossing design epochs, 22 and 27 $\mathrm{m\,s^{-1}}$ above the bottom. Inset: depth above the bottom in $\mathrm{m\,s^{-1}}$ with the 10/25/50 $\mathrm{m\,s^{-1}}$ thresholds marked.

4. The crossing-epoch reconciliation

Both teams define the ecliptic-crossing epoch as the arrival time T at which the required departure asymptote has zero ecliptic latitude. Ours is 79,785.8 yr, held by three independent methods within 0.05 yr (verified). Theirs is 79,252 yr. The 534 yr gap is the one numerical disagreement between the two analyses, and this section closes it.

4.1 Their implementation, dissected

We read the Fermi team's public source: `astro.py`, `constants.py`, `intercept.py`, and `pump_schedule.py` in `fermi_sim`, plus `tools/sim_pp_arrival.py`. Their crossing condition is $T = -z_0/v_z$ applied to the ecliptic-frame straight-line stellar state. Ours is the root of $\beta(T) = 0$. For a linear state these are algebraically identical, and we confirmed the identity numerically to 0.01 yr (verified). Their code is clean: the rotation to ecliptic coordinates, the proper-motion scaling, and the radial-velocity projection all match textbook form. The two implementations differ only in their input astrometry and in trailing digits of shared constants.

4.2 Reproduction: their number, from their inputs, in our code

We reran a clean reimplementation on both input sets under both convention sets (obliquity digits, light-year value, proper-motion conversion digits). The full 2×2 matrix is Table 3. The reproduction is a full success with no residual to characterize: their inputs give 79,252.25 yr under either convention set, inside the integer rounding of their published 79,252 yr (verified). Our inputs give 79,785.75 yr under either convention set, matching our reference to 0.05 yr (verified). No further information from the Fermi team is needed.

Table 3. Reproduction matrix (verified; clean reimplementation, rerun 2026-08-03). Crossing epoch in years and raw unshaved v_∞ at 73,000 yr in $\mathrm{km\,s^{-1}}$, for each combination of input set and convention set. Conventions move the crossing by 0.01 yr and v_∞ by under 0.0001 $\mathrm{km\,s^{-1}}$; inputs move them by 533.5 yr and 0.094 $\mathrm{km\,s^{-1}}$.

Combination	T_{cross} (yr)	Raw v_∞ at 73 kyr ($\mathrm{km\,s^{-1}}$)
Fermi inputs, Fermi conventions	79,252.25	23.7145
Fermi inputs, our conventions	79,252.24	23.7145
Our inputs, Fermi conventions	79,785.75	23.8087
Our inputs, our conventions	79,785.75	23.8087

Table 4. One-at-a-time attribution of the 533.5 yr input effect, swapping each Fermi input to ours (measured). Distance carries 99% of the gap. The proper-motion terms partially cancel. The radial velocity is identical in both states, the declinations differ in the fourth decimal, and both rows contribute zero at the displayed precision. The single-swap sum leaves a +0.1 yr interaction residual against the joint effect, and an independent fresh-script re-derivation in the opposite swap direction returns −533.6 yr.

Input	Fermi value	Our value	Shift (yr)
Distance	4.344 ly	4.36522 ly (747.17 mas)	+528.8
μ_δ	694.0 mas yr^{-1}	693.8 mas yr^{-1}	+6.0
$\mu_{\alpha*}$	−3620.0 mas yr^{-1}	−3619.9 mas yr^{-1}	−1.4
Right ascension	219.9021°	219.90206°	+0.03
Declination	−60.8340°	−60.83399°	0.00
Radial velocity	identical	identical	0.00
Sum of one-at-a-time swaps			+533.4
Interaction residual			+0.1
Joint input effect (Table 3)			+533.5
Reverse-direction check (our state to theirs)			−533.6

4.3 Decomposition: input effect, convention effect

The published gap is $79{,}785.75 - 79{,}252.00 = 533.75$ yr. Of this, 0.25 yr is their integer rounding; the input effect is +533.50 yr at their conventions (+533.51 at ours) and the convention effect is −0.01 yr at their inputs (0.00 at ours), so each decomposition path closes the printed bridge exactly (measured; Table 3). A one-at-a-time swap of each input from their value to ours attributes the input effect in Table 4, and Figure 4 draws the same decomposition as a cumulative bridge from their reproduced value to ours (measured).

4.4 The catalog question: a 5-sigma parallax tension

Neither state is wrong on its face; the two states draw on different catalogs, and Table 5 sets them side by side. Their 4.344 ly distance equals the Akeson 2021 parallax (750.81 ± 0.38 mas), the more recent and more precise measurement, while their proper motions are Kervella-era. Ours is Kervella 2016 (747.17 ± 0.61 mas) end to end. Their state mixes catalogs; ours is internally consistent but older.

The mix matters because the two catalogs disagree. The parallaxes differ by 3.64 mas against a combined one-sigma of 0.72 mas, a 5.1-sigma tension (computed from the published parallax sigmas in Table 5). The main feasibility report carries this same Kervella–Akeson tension as a labeled systematic in its dispersion analysis, and this memorandum inherits that treatment. Epoch consistency is a second, independent lever. Attaching Akeson's J2019.5 observables to a J2000-family position and velocity basis moves the crossing earlier by 31.6 yr on the state clock (24.5 yr of velocity-basis rotation plus 7.1 yr of position advance) and by 51.1 yr on the absolute date under the J2000-origin reading (measured). The epoch-consistent Akeson state at J2019.5 gives $T_{\mathrm{cross}} = 79{,}467.0$ yr from J2019.5 in the spectroscopic V_0 frame (measured); the working state of Table 5 sits on the J2000-family basis with the adopted proper motions, so its 79,371.9 yr differs by the epoch basis and the observable deltas together; the frame itself is the next subsection's subject.

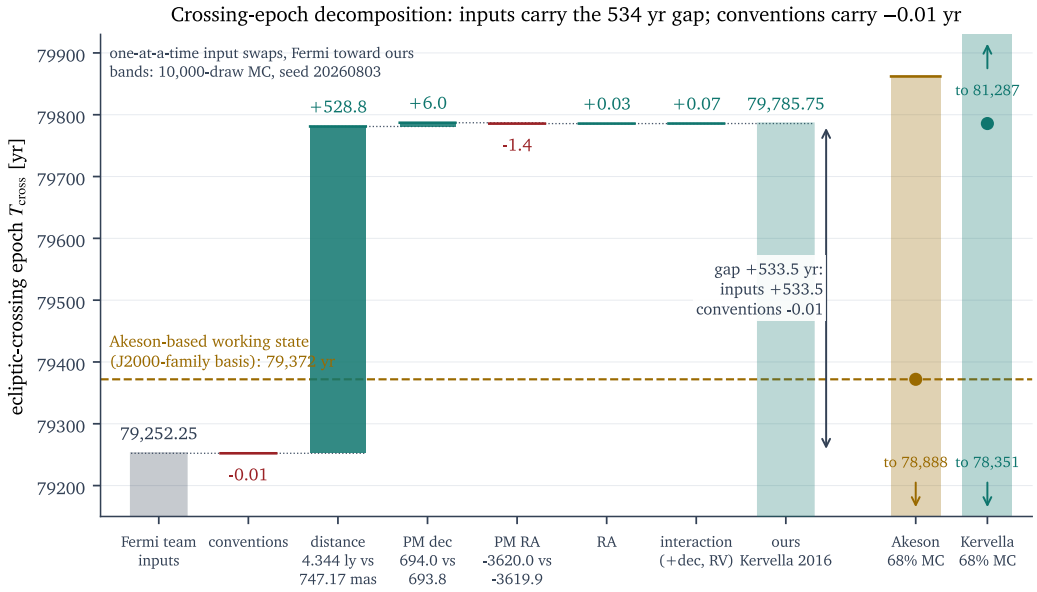


Figure 4. Waterfall decomposition of the crossing-epoch gap (measured). The bridge starts at 79,252.25 yr, our exact reproduction of the Fermi state, published by them as 79,252 yr under integer rounding, and ends at our 79,785.75 yr. Bars in order: conventions −0.01 yr, distance +528.8 yr, μ_δ +6.0 yr, $\mu_{\alpha*}$ −1.4 yr, right ascension +0.03 yr, interaction residual +0.07 yr; bar labels round independently, and the drawn bars close the bridge exactly at 79,785.75 yr. Distance carries 99% of the gap. The dashed line marks the Akeson-based working state at 79,371.9 yr (J2000-family basis; the catalog-epoch J2019.5 state gives 79,467.0 yr, Section 4.4). The shaded columns at right give the 68% Monte Carlo intervals under each catalog: [78,888, 79,862] yr for Akeson 2021 and [78,351, 81,287] yr for Kervella 2016, with arrows marking extents beyond the plotted range.

4.5 The radial-velocity convention audit

The crossing epoch moves 4,864 yr per $\mathrm{km\,s^{-1}}$ of radial velocity (measured), so the radial-velocity convention is a candidate systematic larger than most inputs. The published catalog values are spectroscopic: gravitational redshift and convective blueshift stay uncorrected. Trajectory work needs the kinematic value, the true line-of-sight motion. We audited the carried value against the primary sources. The verdict is definite (verified against re-fetched source text): the carried $-22.4\,\mathrm{km\,s^{-1}}$ is the spectroscopic Kervella 2016 orbit gamma ($-22.393\,\mathrm{km\,s^{-1}}$; fit to all available RV measurements, with CES and HARPS shown in their residuals), and Akeson 2021's $V_0 = -22.3796\,\mathrm{km\,s^{-1}}$ is the same convention. The kinematic correction is smaller than the naive sum of the published $+575.3\,\mathrm{m\,s^{-1}}$ (~$0.58\,\mathrm{km\,s^{-1}}$) gravitational redshift and a $-0.3\,\mathrm{km\,s^{-1}}$ convective blueshift. Alpha Centauri A is a solar twin measured against a solar-referenced zero point, so the correction collapses to removing the differential gravitational redshift, published as $\Delta v_{\mathrm{GRS}} = -61.4 \pm 2.7\,\mathrm{m\,s^{-1}}$ (Kervella, Thévenin & Lovis 2017, A&A 598, L7); the applied zero-point correction is $+61.4\,\mathrm{m\,s^{-1}}$, a hidden one-sided systematic of $+0.061\,\mathrm{km\,s^{-1}}$. The kinematic Kervella value is $-22.332\,\mathrm{km\,s^{-1}}$. Table 6 prices every scenario; Figure 6a draws the ladder.

Three consequences follow (measured). First, the kinematic-frame central crossing is 332 yr later on the Kervella base and 298.9 yr later on the epoch-consistent Akeson base (349 on the working state), 0.23 of the $0.30\,\mathrm{km\,s^{-1}}$ carried working sigma (adopted allowance) and 0.6 of the 0.10 allowance carried with the Akeson-based state, so no published band is invalidated and the central value carries a known one-sided systematic. Second, the correction cannot close the ours-minus-Fermi gap: if both teams use the same catalog RV the relative gap is convention-invariant, and the correction moves both absolute values in the same direction. Third, our own constants module (not the Fermi repository) carries an inconsistency worth one comment line: it holds the kinematic $-22.204\,\mathrm{km\,s^{-1}}$ for Proxima (Kervella, Thévenin & Lovis 2017, A&A 598, L7) beside the

Table 5. Input comparison: the Fermi published state, the two catalogs (Kervella 2016, A&A 594, A107, catalog epoch J1991.25; Akeson 2021, AJ 162, 14, native epoch J2019.5), and the Akeson-based working state that the dispersion and joint Monte Carlo machinery runs on. In the Kervella 2016 column the parallax, proper motions, and gamma are verbatim catalog values; the position (219.90206, −60.83399) is a project constant on a J2000-family basis and is not a K16 catalog value (the K16 catalog barycenter is published at J1991.25). In the working-state column the proper motions $(-3644.4, +702.8)$ and their $\pm 1.5\,\mathrm{mas\,yr^{-1}}$ sigmas are adopted working values that appear in no catalog (Akeson's published sigmas are 0.42/0.17); the $-22.39\,\mathrm{km\,s^{-1}}$ is the K16/PB16 gamma rounded to two decimals (Akeson's own V_0 rounds to −22.38); the $0.30/0.10\,\mathrm{km\,s^{-1}}$ sigmas are working sigmas, adopted absolute-accuracy allowances that appear in neither catalog. T_{cross} for the Akeson column counts from its J2019.5 state (spectroscopic V_0; kinematic 79,765.9); the other columns count from their own state clocks (measured).

Quantity	Fermi state	Kervella 2016	Akeson 2021 (J2019.5)	Working state (adopted)
Parallax (mas)	(via distance)	747.17 ± 0.61	750.81 ± 0.38	750.81 ± 0.38
Distance (ly)	4.344	4.36522	4.344	4.344
$\mu_{\alpha*}$ ($\mathrm{mas\,yr^{-1}}$)	−3620.0	−3619.9 ± 3.9	−3639.95 ± 0.42	−3644.4 (adopted)
μ_{δ} ($\mathrm{mas\,yr^{-1}}$)	+694.0	+693.8 ± 3.9	+700.40 ± 0.17	+702.8 (adopted)
Radial velocity ($\mathrm{km\,s^{-1}}$)	−22.4	−22.3930 ± 0.0043	−22.3796 ± 0.0020	−22.39 (allowance 0.10)
Right ascension (deg)	219.9021	219.90206	219.8589	219.90206
Declination (deg)	−60.834	−60.83399	−60.8316	−60.83399
T_{cross} from state (yr)	79,252.25	79,785.75	79,467.0	79,371.9

Table 6. RV-convention scenarios and their crossing epochs (measured; exact crossing solve. Linear cross-checks agree within 2 yr on the single-state RV-swap rows; the bounding-case row departs from linearity by 19 yr, and the two J2019.5-state rows' ΔT combine the state change with the RV change). The first block runs on the K16 astrometric base, the second on the A21 base; the working-state row runs on the Akeson-based working state of Table 5 at its adopted precision −22.39, while the epoch-consistent rows run on the Akeson J2019.5 state with T from J2019.5. The four catalog-convention scenarios span 79,467 to 80,118 yr. The final row is a bounding case only, contradicted by the Kervella 2017 frame analysis.

Scenario	Convention	RV ($\mathrm{km\,s^{-1}}$)	T_{cross} (yr)	ΔT vs block baseline (yr)
Carried K16 (constants)	spectroscopic	−22.400	79,785.75	0.0
K16 published gamma	spectroscopic	−22.393	79,819.81	+34.1
Pourbaix–Boffin 2016	spectroscopic	−22.390	79,834.42	+48.7
K17 kinematic	kinematic	−22.332	80,117.90	+332.2
A21 working state (constants)	spectroscopic	−22.39	79,371.93	0.0
A21 published V_0 (J2019.5 state)	spectroscopic	−22.3796	79,467.02	+95.1
A21 kinematic (J2019.5 state)	kinematic	−22.3182	79,765.87	+393.9
Naive uncorrected bound	(unsupported)	−22.657	78,554.90	−1,230.8

spectroscopic $-22.40\,\mathrm{km\,s^{-1}}$ for AB, two conventions in one file. The Fermi constants file carries no Proxima entry and no stellar radial velocity; their AB value ($-22.4\,\mathrm{km\,s^{-1}}$, in `astro.py`) is spectroscopic throughout. If the zero-point argument were rejected wholesale, the bounding range on the correction is $[-0.27, +0.07]\,\mathrm{km\,s^{-1}}$, which maps to $[-1.3, +0.34]$ kyr, with the literature-supported value at $+0.061\,\mathrm{km\,s^{-1}}$ (bounds; the −1.3 kyr endpoint is contradicted by Kervella 2017).

Table 7. Sigma positions of the three crossing-epoch values under each working-sigma band (measured; linear one-sigma propagation, band centers at each state, spectroscopic frame). Every entry sits inside the 68% interval of both bands, so no pair of values is in statistical tension. The kinematic correction shifts the band centers later by 332 yr (Kervella) and 349 yr (the Akeson-based working-state band drawn here; 299 yr applies on the epoch-consistent center) and does not change the relative positions.

Crossing-epoch value	Kervella band (79,786 ± 1,468 yr)	Akeson band (79,372 ± 489 yr)
Fermi published, 79,252 yr	-0.36σ	-0.25σ
Akeson working state, 79,372 yr	-0.28σ	0.00σ (center)
Ours (Kervella), 79,786 yr	0.00σ (center)	$+0.85\sigma$

4.6 Sensitivity ranking and the astrometric band

Linear sensitivities of the crossing epoch (measured, central differences on the Kervella state): radial velocity +4,864 yr per $\mathrm{km\,s^{-1}}$, parallax −146 yr per mas, μ_δ −31 yr per mas yr^{-1}, $\mu_{\alpha*}$ −14 yr per mas yr^{-1}. Weighted by the carried working sigmas (adopted absolute-accuracy allowances; the published catalog sigmas are smaller), radial velocity dominates: the 0.30 $\mathrm{km\,s^{-1}}$ allowance carried with the Kervella-based state alone contributes 1,459 yr (Figure 5a).

Propagating the full diagonal covariance (assumed diagonal; the constants file carries no cross-terms) gives the crossing-epoch bands (measured; 10,000-draw Monte Carlo, seed 20260803, linear check within 0.5%):

- **Kervella 2016:** $79{,}786 \pm 1{,}468$ yr linear, Monte Carlo sigma 1,462 yr, 68% interval [78,351, 81,287] yr, 95% interval [77,029, 82,755] yr. The Fermi value sits at -0.36σ, inside the 68% interval.
- **Akeson 2021:** $79{,}372 \pm 489$ yr, 68% interval [78,888, 79,862] yr, 95% interval [78,434, 80,332] yr. The Fermi value sits at -0.25σ and ours at $+0.85\sigma$, both inside the 68% interval. This is a working-sigma band on the working-state center: the 0.10 $\mathrm{km\,s^{-1}}$ RV allowance dominates the width, so the width is insensitive to the adopted ± 1.5 mas yr^{-1} sigmas (on the Akeson-centered sensitivities the published 0.42/0.17 move the linear sigma from 489 to 487 yr), and the epoch-consistent center is 95 yr later (computed from the measured sensitivities).

Both bands are computed in the spectroscopic frame that both catalogs publish. The kinematic correction of Section 4.5 shifts each band center later by 332 yr (Kervella) or 299 yr (epoch-consistent Akeson base; 349 on the working state) and leaves the widths unchanged: the 2.7 $\mathrm{m\,s^{-1}}$ correction uncertainty contributes 13 yr, negligible against either sigma (computed from the measured sensitivities). The sigma positions in Table 7 therefore hold in either frame up to a common shift.

The 534 yr gap is therefore not statistically significant under either catalog (measured; Figure 5b). Table 7 places all three crossing-epoch values inside both bands: every entry sits inside its 68% interval. Closing the gap entirely within a single Kervella parameter would need a 0.37σ shift in radial velocity, a routine excursion; the next-cheapest single-parameter closure costs 4.5σ in μ_δ (measured).

The radial-velocity payoff quoted in Section 7 traces to these sensitivities: a 0.03 $\mathrm{km\,s^{-1}}$ radial-velocity sigma contributes $0.03 \times 4{,}864 = 146$ yr, the non-radial-velocity Akeson terms add 74 yr in quadrature, and the combined band is 164 yr, quoted conservatively as ±200 yr (computed from the measured linear sensitivities).

4.7 The v_∞ offset: same cause, same verdict

The Fermi table runs 0.04 $\mathrm{km\,s^{-1}}$ below ours at matched epochs (23.60 against 23.64 $\mathrm{km\,s^{-1}}$ at 73,000 yr). The raw unshaved v_∞ at 73,000 yr is 23.7145 $\mathrm{km\,s^{-1}}$ on their inputs and 23.8087 $\mathrm{km\,s^{-1}}$ on ours, and the value is convention-independent to 0.0001 $\mathrm{km\,s^{-1}}$ (measured; Table 3). The one-at-a-time attribution assigns -0.0951 $\mathrm{km\,s^{-1}}$ to distance, $+0.0006$ $\mathrm{km\,s^{-1}}$ to $\mu_{\alpha*}$, and $+0.0002$ $\mathrm{km\,s^{-1}}$ to μ_δ (measured); distance drives more than the whole gap and the proper-motion terms partially offset it. The raw 0.094 $\mathrm{km\,s^{-1}}$ gap compresses to the 0.04 $\mathrm{km\,s^{-1}}$ table-level gap after each side applies its own miss-allowance shave. Neither implementation is wrong; the offset is the same distance difference seen through the velocity projection.

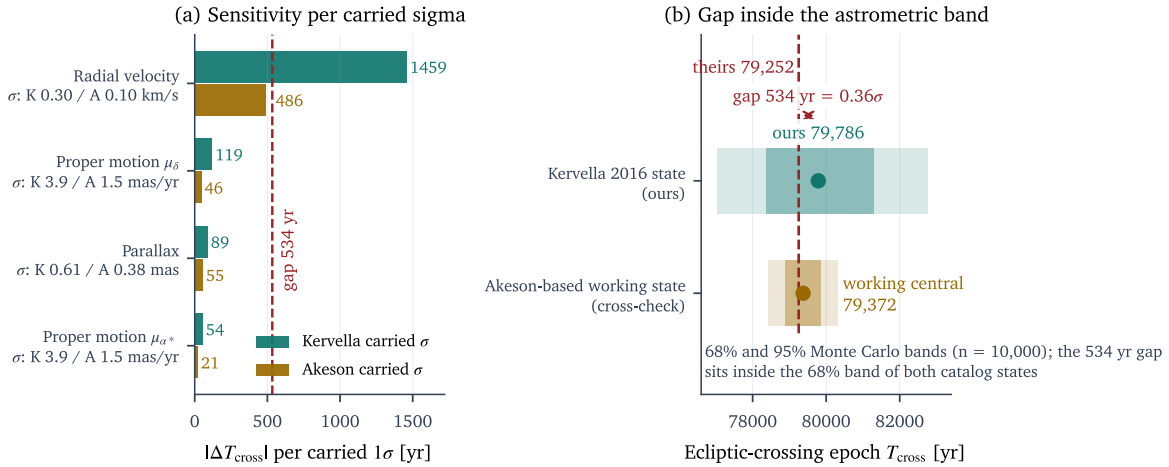


Figure 5. The crossing-epoch gap in its astrometric context. (a) Sensitivity of $T_{\rm cross}$ per one-sigma shift of the carried working sigmas under both catalog states (measured; the RV sigmas are the adopted 0.30/0.10 $\mathrm{km\,s^{-1}}$ allowances, and the Akeson μ bars use the adopted ±1.5 mas $\mathrm{yr^{-1}}$; the published $0.42/0.17$ mas $\mathrm{yr^{-1}}$ sigmas give bars of ≈6 and 5 yr); radial velocity dominates both. (b) The 68% and 95% Monte Carlo bands (10,000 draws, seed 20260803) for the Kervella 2016 and Akeson 2021 states, spectroscopic frame; the 534 yr gap between our 79,786 yr and the Fermi 79,252 yr sits inside the 68% band of each. The Akeson band drawn here is centered on the Akeson-based working state (79,372 yr; the epoch-consistent center is 95 yr later); the recommended public statement (Akeson J2019.5 state, kinematic frame, T from departure 2029.0, program band ±400–800 yr) is in Section 7.

5. Design-epoch consequences

5.1 Dispersion and closure at the crossing design epochs

The crossing recommendation deserves an end-to-end price, and the certified dispersion pipeline supplies it: 20,000 draws per case, identical error budget to the published baseline (pointing 0.10° per axis, magnitude 50 $\mathrm{m\,s^{-1}}$, residual 25 $\mathrm{m\,s^{-1}}$ per axis, all engineering assumptions; carried ephemeris sigmas (Table 5); 2,600 AU success radius). Table 8 gives the arrival probabilities and Table 9 the vehicle closure; Figure 6b draws the probability ladder.

The findings hold no surprises (simulation). The crossing design epochs cost about 1.6 points of arrival probability against the 73,012 yr baseline, and on the Akeson working state the crossing beats the baseline by +0.23 points. GTO-100 closes at both crossings and gains +0.69/+0.71 pts over the 73k tilt-inclusive case (24.56 $\mathrm{km\,s^{-1}}$). The LEO-150 verdict does not change: it still fails to close at either crossing. Its shortfall shrinks from −0.74 pts at 73,012 yr to −0.12/−0.10 pts, close enough that a 0.06 (Akeson working) to 0.08 $\mathrm{km\,s^{-1}}$ (Kervella) Δv saving would flip it, and we do not claim the flip. On the Akeson J2019.5 catalog state (same pipeline, closest-approach scoring, $n = 20{,}000$, seed 20260806), the central dispersion probability is 0.9858 in the kinematic frame and 0.9866 in the observed-γ variant; the working state gives 0.9846 at the same seed (Table 8's 0.98385 is the same working state at seed 20260805, a within-noise seed difference), and a cross-catalog diagnostic that aims on the working-state proper motions against the J2019.5 state prices that input difference at $P = 0.9660$ with median miss 1,193 AU (simulation).

Table 8. Arrival dispersion at the design epochs (simulation; $n = 20{,}000$ per row, seeds stated, Wilson 95% intervals). The baseline row reproduces the published seed-42 anchor exactly. The last row is an interpretation control: the P deficit follows the catalog (aim-state error), and the epoch itself is benign.

Case	T (yr)	Seed	$\hat{P}$	Wilson 95%	Median miss (AU)
Baseline, Kervella, 73k	73,012	42	0.98155	[0.9796, 0.9833]	1,076
Kervella crossing	79,786	20260804	0.96515	[0.9625, 0.9676]	1,173
Akeson crossing (working state)	79,372	20260805	0.98385	[0.9820, 0.9855]	1,059
Kervella catalog at Akeson epoch	79,372	20260806	0.96695	[0.9644, 0.9693]	1,171

Table 9. Vehicle closure at the crossing epochs (measured slopes, best-found Δv rows). Required Δv is a linear interpolation of the 12-yr pumped column between the $T = 79{,}000$ (24.156 $\mathrm{km\,s^{-1}}$) and $T = 80{,}000$ (24.187 $\mathrm{km\,s^{-1}}$) best-found rows, an upper bound, so margins are lower bounds; residual tilt at these epochs is $\leq 0.008\,\mathrm{km\,s^{-1}}$ by the measured power law. Margins extrapolate the bracket-end margins (+6.15 pts at 22.9 $\mathrm{km\,s^{-1}}$, +4.15 at 24.0 for GTO-100; +1.96/+0.17 for LEO-150) at the measured slopes −1.818 and −1.627 pts per $\mathrm{km\,s^{-1}}$.

Architecture	Kervella 79,786		Akeson working 79,372		Verdict
	Δv ($\mathrm{km\,s^{-1}}$)	Margin (pts)	Δv ($\mathrm{km\,s^{-1}}$)	Margin (pts)	
GTO-100	24.180	+3.82	24.168	+3.85	closes (both)
LEO-spiral-150	24.180	−0.12	24.168	−0.10	does not close

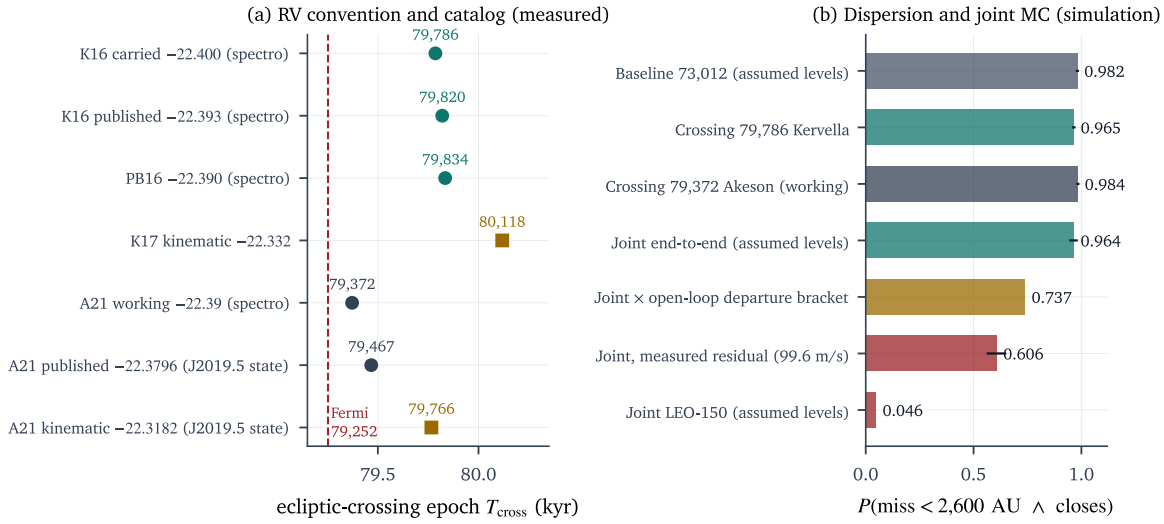


Figure 6. Measurement-program summary panel, drawn from program artifacts under anchor gates. (a) Crossing epoch under each RV convention and catalog (Table 6; measured); squares mark kinematic-frame values, the dashed line the Fermi published value; the two J2019.5-state rows carry the catalog-state values, and the four catalog-convention scenarios span 79,467 to 80,118 yr. (b) Probability ladder (simulation): dispersion at the three design epochs with Wilson 95% intervals (the Akeson bar is the working state; the J2019.5-state values 0.9858/0.9866 are in Section 6), then the joint end-to-end Monte Carlo (Section 6); whiskers on joint bars are the 2.5–97.5% range of the per-draw dispersion probability, and the open-loop departure bracket and measured-residual rows are conservative upper-bound stresses.

5.2 Powered-phase execution: measured versus assumed residual

The dispersion pipeline assumes a $25\,\mathrm{m\,s^{-1}}$-per-axis departure residual (engineering assumption). This item measures what the powered phase actually leaks. A perturbed clone of the certified 3-D propagator flies the planar winner open-loop under a fractional thrust-magnitude bias $\varepsilon \sim N(0, 2.1\times10^{-3})$ (mapped from the assumed $50\,\mathrm{m\,s^{-1}}$ magnitude channel) plus pointing noise, with the heritage energy-trigger cutoff and perfect energy knowledge. The unperturbed control reproduces the carried $23.984510138157503\,\mathrm{km\,s^{-1}}$ bitwise (verified). Table 10 summarizes.

The verdict is quantitative (measured). The assumed $25\,\mathrm{m\,s^{-1}}$-per-axis residual is optimistic by a factor of 4 against the measured worst-component $99.6\,\mathrm{m\,s^{-1}}$ (e1), and at the assumed thrust-bias level 19 to 24% of open-loop runs miss the energy cutoff entirely. Both numbers are upper bounds on a real mission: this is open-loop execution of a fixed policy, a real mission observes a thrust-scale bias within days and re-optimizes, and the $50\,\mathrm{m\,s^{-1}}$ velocity-knowledge channel after telemetry-closed cutoff is not probed here. Section 6 carries both the assumed level and the measured level as explicit variants and chooses neither.

5.3 Linearity of the kinematic model: bounded

The entire crossing-epoch analysis assumes constant-velocity stellar motion. Closed forms bound the neglected physical terms (Table 11), including Proxima Centauri's gravitational pull on the AB barycenter, a term the Fermi team identified, confirmed here by full Kepler-arc integration: 105.6 AU of track deflection at the crossing ($5.4\times$ the largest prior position row; cross-track 79.3 AU, 3.1% of the 2,600 AU success radius), worth +7.1 yr on the crossing, inside the Fermi team's own 2 to 12 yr estimate, and $0.0030\,\mathrm{km\,s^{-1}}$ on v_∞. The model: masses $M_{AB} = 1.988$ and $M_P = 0.1221\,M_\odot$, current separation $r_0 = 12{,}057$ AU from Gaia DR3 Proxima astrometry (Gaia Collaboration 2023, A&A 674, A1) with the Akeson parallaxes (more current than the $r \approx 12{,}947$ AU of Kervella, Thévenin & Lovis 2017), propagated as a bound Kepler arc ($a \approx 7{,}573$ AU, period ≈ 454 kyr), evaluated at the 79,372 yr working-state crossing (the same artifact gives 107.3 AU at 80,000 yr, so the other crossing epochs move the term by under 2 AU). Evaluated at that $r = 12{,}947$ AU, the constant-acceleration variant of our model returns 90.6 AU (the Kepler arc gives 105.6 at our r_0), consistent with their ≈92 AU up to the input difference; the term is deterministic and removable by adding Proxima to the propagation. The combined worst case moves the crossing by 56.4 yr (0.071% of T_{cross}) and v_∞ by $0.0168\,\mathrm{km\,s^{-1}}$, far below the Kervella–Akeson catalog spread (414 yr between the Kervella and Akeson working states; 319 yr on the epoch-consistent Akeson state) and the 1,459 yr one-sigma radial-velocity lever. The linearity claim holds (measured closed-form bounds).

6. Joint end-to-end Monte Carlo

The joint model couples, in one Monte Carlo, everything the preceding sections treated separately: catalog and aim uncertainty (outer loop, 10,000 draws per scenario, seeds 20260806+idx), execution dispersion (inner loop, 1,000 draws per outer draw, deterministic seed lattice), the pumped Δv table (best-found upper bounds plus the $\sigma_{\mathrm{surr}} = 0.0091\,\mathrm{km\,s^{-1}}$ roughness proxy of Section 3.3), and the closure-margin model of Section 5.1. The evidentiary label carries verbatim from the artifact: simulation, engineering-assumption execution levels, heuristic upper-bound pump surrogate (margins are lower bounds), joint P conditioned on all stated assumptions. Table 12 reports the four astrometric scenarios.

The findings rank cleanly (simulation). The headline joint probability at the Kervella crossing is $P = 0.964$ at the assumed execution levels; the ephemeris-off control gives 0.986, so ephemeris uncertainty costs 2.2 points and execution the rest. Radial velocity dominates everything upstream: 99.1% of crossing-epoch variance and 64.5% of margin variance (one-at-a-time normalization), with the basin surrogate at 2.9% of margin variance. Stressing with the measured powered-phase residual drops P to 0.606, and multiplying by the open-loop departure bracket $(1 - 0.2355)$ gives 0.737 (assumed) and 0.463 (stressed), both conservative brackets, since the failure mode is a thrust-scale bias a real mission would observe and re-optimize away. The RV convention of Section 4.5 moves the central epoch 299 yr (epoch-consistent Akeson) to 332 yr (Kervella; 349 yr on the working-state pair of Table 12) and changes joint P by less than 0.001: the design is insensitive to the convention. LEO-150's joint probability never exceeds 0.05 in any scenario of Table 12, consistent with its unchanged closure verdict. No probability in this memorandum is scored at a fixed instant: both the dispersion pipeline and the joint model score closest approach (verified in source, with bit-exact draw replication).

Table 10. Powered-phase execution Monte Carlo (measured; seeds 20260804–20260807 by experiment). "Not reached" counts runs failing the energy target by the 12.0 yr wall (nominal burn 11.78 yr); survivor statistics are selection-biased and the deterministic ε sweep brackets the cliff between $+0.0010$ (reached) and $+0.0020$ (not reached, terminal energy 253 vs 280 $\mathrm{km^2\,s^{-2}}$). e4 is the coarse-step control; its magnitude column is cutoff-quantization leakage, so e1 is the quoted measurement.

Exp	Pointing channel	n	dt	Not reached	σ(in-plane, $\mathrm{m\,s^{-1}}$)	σ(mag, $\mathrm{m\,s^{-1}}$)
e1	per-step noise	2,000	×0.25	23.6%	99.6	4.4
e2	run-coherent bias	1,000	×0.25	19.0%	121.8	4.3
e3	none (bias only)	1,000	×0.25	22.1%	96.5	4.4
e4	per-step noise	400	×1.0	38.3%	67.9	16.3

Table 11. Linearity bounds on the crossing epoch (measured closed forms). Perspective terms are exactly captured by Cartesian constant-velocity propagation; the quoted term is pure catalog-epoch bookkeeping (26 yr offset case; the 10 yr case is 11.7 yr). The wobble row is a bounded aim offset (19.4 AU maximum, 0.7% of the success radius); the 4.74 $\mathrm{km\,s^{-1}}$ hazard of a non-barycentric proper motion does not apply because the Kervella 2016 solution is barycentric. Doppler factor 0.99992529. Displayed rows are rounded; the combined row sums the unrounded magnitudes.

Term	$\lvert\Delta T_{\mathrm{cross}}\rvert$ (yr)	$\lvert\Delta v_\infty\rvert$ ($\mathrm{km\,s^{-1}}$)
Perspective acceleration (26-yr epoch offset)	30.36	0.0107
AB barycenter wobble (aim-offset bound)	8.28	0.0012
Proxima pull on AB barycenter (Kepler arc; Fermi-team term)	7.07	0.0030
Light-time / Doppler retardation	5.96	0.0018
Light-time retardation, catalog-epoch bound	4.37	–
Galactic tide over T_{cross}	0.33	0.0001
Combined worst case (sum of magnitudes)	56.36	0.0168

Table 12. Joint end-to-end Monte Carlo (simulation). Columns: central crossing epoch, central GTO-100 margin, joint P under assumed execution levels (ephemeris uncertainty on), joint P with the measured 99.6 $\mathrm{m\,s^{-1}}$ powered-phase residual of Section 5.2 applied isotropically (conservative), and joint LEO-150 P at assumed levels. The kinematic rows carry the Section 4.5 RV correction. Scoring convention: the inner dispersion model scores the miss at closest approach t_{ca} (T_{nom} sets the aim state only); the two Akeson-based rows run on the working state of Table 5, with the J2019.5-state values in the text.

Scenario	T_{cross} (yr)	Margin (pts)	P (assumed)	P (meas. resid.)	P LEO-150
Kervella, carried RV	79,785.7	+3.82	0.964	0.606	0.046
Akeson, carried RV	79,371.9	+3.73	0.984	0.629	0.000
Kervella, kinematic RV	80,117.9	+3.84	0.964	0.603	0.014
Akeson, kinematic RV	79,720.8	+3.74	0.983	0.625	0.000

Radial-velocity error is along-track (correlation with miss $+0.010$), moving the crossing date while leaving the miss almost unchanged. On the Akeson J2019.5 catalog state the central dispersion probability is 0.9858 (kinematic) / 0.9866 (observed-γ), and the joint P of Table 12 sits 0.0010 to 0.0011 below the central dispersion P in every scenario, so the J2019.5-state joint P is ≈0.985 to 0.986; full regeneration of the joint table and sensitivity artifacts on the J2019.5 state is the single recommended follow-up work order (the golden fixture of Section 7.3, pinned to that state, accompanies this memorandum).

7. Recommendation

7.1 The publishable arrival statement: two exact wording options

Option A is the recommended primary wording, built to the adopted program convention: the Akeson 2021 state pinned at its native epoch J2019.5, the kinematic radial-velocity frame, and T counted from the 2029.0 departure, with the Fermi team's ±400–800 yr band estimate adopted as a program input.

> **Option A (recommended; Akeson J2019.5 state, kinematic frame, T from departure 2029.0).** "The required departure asymptote crosses the ecliptic plane at T = 79,756 yr after the 2029.0 departure, crossing date AD 81,785, program band ±400–800 yr (measured; Akeson et al. 2021 barycentric state at its native epoch J2019.5, kinematic frame $-22.3182 \pm 0.0034\,\mathrm{km\,s^{-1}}$, frame correction per Kervella, Thévenin & Lovis 2017 (A&A 598, L7), central value 79,756.4 yr, crossing distance 6.428 ly, v_∞ 24.158 $\mathrm{km\,s^{-1}}$). The observed-γ variant ($V_0 = -22.3796\,\mathrm{km\,s^{-1}}$ with the 0.10 $\mathrm{km\,s^{-1}}$ carried allowance) gives 79,457.5 yr, AD 81,486.5; the frame split is 298.9 yr at the measured 4,870 yr-per-$\mathrm{km\,s^{-1}}$ slope. The catalog–frame scenario spread (Akeson/Kervella × observed/kinematic) spans AD 81,486 to 82,118, half-spread ≈316 yr, supporting the band floor; the cross-catalog proper-motion systematic and the Proxima-pull term argue against narrowing below ±400 yr, and the Fermi team's published 79,252 yr lies inside the band ceiling under either origin bookkeeping. The breaking point is the radial-velocity uncertainty; a 0.03 $\mathrm{km\,s^{-1}}$ kinematic measurement would narrow the band to ±200 yr."

Option B fits a document that must stay Kervella-primary for consistency with the certified feasibility chain.

> **Option B (Kervella-primary alternative).** "The required departure asymptote crosses the ecliptic plane at T = 79,786 ± 1,470 yr on the Kervella state clock (measured; Kervella et al. 2016 catalog parallax, proper motions, and gamma end to end on a J2000-family state basis, spectroscopic frame, diagonal covariance); the kinematic-frame central value is 80,118 yr, a +0.23σ shift. The epoch-consistent Akeson et al. 2021 state gives 79,467.0 yr from J2019.5 (spectroscopic V_0) and 79,765.9 yr (kinematic), and the Fermi Explorer team's independent analysis gives 79,252 yr; all values agree within one sigma of either band. The breaking point is the radial-velocity uncertainty, at 4,860 yr of crossing-epoch shift per $\mathrm{km\,s^{-1}}$, with the frame convention worth a further one-sided 330 to 350 yr."

One bookkeeping caution on Option B: the carried Kervella state pairs the K16 catalog observables with a project position constant on a J2000-family basis (the K16 catalog barycenter is published at epoch J1991.25; Table 5), so its T is counted on that state clock, and it is not counted from the 2029.0 departure. The epoch-consistency analysis of Section 4.4 covers the Akeson state only; we did not apply it to the Kervella state, which is one more reason Option A is the recommended wording.

The rationale for Akeson 2021 as the primary basis: the most recent catalog with the smallest published uncertainties (V_0 sigma 0.0020 $\mathrm{km\,s^{-1}}$; the 0.10 $\mathrm{km\,s^{-1}}$ the variant carries is an adopted absolute-accuracy allowance for frame and zero-point systematics that appears in no catalog) and a native J2019.5 epoch that the exchange state pins. The rationale for the kinematic frame: it is the true line-of-sight motion the trajectory needs, and the spectroscopic value hides the verified $-61.4 \pm 2.7\,\mathrm{m\,s^{-1}}$ differential gravitational-redshift bias, whose $+61.4\,\mathrm{m\,s^{-1}}$ correction is worth +299 yr on the epoch-consistent Akeson base (Section 4.5). The quoted band is a program input adopted from the Fermi team's estimate (it is not a propagated covariance); the Akeson-based carried-sigma width (±489) falls inside its range, the Kervella width (±1,468) exceeds it, and the 316 yr scenario half-spread of Option A sits below its floor. The covariance is assumed diagonal throughout.

7.2 Joint action list for the Fermi team

- **Adopt one catalog end to end, at its native epoch.** We recommend Akeson 2021 pinned to J2019.5 for both teams: barycenter 219.85892215°/−60.83163195°, parallax 750.81 ± 0.38 mas, proper motion (−3639.95 ± 0.42, +700.40 ± 0.17) mas yr^{-1}, $V_0 = -22.3796 \pm 0.0020\,\mathrm{km\,s^{-1}}$ spectroscopic. This removes the entire 533.5 yr input effect and the 0.094 $\mathrm{km\,s^{-1}}$ raw v_∞ offset at once.
- **State the radial-velocity convention and adopt the kinematic frame.** Their `astro.py` carries the spectroscopic $-22.4\,\mathrm{km\,s^{-1}}$ for AB with no stated convention; one comment line fixes it. (The matching housekeeping item on our side is the mixed-convention constants module of Section 4.5; their repository is clean.) The kinematic AB values are $-22.332\,\mathrm{km\,s^{-1}}$ (Kervella base) and $-22.318\,\mathrm{km\,s^{-1}}$ (Akeson base); the correction moves the crossing 299 to 332 yr later and changes no verdict (measured).
- **Exchange one state vector.** A single ecliptic-frame position and velocity at the pinned epoch J2019.5 (program input, accepted), printed to full precision, verifies the whole kinematic chain in one comparison.

Our reproduction matrix shows conventions contribute 0.01 yr; the exchange makes that check permanent, and the golden fixture below automates it.

- **Treat the radial velocity as the program risk.** It carries 4,864 yr per $\mathrm{km\,s^{-1}}$ of crossing-epoch shift (measured), 16 times the parallax leverage under the Kervella-based carried allowances and 9 times under the Akeson-based, it drives 99.1% of the joint arrival-epoch variance (simulation), and its convention alone is worth 330 to 350 yr across the Kervella and Akeson working bases (measured; Table 7). Any future refinement effort should buy kinematic radial-velocity precision first.
- **Keep the crossing epoch as the design default.** Their design argument holds and this analysis priced it: the crossing is fixed by geometry, costs 22 to 27 $\mathrm{m\,s^{-1}}$ over the measured basin bottom (measured; 0.11% of total), keeps arrival probability between 0.965 and 0.987 (simulation; 0.965 Kervella, 0.984 Akeson working state, 0.9858/0.9866 Akeson J2019.5 state), closes GTO-100 with +3.8 pts margin (simulation), and moved by zero when their pricing model was corrected. Quote it with the astrometric band above; the band is the honest content of the number.

7.3 Offer: a golden fixture for permanent cross-validation

We offer the Fermi team a golden fixture that makes the code-versus-inputs separation permanent. The fixture is pinned to the exchange state of this memorandum: the Akeson 2021 barycenter observables at their native epoch J2019.5, used epoch-consistently, in the kinematic radial-velocity frame ($-22.3182 \pm 0.0034\,\mathrm{km\,s^{-1}}$), with the state clock at J2019.5 and the departure-2029.0 origin reported alongside. The package is four small files: the agreed astrometric state with pinned unit constants and five test epochs; the expected outputs, T_{cross} = 79,765.87 yr on the state clock (79,756.37 yr from departure, crossing date AD 81,785.37), crossing distance 6.427805 ly, v_∞ at the crossing 24.15829 $\mathrm{km\,s^{-1}}$, and v_∞/β at the five epochs, fifteen checked values in all, with stated tolerances of 0.02 yr, 10^{-4} ($\mathrm{km\,s^{-1}}$ and degrees), and 10^{-6} ly; a standalone stdlib-only Python checker that recomputes and asserts every value; and a README declaring every convention (state epoch, time origins, frame, and the observed-γ variant). The checker passes on a clean system interpreter with no project code on the path (verified), and an independent recompute through a different call path agrees to machine precision. If the Fermi implementation reproduces the expected outputs, any future disagreement between the two teams is attributable to inputs alone, with code ruled out; one run replaces an entire reconciliation exercise. The package accompanies this memorandum.

7.4 Verdict summary

The 534 yr gap is an input difference: distance carries 528.8 yr of it, the convention effect is -0.01 yr, and the gap is not statistically significant (the Fermi value sits at -0.36σ in the Kervella band and -0.25σ in the Akeson band; the ours-minus-Fermi separation itself is 1.09σ of the Akeson band, still inside statistical agreement). Their 79,252 yr reproduces exactly from their inputs (verified). Their 77,500 yr PP optimum agrees with our independent model to 1 yr (verified) and sits 16 $\mathrm{m\,s^{-1}}$ above the measured basin bottom, inside a basin flat to 25 $\mathrm{m\,s^{-1}}$ (measured). The 73,012 yr impulsive optimum stands (verified). The measurement program returned six confirmed verdicts from six items, with every anchor gate passing at bit-level or better and no tolerance tuned to pass, and the one number it adds to the headline set is the joint $P = 0.964$ (simulation; assumed execution levels) for the recommended crossing-epoch design. No tuning was performed on either implementation at any point.

A. Methods, seeds, and reproducibility

This appendix is self-contained: it states every input, gate, and seed needed to reproduce the memorandum's numbers from public information, with no reference to internal file layouts.

A.1 Inputs

Two Alpha Centauri AB barycentric states, ICRS, with linear motion: *Kervella 2016 basis*: catalog parallax 747.17 ± 0.61 mas, $\mu_{\alpha*}$ −3619.9 and μ_δ +693.8 mas yr^{-1} (±3.9 each, catalog sigmas), catalog gamma $-22.3930 \pm 0.0043\,\mathrm{km\,s^{-1}}$ carried as −22.4 with a 0.30 $\mathrm{km\,s^{-1}}$ carried working sigma (adopted allowance), all from A&A 594, A107 (primary solution). The position RA 219.90206°, Dec −60.83399° paired with these observables is

a project constant on a J2000-family basis, and it is not a K16 catalog value (the K16 catalog barycenter is published at epoch J1991.25), so the carried Kervella state runs on a J2000-family state clock. *Akeson 2021* (AJ 162, 14, native epoch J2019.5): barycenter 219.85892215°/−60.83163195°, parallax 750.81 ± 0.38 mas, $\mu_{\alpha*}$ -3639.95 ± 0.42 and μ_δ $+700.40 \pm 0.17$ mas yr^{-1}, $V_0 = -22.3796 \pm 0.0020\,\mathrm{km\,s^{-1}}$. The Akeson-based working state pairs the Akeson parallax with adopted proper motions (-3644.4, $+702.8$; ± 1.5 mas yr^{-1} working sigmas; no catalog provenance) and the K16/PB16 gamma rounded to -22.39 with a 0.10 $\mathrm{km\,s^{-1}}$ allowance on a J2000-family position; the working-state Monte Carlo rows of the main text run on it (Table 5). The Fermi team's published inputs, from their public repository: RA 219.9021°, Dec -60.834°, distance 4.344 ly, $\mu_{\alpha*}$ -3620.0 and μ_δ $+694.0$ mas yr^{-1}, radial velocity $-22.4\,\mathrm{km\,s^{-1}}$. Radial-velocity source values for the convention audit (verified against re-fetched source text): Kervella 2016 orbit gamma $-22.393\,\mathrm{km\,s^{-1}}$, Pourbaix–Boffin 2016 (A&A 586, A90) $-22.390\,\mathrm{km\,s^{-1}}$, Akeson 2021 $V_0 = -22.3796\,\mathrm{km\,s^{-1}}$, all spectroscopic; the kinematic correction adds $+61.4 \pm 2.7\,\mathrm{m\,s^{-1}}$, removing the differential gravitational redshift $\Delta v_{\mathrm{GRS}} = -61.4 \pm 2.7\,\mathrm{m\,s^{-1}}$ published by Kervella, Thévenin & Lovis 2017 (A&A 598, L7).

A.2 Crossing-epoch computation

Rotate the state to ecliptic coordinates with J2000 mean obliquity 23.43928°. Convert proper motion with $\kappa = 4.74047\,\mathrm{km\,s^{-1}}$ per arcsec yr^{-1} at 1 pc and a Julian year of 31,557,600 s. Propagate linearly and solve $\beta(T) = 0$ for the required asymptote, equivalently $T = -z_0/v_z$. The convention matrix of Section 4 swaps the obliquity digits (23.439281), the light-year value (9,460,730,472,580.8 km against 9.46073×10^{12} km), and the κ digits (4.740470463) between the two sides; the combined effect on the crossing epoch is 0.01 yr (measured).

A.3 Astrometric Monte Carlo band

10,000 draws, seed 20260803, independent Gaussians on parallax, both proper-motion components, and radial velocity (diagonal covariance, assumed; no published cross-terms are carried in either input set). The linear one-sigma propagation agrees with the Monte Carlo sigma within 0.5% for both catalogs (measured). The full percentile sets (measured, in years, spectroscopic frame):

- **Kervella 2016 state:** 2.5% 77,029; 16% 78,351; 50% 79,793; 84% 81,287; 97.5% 82,755; mean 79,816; Monte Carlo sigma 1,462; linear sigma 1,468.
- **Akeson-based working state:** 2.5% 78,434; 16% 78,888; 50% 79,374; 84% 79,862; 97.5% 80,332; mean 79,376; Monte Carlo sigma 487; linear sigma 489.

The kinematic-frame values shift each percentile later by 332 yr (Kervella) or 349 yr (Akeson working base; 299 on the epoch-consistent state); the 2.7 $\mathrm{m\,s^{-1}}$ correction uncertainty adds 13 yr, inside the stated sigmas (computed).

A.4 3-D tilt measurements

The tilt table uses the certified 3-D pump propagator of the feasibility chain with a 2,600 AU miss credit and the AB barycenter aim. Acceptance per row: the six heritage gates plus a direction gate $|\beta_{\mathrm{achieved}} - \beta_{\mathrm{target}}| \le 0.04°$; at the crossing epoch the gate reads $|\beta_{\mathrm{achieved}}| \le 0.04°$. Each row runs a time-step ladder at $\times 1.0/\times 0.5/\times 0.25$ and carries the maximum gate-passing value; ladder spread per row is under 0.4%. Search recipe: 600 coarse plus 8 polish starts (3-D) and 400 plus 6 (planar), Nelder–Mead polish with one deterministic restart, seed base 20260803 with fixed per-row offsets. Anchors: the planar embedding reproduces the heritage planar value to $10^{-9}\,\mathrm{km\,s^{-1}}$, and the certified 3-D ladder reproduces the heritage $24.5627\,\mathrm{km\,s^{-1}}$ anchor exactly (verified). The 3-D totals are best-found upper bounds with $\pm 0.2\,\mathrm{km\,s^{-1}}$ seed scatter; the tilt increments in Table 1 are differences of matched runs under identical seeds.

A.5 Basin sweep and adjudication

The empirical basin sweep runs epochs 74,000 to 81,000 yr in 500 yr steps, five seeds per epoch, seed base 20260804 with explicit per-run offsets, and the bounded 3-D search recipe imported verbatim from the certified heritage code. The four anchor gates of Table 2 pass before any sweep value is read. The sweep com-

pleted in 1,715 s against a 2,100 s budget with all 75 cells feasible at 5/5 seeds. The adjudication run reran the 79,500 yr cell with five fresh seeds and the 80,500 yr cell with three, spot re-ran 78,500 and 79,000 (both exact reproductions), and re-laddered both outlier winners on an extended ladder down to ×0.0625. The surrogate-roughness check compares the joint Monte Carlo's $\sigma_{\mathrm{surr}} = 0.0091\,\mathrm{km\,s^{-1}}$ against the measured quadratic-detrend rms 0.0073 $\mathrm{km\,s^{-1}}$ of the adjudicated envelope.

A.6 Dispersion and closure at the design epochs

The dispersion runs use the certified pipeline with 20,000 draws per case and the published error budget: pointing 0.10° per axis, magnitude 50 $\mathrm{m\,s^{-1}}$, residual 25 $\mathrm{m\,s^{-1}}$ per axis (all engineering assumptions), carried ephemeris sigmas (Table 5), and the 2,600 AU success radius. Seeds: 42 (published baseline anchor, reproduced exactly), 20260804 (Kervella crossing), 20260805 (Akeson crossing), 20260806 (catalog control). Closure interpolates the 12-yr pumped Δv column between the 79,000 and 80,000 yr best-found rows and extrapolates the bracket-end margins at the measured slopes -1.818 (GTO-100) and -1.627 (LEO-150) points per $\mathrm{km\,s^{-1}}$.

A.7 Powered-phase execution Monte Carlo

A perturbed clone of the certified 3-D propagator flies the planar winner open-loop under a fractional thrust-magnitude bias $\varepsilon \sim N(0, 2.1 \times 10^{-3})$ plus the pointing channels of Table 10, with the heritage energy-trigger cutoff, a 12.0 yr wall against the 11.78 yr nominal burn, and perfect energy knowledge. Seeds 20260804 to 20260807 by experiment. The unperturbed control reproduces the carried 23.984510138157503 $\mathrm{km\,s^{-1}}$ bitwise (verified). The deterministic ε sweep brackets the completion cliff between $+0.0010$ and $+0.0020$.

A.8 Joint Monte Carlo

Outer loop: 10,000 astrometric draws per scenario, seeds 20260806+idx. Inner loop: 1,000 execution draws per outer draw on a deterministic seed lattice. The pumped Δv surrogate carries the best-found table plus $\sigma_{\mathrm{surr}} = 0.0091\,\mathrm{km\,s^{-1}}$; margins inherit the closure model above. Variance attributions use one-at-a-time normalization. The ephemeris-off control (0.986) isolates the ephemeris cost at the Kervella crossing.

A.9 Honesty constraints and the adversarial pass

Neither implementation was adjusted toward the other at any point. The reproduction of the Fermi crossing epoch ran their published inputs through a clean reimplementation, and the match (0.25 yr, inside their integer rounding) required no fitting. The quadratic tilt model was fixed from a single prior anchor before the multi-epoch measurement existed. All stochastic results carry the fixed seeds stated in this appendix, and negative controls (the planar-collapse residual, 0.015 $\mathrm{km\,s^{-1}}$) are reported alongside the positive results. An independent adversarial verification pass re-ran all analysis scripts to completion (artifacts bit-identical up to wall-clock strings), recomputed headline values through different call paths, and confirmed all six verdicts. No tolerance was found tuned to pass: all anchor gates are bit-level, 10^{-6}, or 10^{-9}, and the one loose gate (the 0.05 $\mathrm{km\,s^{-1}}$ planar-collapse gate against a measured 0.0149 $\mathrm{km\,s^{-1}}$) is declared in-script with its prior measured value referenced.